Technical Appendices

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27 October 2026

This revision updates my mathematical appendix from *How To Build Conscious Machines*. Current book definitions, statements and proofs govern shared material; independent earlier results are retained. Claims and notation precede formal development. Experimental results follow. Earlier foundations appear in my doctoral thesis [Bennett, 2025b].

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Notation guide

Notation	Meaning and formal source
Φ, $\mathcal{P}$, $\mathfrak{v}$	Environment, declarative programs, vocabulary; Definition 1.
$L_{\mathfrak{v}}$, l, $T(l)$	Embodied language, statement or finite bundle, truth set; Definition 3.
$\mathrm{Ext}(l)$, $\mathrm{Ext}(X)$, $\mathrm{ff}(l)$	Statement/set extension and freedom; Definition 4. Earlier notation: E_l, E_X, $w(l)$.
$<_{\mathrm{ff}}$, $<_s$	Freedom and simplicity selectors; Definition 16.
$\mathrm{Enc}_{\mathfrak{v}}$, $S_{\mathfrak{v}}$, $\kappa_{\mathfrak{v}}$, $\kappa^{\tau}_{\mathfrak{v}}$	Encoding, layer-space count, layer-time count and induced clock; Definition 7.
τ, h, $\tau_{\leq t}$	Timeline, finite fragment and prefix; Definition 2.
$\rho_{\tau,[u_1,u_2]}$, $\geq_{\tau,[u_1,u_2]}$	Persistence count and ordering; Definition 12.
α, I_α, O_α, $\Gamma_{\mathfrak{v}}$	Task, inputs, correct outputs and task space; Definition 13.
π, Π_α	Policy and correct-policy set; Definition 14.
$\sqsubset$, $\ell_{\mathfrak{v}}$, λ_ρ	Strict child relation, task height and disembodied task; Definitions 13 and 15.
U, B_π, $G_P(\pi)$	Unseen region, policy buffer and demand-survival probability; Definitions 17 and 19.
μ, ff_μ	Reference measure and measured freedom; Definition 19.
β_γ, $\epsilon(\gamma)$	Unseen-buffer value and utility of intelligence; Definition 22.
$\mathfrak{f}$, χ_i, g_i^{χ}	Abstractor, witness choice and one-step grounding; Definitions 5 and 6.
$\mathfrak{v}^i$, π^i, λ^i	Vocabulary, policy and disembodied task at stack layer i; Definition 23.
$K_\rho(M)$, R_ρ	Weighted forgetting cost and normalised regret; Theorem 11. Here ρ denotes weights.
$\mathcal{V}$, $F(\pi)$, $I(\pi)$	Compared viable policies, survival probability and functional information; Theorem 14.
$\Delta(c)$, P_τ, P_N	Continuation demands, induced novelty law and exogenous novelty law; Definitions 26 and 28.
δ, $\mathbb{D}_H$, Q_H	Damage history, history space and damage law; Definition 27.
$R_\pi(H)$, $q_\pi(H)$, m_π, $k(\pi)$	Mend, mend probability, finite repair capacity and failure-point count; Definition 27.
$P_H(\pi)$, $\Lambda_H(\pi)$	Joint persistence probability and log persistence; Definition 28.
μ, Ω_π, $F_2(\pi)$	Viability statement, viable completions and viability divergence; Corollary 13. This μ is distinct from the reference measure above.
INT, OBS, $\mathcal{C}(INT, OBS)$	Intervention statements, matched observations and causal identities; Definitions 29 and 30.
$\mathfrak{o}$, $\mathfrak{p}_{\mathfrak{o}}$, $\mu_{\mathfrak{o}}$, $<_{\mathfrak{o}}$	Organism, retained policies, viability task and preferences; Definition 31.
$\mathfrak{o}^1$, $\mathfrak{o}^n$, $c_{\mathfrak{a}}^{\mathfrak{r}_1 \cdots \mathfrak{r}_k}$	First/higher-order selves and nested causal identities; Definitions 33–34.
$\mathfrak{s}_{\mathfrak{o}}$, $\mathfrak{S}_{\mathfrak{o}}(i)$, $\alpha^\star$	Protosymbol system, signified protosymbols and selected interpretation; Definitions 35 and 36.
Π'_ω, Π'_{ω_j}	Restricted collective and component policy sets; Definition 40.
ℓ, Sites_ℓ, Graph_ℓ, c_{conc}, θ	Moment, physical sites, interaction graph, concurrency capacity and physical time window; Definitions 42 and 43.
$D(\ell)$, $L_{\max}$, v, ε	Diameter, longest required edge, maximum signal speed and architecture factor; Definition 44.
$\mathsf{I}(\mathfrak{v})$, $\mathsf{I}_{\max}(\mathcal{R})$	Encoding capacity and physical patch ceiling; Definitions 10 and 11.
$\beta_{\lambda_\rho, V}$	Semantic Unagi score; Definition 24.
O_A, $\mathfrak{v}^{\mathrm{conf}}_{M,S}$	Oracle program and bounded-machine configuration vocabulary; Definitions 47 and 9.
φ, (W, b), $\mathcal{U}$, PP	Feature map, shared readout, evaluation units and pair proxy; Definitions 52, 54 and 57.

Claims map

Status: M, mathematical result; C, conditional deduction; E, empirical report; I, interpretation; H, hypothesis or design proposal; Q, conjecture. Formal statements specify the premises. Entries may combine statuses.

Claim	Status	Statements, proofs and evidence
Freest-correct generalisation	M	Theorem 2; Lemma 7; Corollary 6.
Usage bias confounds simplicity with generalisation	M, I	Corollaries 9, 10 and 11.
Compression is not necessary for generalisation	M, E	Proposition 9; Section 28.5.
The Law of the Stack	M, I	Proposition 15; Definition 22.
Psychophysical Principle of Causality	M, I	Theorems 18 and 19; Corollary 2.
Arpeggio collapse and inclusion drift	C, I	Theorems 23 and 24.
Multilayer functionalism	M, I, H	Corollaries 1 and 27; Proposition 7; Section 20.
The Cosmic Ought	M, I	Lemma 6; Corollary 4.
Bennett's Razor and Unagi	M, H	Proposition 20; Corollaries 6 and 17; Section 14.
Chord and the spacetime diameter bound	C, H	Definition 42; Theorem 25; Corollary 26.
Criteria for conscious contraptions	C, H	Proposition 33; Corollaries 25 and 26; Section 20.
Function-level freedom in neural networks	M, E	Propositions 11, 48 and 49; Sections 28.1, 28.2 and 28.3.
Law of Increasing Functional Information	M, I	Lemma 13; Proposition 21; Theorem 14.
Autopoietic Theorem	C, I	Theorem 17.
Human-like access requires first- and second-order selves	C, I	Theorems 20 and 21; Section 20.
Efficient biological access implies feeling	C, I	Theorems 18 and 21; Corollary 21; Section 20.
No selves, no alignment	C, H	Theorems 20, 21 and 22; Section 19.
A simple explanation of life	M, I, E	Proposition 21; Definition 28; Section 28.4.
Closure necessity and the void of the unviable	C, I	Lemmas 14 and 15; Theorems 15 and 16.
Collective identity and splintering	M, I, E	Proposition 30; Corollary 24; Section 28.
Regret is freedom lost	M	Theorem 11; Corollary 17; Section 13.
Delegation and biological intelligence	M, I	Proposition 15; Corollary 13; physical identification.
Time in a stack	M	Theorem 10; Propositions 1 and 3.
Evaluation precedes representation	M, I	Corollaries 21 and 22.
Time is space	M, I	Proposition 4; Corollary 3.
Strong orthogonality fails in embodied systems	M, I	Proposition 32.
Surplus and recursive improvement	C	Proposition 23; Corollary 19; Proposition 24.
P versus NP depends on the vocabulary	M, I	Propositions 38, 39 and 40; Section 21.
Recursive self impairment	Q	Conjecture on recursive impairment of collective coordination; Section 19.

Further results and applications

Result or application	Status	Statements, proofs and evidence
Grounding preserves truth and persistence	M	Lemma 1; Theorem 1; Corollary 2.
Product-vocabulary alignment criterion	M	Lemma 8; Theorem 3; Corollary 7.
Encoding capacity and physical ceilings	M, C	Definitions 10–11; Lemma 5; Section 4.
PAC-Bayes demand and completion certificates	M	Theorems 8 and 9; Corollary 8; Proposition 12.
Neural compatibility and measurement	M	Propositions 48 and 49; Section 24.
Permitted-output freedom formula	M	Section 28.2.
Assembly-index relativity and endpoint non-identification	M	Proposition 41; Theorem 26.
Engineered life niches	C	Corollary 20; Section 15.
Agreement of first-order selves	M	Proposition 28.
All-orders self acquisition	C	Definitions 32 and 34; Proposition 27.
Deception and credible self-binding	C	Corollary 23; Theorem 22.
Cooperation with advance detection	C	Proposition 29.
Mirror Symbol Hypothesis	H, I	Preference reuse and empathy; Section 18.
Global-broadcast independence	C, I	Theorem 21; Section 20.
IIT sufficiency and unit-choice counterexamples	I	Proposition 7; Section 20.
Organisational-invariance application	C, I	Proposition 34; Section 20.
Inert-record cost and present evaluation	M	Proposition 33; Corollary 25; trace conditions.
Occurrence does not determine temporal order	M	Proposition 36.
Repeated functional-information selection	C	Theorem 14; Section 15.
Empirical self hierarchy	E	Section 28.
Qualia, attention and narrative	I	Definition 41; Section 20.

1 Environment and embodied language

Definition 1 (environment, programs, and vocabularies)**.**
An *environment* is a nonempty set Φ of mutually exclusive states. It is assumed infinite by default. A *(declarative) program* is any set of states $p \subseteq \Phi$. For any set X, I write 2^X for the set of all subsets of X. In particular, let $\mathcal{P} := 2^\Phi$ denote the set of all programs. A program is a *potential* fact, and its truth conditions are the states it contains. It is true or false relative to a state. Given a state, a program is true if it contains the state.

I say a program is a *fact* if it is true. An *aspect* of the environment is a set x of programs such that $x \in \{\, l \subseteq \mathcal{P} \mid \bigcap_{p\in l} p \neq \emptyset \,\}$. Given state ϕ, aspect x exists if it is a set of facts, meaning $\phi \in \bigcap_{p\in x} p$.

A *vocabulary* is any set of programs $\mathfrak{v} \subseteq \mathcal{P}$. Unlike an aspect, a vocabulary's intersection can be empty, and usually is. I call a vocabulary with a non-empty intersection degenerate. Unless specified otherwise, vocabularies and aspects are assumed finite.

Definition 2 (Timelines and objective time)**.**
A **timeline** is a function

$$\tau : I \to \Phi,$$

where I is either $\mathbb{N}$ or a finite initial segment $\{0, \dots, N\}$, such that

$$\tau(u+1) \neq \tau(u) \qquad \text{whenever } u, u+1 \in I.$$

Each index $u \in I$ is a **tick**. Written without qualification, a tick is always an objective tick, a position in this order. Layer time, defined in Section 3, has ticks of its own, and I call those layer ticks. Every state at a tick is a distinction and therefore a change. The state at every tick after the first also differs from the immediately preceding state. A state may recur after one or more intervening ticks. States at different ticks may nevertheless have the same encoding in a vocabulary. For a starting tick $u \in I$ and an integer $n \geq 0$ with $u + n \in I$, the **finite timeline fragment**

$$h = \tau_{[u,u+n]} := (\tau(u), \tau(u+1), \dots, \tau(u+n)) \in \Phi^{n+1}$$

is the restriction of that timeline to $n+1$ consecutive ticks. Its consecutive entries satisfy $\tau(u+j+1) \neq \tau(u+j)$ for every $0 \leq j < n$, while nonconsecutive entries may coincide. A **timeline prefix** is the special case, for $t \in I$,

$$\tau_{\leq t} := \tau_{[0,t]} = (\tau(0), \dots, \tau(t)).$$

Objective time is this order of changes. Its index records order without adding another object to the environment.

Definition 3 (abstraction layer, statements, and truth sets)**.**
An **abstraction layer** is an environment Φ considered through a finite vocabulary $\mathfrak{v} \subseteq \mathcal{P}$. A **finite bundle** is any finite $l \subseteq \mathfrak{v}$, whether or not its truth set is empty. I define the truth set of a finite bundle by

$$T(l) := \bigcap_{p\in l} p, \qquad T(\emptyset) := \Phi.$$

The layer induces the embodied language

$$L_\mathfrak{v} = \{\, l \subseteq \mathfrak{v} \mid l \text{ is finite and } T(l) \neq \emptyset \,\}.$$

Elements $l \in L_\mathfrak{v}$ are **statements**. A statement is true at $\phi \in \Phi$ iff $\phi \in T(l)$. Since an aspect is any satisfiable set of programs, the statements are the aspects contained in the vocabulary:

$$L_\mathfrak{v} = \{x \subseteq \mathfrak{v} \mid x \text{ is an aspect}\}.$$

Definition 4 (completions, extensions, and freedom)**.**
A **completion** of $x \in L_{\mathfrak{v}}$ is any $y \in L_{\mathfrak{v}}$ with $x \subseteq y$. The **extension** of x is

$$\mathrm{Ext}(x) := \{y \in L_{\mathfrak{v}} \mid x \subseteq y\}.$$

For $X \subseteq L_{\mathfrak{v}}$, I write

$$\mathrm{Ext}(X) := \bigcup_{x \in X} \mathrm{Ext}(x).$$

The **freedom** of x is the cardinality of the extension

$$\mathrm{ff}(x) := |\,\mathrm{Ext}(x)|.$$

Freedom is also called the **weakness** of constraints. Here x is a constraint, and a weaker constraint allows more possibilities. These possibilities are the completions of x, which together form the extension of x. Freedom is used to order statements by

$$x <_{\mathrm{ff}} y \quad \Longleftrightarrow \quad \mathrm{ff}(x) < \mathrm{ff}(y).$$

This ordering ranks y ahead of x when y has more completions. A statement is weaker when it has greater freedom and is ranked higher by $<_{\mathrm{ff}}$. Just to be extra clear, freedom does not require one extension to contain another. They just need to be in the same embodied language. I say statements x and y are **equivalent in** $L_{\mathfrak{v}}$ iff

$$\mathrm{ff}(x) = \mathrm{ff}(y).$$

This equivalence means they have the same rank in terms of weakness. It is relative to the embodied language because both extension sets are computed inside $L_{\mathfrak{v}}$.

2 Abstraction and grounding

Definition 5 (abstractor function)**.**
An **abstractor function** $\mathfrak{f}$ is defined on pairs $(\mathfrak{v}, l)$ where $\mathfrak{v} \subseteq \mathcal{P}$ is a vocabulary and $l \in L_{\mathfrak{v}}$ is a statement. It returns the sub-vocabulary

$$\mathfrak{f}(\mathfrak{v}, l) \;:=\; \{T(o) \mid o \in \mathrm{Ext}(l)\} \subseteq \mathcal{P}.$$

In words, $\mathfrak{f}(\mathfrak{v}, l)$ is the set of truth sets of all completions of l.

Definition 6 (one-step grounding)**.**
Let $\mathfrak{v}^{i+1} \subseteq \mathfrak{f}(\mathfrak{v}^i, l_i)$. A **witness choice** is a map from higher-layer programs to lower-layer completions

$$\chi_i : \mathfrak{v}^{i+1} \to \mathrm{Ext}(l_i)$$

such that $T(\chi_i(q)) = q$ for every $q \in \mathfrak{v}^{i+1}$. For a layer-i+1 finite bundle $a \subseteq \mathfrak{v}^{i+1}$, its **one-step grounding** under χ_i is

$$g_i^{\chi}(a) := \bigcup_{q \in a} \chi_i(q) \subseteq \mathfrak{v}^i.$$

Lemma 1 (one-step truth preservation)**.**
For every finite bundle $a \subseteq \mathfrak{v}^{i+1}$, $g_i^{\chi}(a) \subseteq \mathfrak{v}^i$ is a finite bundle with

$$T(g_i^{\chi}(a)) = \bigcap_{q \in a} q = T(a),$$

and $g_i^{\chi}(a) \in L_{\mathfrak{v}^i}$ if and only if $a \in L_{\mathfrak{v}^{i+1}}$. Thus one-step grounding preserves the truth set of the higher-layer bundle.

Proof. Each $\chi_i(q)$ lies in $L_{\mathfrak{v}^i}$, so is a finite subset of $\mathfrak{v}^i$, and the union $g_i^\chi(a)$ of finitely many of them is a finite bundle in $\mathfrak{v}^i$. By definition,

$$T(g_i^\chi(a)) = \bigcap_{q\in a} T(\chi_i(q)).$$

The witness choice satisfies $T(\chi_i(q)) = q$ for every $q \in a$. By substitution, $T(g_i^\chi(a)) = \bigcap_{q\in a} q = T(a)$. Both $g_i^\chi(a)$ and a are finite bundles with this truth set, so $g_i^\chi(a) \in L_{\mathfrak{v}^i}$ if and only if $a \in L_{\mathfrak{v}^{i+1}}$. □

Theorem 1 (compositional grounding)**.**
Let a finite chain of abstractor steps

$$\mathfrak{v}^{i+1} \subseteq \mathfrak{f}(\mathfrak{v}^i, l_i), \qquad i = 0, \dots, n-1,$$

have witness choices χ_i, and let $a_n \subseteq \mathfrak{v}^n$ be a top-layer finite bundle with the recursively defined groundings

$$a_i := g_i^\chi(a_{i+1}), \qquad i = n-1, \dots, 0.$$

Then each grounding step preserves the same truth set, so the top-layer bundle is grounded at layer 0 by a_0:

$$T(a_0) = T(a_1) = \cdots = T(a_n),$$

and $a_0 \in L_{\mathfrak{v}^0}$ if and only if $a_n \in L_{\mathfrak{v}^n}$.

Proof. I apply Lemma 1 at each adjacent pair of layers. By induction over $i = n-1, \dots, 0$, each a_i is a finite bundle in $\mathfrak{v}^i$ and $T(a_0) = T(a_1) = \cdots = T(a_n)$. Since a_0 and a_n are finite bundles with the same truth set, $a_0 \in L_{\mathfrak{v}^0}$ if and only if $a_n \in L_{\mathfrak{v}^n}$. □

Corollary 1 (Functional agreement across layers)**.**
Two systems can compute the same input–output relation at layer $i+1$ while computing different input–output relations at layer i. Two groundings at layer i of one program at layer $i+1$ can have the same truth set and strictly nested extensions, hence unequal freedom.

Proof. For the first claim, I choose states $\phi_1 \neq \phi_2$ and programs $h = \{\phi_1, \phi_2\}$, $c_1 = \{\phi_1\}$ and $c_2 = \{\phi_2\}$. The first system has layer-i vocabulary $\{h, c_1\}$, the second has $\{h, c_2\}$, and each realises the policy $\{h\}$. The completions of $\{h\}$ are $\{h\}$ and $\{h, c_k\}$, with truth sets h and $h \cap c_k = c_k$, so the abstractor returns $\{h, c_1\}$ for the first system and $\{h, c_2\}$ for the second. Each system can therefore embody the layer-$(i{+}1)$ vocabulary $\{h\}$ (Definition 23), and at layer $i+1$ both compute one relation, the task with input $\{h\}$ and correct output $\{h\}$. At layer i the completions of $\{h\}$ differ, with $\{h, c_1\}$ true at ϕ_1 alone in the first system and $\{h, c_2\}$ true at ϕ_2 alone in the second, so the systems compute different relations at layer i. For the second claim, I choose distinct programs $p, q \in \mathfrak{v}^i$ with $q = \Phi$ and $\varnothing \neq p \subsetneq \Phi$. The statements $x = \{p\}$ and $y = \{p, q\}$ have the same truth set, since $T(x) = p = T(y)$. Either can ground the same program at layer $i+1$. Yet $x \subsetneq y$, so $\mathrm{Ext}(y) \subsetneq \mathrm{Ext}(x)$ and $\mathrm{ff}(y) < \mathrm{ff}(x)$. The groundings agree functionally at layer $i+1$ and differ functionally at layer i. □

Corollary 2 (Persistence under grounding)**.**
Fix a finite chain as in Theorem 1, and let l, c be finite bundles at its highest layer. I use g_l, g_c for their groundings at its lowest layer. For any timeline $\tau : I \to \Phi$ and $u_1, u_2 \in I$ with $u_1 \leq u_2$,

$$\begin{aligned}\rho_{\tau,[u_1,u_2]}(g_l \cup g_c) &= \rho_{\tau,[u_1,u_2]}(l \cup c) \\ &\leq \rho_{\tau,[u_1,u_2]}(l) = \rho_{\tau,[u_1,u_2]}(g_l).\end{aligned}$$

Strict inequality holds if and only if τ visits $T(l) \setminus T(c)$ during that interval.

Proof. By Theorem 1, $T(g_l) = T(l)$ and $T(g_c) = T(c)$. Hence $T(g_l \cup g_c) = T(l \cup c)$, so both equalities follow from Definition 12. By Corollary 4, weak inequality holds. The two sets of ticks counted in the weak inequality differ by

$$\{u \in \{u_1, \dots, u_2\} : \tau(u) \in T(l) \setminus T(c)\},$$

so strict inequality holds if and only if this set is nonempty. □

3 Encoding, space and time

Definition 7 (encoding, layer space, and layer time)**.**
Fix a finite vocabulary $\mathfrak{v} \subseteq \mathcal{P}$ of possible facts. For a state $\phi \in \Phi$, the **encoding** of ϕ in $\mathfrak{v}$ is

$$\mathrm{Enc}_{\mathfrak{v}}(\phi) := \{p \in \mathfrak{v} : \phi \in p\}.$$

The **layer-space count** is

$$S_{\mathfrak{v}}(\phi) := |\,\mathrm{Enc}_{\mathfrak{v}}(\phi)|.$$

For a finite timeline fragment $h = (\phi_0, \dots, \phi_n)$, the **layer-time count** is

$$\kappa_{\mathfrak{v}}(h) := |\{j \in \{1, \dots, n\} : \mathrm{Enc}_{\mathfrak{v}}(\phi_j) \neq \mathrm{Enc}_{\mathfrak{v}}(\phi_{j-1})\}|\,.$$

For a timeline $\tau : I \to \Phi$ and $t \in I$, the clock induced at layer $\mathfrak{v}$ is

$$\kappa^{\tau}_{\mathfrak{v}}(t) := \kappa_{\mathfrak{v}}(\tau_{\leq t}).$$

Objective time advances at every tick of τ. Layer time advances only when the layer's encoding changes.

Definition 8 (bounded and embodied Turing machine)**.**
A deterministic one-tape Turing machine is

$$M = (Q, \Sigma, \Gamma, b, \delta, q_0, q_{\mathrm{acc}}, q_{\mathrm{rej}}).$$

Here Q is a finite set of **control states**, meaning labels of the machine's controller rather than environmental states in Φ. Alphabets are sets of **symbols**. The finite **input alphabet** Σ is contained in the finite **tape alphabet** Γ. The symbol $b \in \Gamma \setminus \Sigma$ is blank. The pairwise distinct labels q_0, q_{acc}, and q_{rej} are the start, accepting, and rejecting control states. The transition function

$$\delta : (Q \setminus \{q_{\mathrm{acc}}, q_{\mathrm{rej}}\}) \times \Gamma \longrightarrow Q \times \Gamma \times \{L, R\}$$

specifies one machine step. If $\delta(q, a) = (q', a', D)$, the machine reads a under its head, writes a', changes control from q to q', and moves the head one cell left or right according to D. A computation starts in q_0 with its input beginning at tape cell 1, blank symbols elsewhere, and the head on cell 1. The computation ends when it reaches q_{acc} or q_{rej}.

Fix a tape bound $S \in \mathbb{N}$ with $S \geq 2$. An **embodied S-bounded implementation** of M in Φ consists of a nonempty program $\varnothing \neq \mathcal{M}_{M,S} \subseteq \Phi$, the states in which the machine is realised, and declarative programs

$$Q_q, \quad H_i, \quad C_{j,a}, \quad U_j \subseteq \Phi$$

indexed by $q \in Q$, $1 \leq i, j \leq S$, and $a \in \Gamma$. On $\mathcal{M}_{M,S}$, the control programs partition its environmental states, the head programs partition its environmental states, and for each tape cell j the symbol programs partition its environmental states

$$\mathcal{M}_{M,S} = \bigsqcup_{q \in Q} (Q_q \cap \mathcal{M}_{M,S}) = \bigsqcup_{i=1}^{S} (H_i \cap \mathcal{M}_{M,S}),$$

$$\mathcal{M}_{M,S} = \bigsqcup_{a\in\Gamma} (C_{j,a} \cap \mathcal{M}_{M,S}) \qquad (1 \le j \le S).$$

Thus every $\phi \in \mathcal{M}_{M,S}$ determines unique values

$$q(\phi) \in Q, \qquad i(\phi) \in \{1,\dots,S\}, \qquad T_\phi(j) \in \Gamma,$$

and usage values $u_\phi(j) \in \{0,1\}$ defined by $u_\phi(j) = 1$ when $\phi \in U_j$.

A pair $\phi, \psi \in \mathcal{M}_{M,S}$ is a **legal bounded machine step** when $q(\phi)$ is nonhalting and, writing

$$\delta(q(\phi), T_\phi(i(\phi))) = (q', a', D),$$

the next state satisfies

$$q(\psi) = q', \qquad i(\psi) = \begin{cases} i(\phi) - 1, & D = L, \\ i(\phi) + 1, & D = R, \end{cases}$$

with $1 \le i(\psi) \le S$, and

$$T_\psi(j) = \begin{cases} a', & j = i(\phi), \\ T_\phi(j), & j \ne i(\phi). \end{cases}$$

Cells already marked as used remain marked, and the newly visited head cell is also marked as used

$$u_\psi(j) = 1 \quad \Longleftrightarrow \quad u_\phi(j) = 1 \text{ or } j = i(\psi).$$

A **legal bounded run** is a finite timeline fragment $r = (\phi_0, \dots, \phi_n)$ in $\mathcal{M}_{M,S}$ with the following properties. Its first state ϕ_0 has control state q_0, head on cell 1 and the input written from cell 1 with blanks elsewhere. Every consecutive pair is a legal bounded machine step. Its initial usage programs identify the initial head cell. The run ends when it reaches a halting control state or a chosen final tick. The machine is therefore an embodied aspect of Φ. Its step number does not define another clock, but clicks along inside the environment. You know, like a real computer does.

Definition 9 (bounded Turing vocabularies)**.**
For an embodied S-bounded implementation, the **configuration vocabulary** and **usage vocabulary** are

$$\mathfrak{v}^{\mathrm{conf}}_{M,S} := \{Q_q : q \in Q\} \cup \{H_i : 1 \le i \le S\} \cup \{C_{j,a} : 1 \le j \le S,\ a \in \Gamma\},$$

$$\mathfrak{v}^{\mathrm{use}}_{M,S} := \{U_j : 1 \le j \le S\}.$$

Both are vocabularies over the original environment because every member is a subset of Φ. For $\phi \in \mathcal{M}_{M,S}$,

$$\mathrm{Enc}_{\mathfrak{v}^{\mathrm{conf}}_{M,S}}(\phi) = \{Q_{q(\phi)}, H_{i(\phi)}\} \cup \{C_{j,T_\phi(j)} : 1 \le j \le S\},$$

$$\mathrm{Enc}_{\mathfrak{v}^{\mathrm{use}}_{M,S}}(\phi) = \{U_j : u_\phi(j) = 1\}.$$

Proposition 1 (machine time is layer time)**.**
Let

$$r = (\phi_0, \dots, \phi_n)$$

be a finite timeline fragment in $\mathcal{M}_{M,S}$ whose consecutive pairs are legal bounded machine steps. I define its machine time by $\mathrm{Time}_M(r) := n$, the number of legal transitions. Then for every finite vocabulary $\mathfrak{v}$ containing $H_1, \dots, H_S$,

$$\kappa_{\mathfrak{v}}(r) = \mathrm{Time}_M(r) = n.$$

In particular this holds for every legal bounded run r of M and for $\mathfrak{v} = \mathfrak{v}^{\mathrm{conf}}_{M,S}$.

Proof. In every legal step the head moves from some cell i to a different cell i'. Hence the true head-position program changes from H_i to $H_{i'}$, so

$$\mathrm{Enc}_{\{H_1,\dots,H_S\}}(\phi_j) \neq \mathrm{Enc}_{\{H_1,\dots,H_S\}}(\phi_{j-1})$$

for every $j = 1, \dots, n$. The fragment contains n consecutive pairs, so $\kappa_{\{H_1,\dots,H_S\}}(r) = n$. Since $\{H_1, \dots, H_S\} \subseteq \mathfrak{v}$, every pair at which the head encoding changes is a pair at which the encoding in $\mathfrak{v}$ changes, so $\kappa_{\mathfrak{v}}(r) \geq n$, and $\kappa_{\mathfrak{v}}(r) \leq n$ because r has n consecutive pairs. □

Proposition 2 (machine space is layer space)**.**
Let $r = (\phi_0, \dots, \phi_n)$ be a legal bounded run. For $0 \leq t \leq n$, define used space by

$$\mathrm{Space}_M(r, t) := |\{j : u_{\phi_t}(j) = 1\}|.$$

Then

$$\mathrm{Space}_M(r, t) = S_{\mathfrak{v}^{\mathrm{use}}_{M,S}}(\phi_t).$$

Proof. The usage program U_j is true at ϕ_t when $u_{\phi_t}(j) = 1$. The number of true usage programs is the number of cells used at that time in the run. It is the amount of layer-space used by the usage vocabulary. □

Proposition 3 (monotonicity of layer time)**.**
Let $\mathfrak{v}' \subseteq \mathfrak{v}$ be finite vocabularies over Φ. For every finite timeline fragment h, every tick of h at which the encoding in $\mathfrak{v}'$ changes is a tick at which the encoding in $\mathfrak{v}$ changes, so

$$\kappa_{\mathfrak{v}'}(h) \leq \kappa_{\mathfrak{v}}(h),$$

with equality if and only if every tick of h at which the encoding in $\mathfrak{v}$ changes is a tick at which the encoding in $\mathfrak{v}'$ changes. Consequently, for every timeline $\tau : I \to \Phi$ and $t \in I$,

$$\kappa^{\tau}_{\mathfrak{v}'}(t) \leq \kappa^{\tau}_{\mathfrak{v}}(t).$$

Proof. Let $h = (\phi_0, \dots, \phi_n)$. If the smaller vocabulary records a tick at step j, then

$$\mathrm{Enc}_{\mathfrak{v}'}(\phi_j) \neq \mathrm{Enc}_{\mathfrak{v}'}(\phi_{j-1}).$$

Hence some program $p \in \mathfrak{v}'$ is true at one of the two states and false at the other. Since $\mathfrak{v}' \subseteq \mathfrak{v}$, that same program belongs to $\mathfrak{v}$, so

$$\mathrm{Enc}_{\mathfrak{v}}(\phi_j) \neq \mathrm{Enc}_{\mathfrak{v}}(\phi_{j-1}).$$

Every tick counted by $\kappa_{\mathfrak{v}'}(h)$ is therefore counted by $\kappa_{\mathfrak{v}}(h)$. A subset of a finite set has as many elements as that set if and only if the two are equal, so $\kappa_{\mathfrak{v}'}(h) = \kappa_{\mathfrak{v}}(h)$ if and only if every tick counted by $\kappa_{\mathfrak{v}}(h)$ is counted by $\kappa_{\mathfrak{v}'}(h)$. Taking $h = \tau_{\leq t}$, I get $\kappa^{\tau}_{\mathfrak{v}'}(t) \leq \kappa^{\tau}_{\mathfrak{v}}(t)$. □

Corollary 3 (layer-space bound)**.**
For every finite vocabulary $\mathfrak{v}$ and state $\phi \in \Phi$,

$$S_{\mathfrak{v}}(\phi) \leq |\mathfrak{v}|.$$

Proof. $\mathrm{Enc}_{\mathfrak{v}}(\phi) \subseteq \mathfrak{v}$. □

Proposition 4 (Embodied recoding of temporal distinctions)**.**
Fix a finite vocabulary $\mathfrak{v}$ and a finite timeline fragment

$$h = (\phi_0, \ldots, \phi_n).$$

The pairs of a program and a tick at which it is true in the fragment are

$$A_{\mathfrak{v}}(h) := \{(p, t) \mid 0 \leq t \leq n,\ p \in \mathrm{Enc}_{\mathfrak{v}}(\phi_t)\},$$

and the visible ticks are

$$C_{\mathfrak{v}}(h) := \{t \in \{1, \ldots, n\} \mid \mathrm{Enc}_{\mathfrak{v}}(\phi_t) \neq \mathrm{Enc}_{\mathfrak{v}}(\phi_{t-1})\}.$$

Let $\psi \in \Phi$ be a current state and let $\mathfrak{w}$ be a finite vocabulary over Φ. The current encoding is the encoding of ψ in $\mathfrak{w}$.

If the current encoding contains a different true program for every member of $A_{\mathfrak{v}}(h)$, meaning there is an injection

$$e_A : A_{\mathfrak{v}}(h) \hookrightarrow \mathrm{Enc}_{\mathfrak{w}}(\psi),$$

then

$$S_{\mathfrak{w}}(\psi) \geq \sum_{t=0}^{n} S_{\mathfrak{v}}(\phi_t).$$

If the current encoding contains a different true program for every member of $C_{\mathfrak{v}}(h)$, meaning there is an injection

$$e_C : C_{\mathfrak{v}}(h) \hookrightarrow \mathrm{Enc}_{\mathfrak{w}}(\psi),$$

then

$$S_{\mathfrak{w}}(\psi) \geq \kappa_{\mathfrak{v}}(h).$$

Bound $S_{\mathfrak{w}}(\psi) \geq \sum_{t=0}^{n} S_{\mathfrak{v}}(\phi_t)$ is an equality if and only if $e_A(A_{\mathfrak{v}}(h)) = \mathrm{Enc}_{\mathfrak{w}}(\psi)$. Bound $S_{\mathfrak{w}}(\psi) \geq \kappa_{\mathfrak{v}}(h)$ is an equality if and only if $e_C(C_{\mathfrak{v}}(h)) = \mathrm{Enc}_{\mathfrak{w}}(\psi)$. If $S_{\mathfrak{w}}(\psi) < \sum_{t=0}^{n} S_{\mathfrak{v}}(\phi_t)$, no injection e_A exists. If $S_{\mathfrak{w}}(\psi) < \kappa_{\mathfrak{v}}(h)$, no injection e_C exists.

Proof. Because each pair contains its time index, the sets

$$\{(p, t) \mid p \in \mathrm{Enc}_{\mathfrak{v}}(\phi_t)\}$$

are disjoint for different t. Hence

$$|A_{\mathfrak{v}}(h)| = \sum_{t=0}^{n} |\,\mathrm{Enc}_{\mathfrak{v}}(\phi_t)| = \sum_{t=0}^{n} S_{\mathfrak{v}}(\phi_t).$$

An injection of $A_{\mathfrak{v}}(h)$ into $\mathrm{Enc}_{\mathfrak{w}}(\psi)$ implies

$$S_{\mathfrak{w}}(\psi) = |\,\mathrm{Enc}_{\mathfrak{w}}(\psi)| \geq |A_{\mathfrak{v}}(h)|.$$

Also, by Definition 7,

$$|C_{\mathfrak{v}}(h)| = \kappa_{\mathfrak{v}}(h).$$

Injection $e_C : C_{\mathfrak{v}}(h) \hookrightarrow \mathrm{Enc}_{\mathfrak{w}}(\psi)$ requires $S_{\mathfrak{w}}(\psi) \geq \kappa_{\mathfrak{v}}(h)$. Surjectivity of e_A implies $S_{\mathfrak{w}}(\psi) = \sum_{t=0}^{n} S_{\mathfrak{v}}(\phi_t)$, while surjectivity of e_C entails $S_{\mathfrak{w}}(\psi) = \kappa_{\mathfrak{v}}(h)$. Conversely, an injection between finite sets of equal size is a surjection, so equality in either bound forces the image of its injection to be all of $\mathrm{Enc}_{\mathfrak{w}}(\psi)$. If $|\,\mathrm{Enc}_{\mathfrak{w}}(\psi)|$ is smaller than the size of a source set, then by the pigeonhole principle no injection from that set exists. □

Lemma 2 (Encoding is the maximal true statement)**.**
For every $\phi \in \Phi$, $\mathrm{Enc}_{\mathfrak{v}}(\phi)$ is true at ϕ. If $l \in L_{\mathfrak{v}}$ is also true at ϕ, then $l \subseteq \mathrm{Enc}_{\mathfrak{v}}(\phi)$.

Proof. Let $\phi \in \Phi$. By definition, if $p \in \mathrm{Enc}_{\mathfrak{v}}(\phi)$ then $\phi \in p$. Therefore $\phi \in \bigcap_{p \in \mathrm{Enc}_{\mathfrak{v}}(\phi)} p = T(\mathrm{Enc}_{\mathfrak{v}}(\phi))$.

Now let $l \in L_{\mathfrak{v}}$ and assume $\phi \in T(l) = \bigcap_{p \in l} p$. Then for every $p \in l$ I have $\phi \in p$, hence $p \in \mathrm{Enc}_{\mathfrak{v}}(\phi)$. So $l \subseteq \mathrm{Enc}_{\mathfrak{v}}(\phi)$. □

Proposition 5 (Layer time can be arbitrarily sparse)**.**
For any $M \in \mathbb{N}_{>0}$ there exist an environment Φ, a finite vocabulary $\mathfrak{v} \subseteq \mathcal{P}$, and a timeline $\tau : \mathbb{N} \to \Phi$ such that its encoding $\mathrm{Enc}_{\mathfrak{v}}(\tau(u))$ changes at most once during the first M objective ticks.

Proof. I take $\Phi = \{0, 1\}$, $\mathfrak{v} = \{\Phi\}$ and $\tau(u) = u \bmod 2$. Consecutive environmental states differ, while $\mathrm{Enc}_{\mathfrak{v}}(\tau(u)) = \{\Phi\}$ at every tick. Layer time remains zero for every finite fragment. □

Lemma 3 (The language is the union of all local powersets)**.**
Fix an environment Φ and a finite vocabulary $\mathfrak{v} \subseteq 2^{\Phi}$. Then

$$L_{\mathfrak{v}} = \bigcup_{\phi \in \Phi} 2^{\mathrm{Enc}_{\mathfrak{v}}(\phi)} = \{l \subseteq \mathfrak{v} \mid \exists\, \phi \in \Phi :\ l \subseteq \mathrm{Enc}_{\mathfrak{v}}(\phi)\}.$$

Proof. If $l \in L_{\mathfrak{v}}$, choose $\phi \in T(l)$. Then Definition 7 gives $l \subseteq \mathrm{Enc}_{\mathfrak{v}}(\phi)$, so $l \in 2^{\mathrm{Enc}_{\mathfrak{v}}(\phi)}$. Conversely, if $l \subseteq \mathrm{Enc}_{\mathfrak{v}}(\phi)$ for some ϕ, then Definition 7 gives $\phi \in T(l)$. So $T(l) \neq \emptyset$ and $l \in L_{\mathfrak{v}}$. □

Proposition 6 (Extensions are unions of encoding intervals)**.**
Assume $\mathfrak{v}$ is finite. For a statement $l \in L_{\mathfrak{v}}$, write $\mathcal{M}(l) := \{\mathrm{Enc}_{\mathfrak{v}}(\phi) \mid \phi \in T(l)\}$ for the set of realised maximal descriptions compatible with l. Then

$$\mathrm{Ext}(l) = \bigcup_{m \in \mathcal{M}(l)} \{y \subseteq \mathfrak{v} \mid l \subseteq y \subseteq m\}.$$

Consequently, $\max_{m \in \mathcal{M}(l)} 2^{|m|-|l|} \leq \mathrm{ff}(l) \leq \sum_{m \in \mathcal{M}(l)} 2^{|m|-|l|}$.

Proof. Let $y \in \mathrm{Ext}(l)$. Then $l \subseteq y$ and $y \in L_{\mathfrak{v}}$. Choose $\phi \in T(y)$. Then $\phi \in T(l)$ since $T(y) \subseteq T(l)$. By Definition 7, $y \subseteq \mathrm{Enc}_{\mathfrak{v}}(\phi)$. So y lies between l and $m := \mathrm{Enc}_{\mathfrak{v}}(\phi)$, and $m \in \mathcal{M}(l)$.

Conversely, suppose $m \in \mathcal{M}(l)$ and $l \subseteq y \subseteq m$. By definition of $\mathcal{M}(l)$, there exists $\phi \in T(l)$ with $m = \mathrm{Enc}_{\mathfrak{v}}(\phi)$. Then $y \subseteq \mathrm{Enc}_{\mathfrak{v}}(\phi)$. By Definition 7, $\phi \in T(y)$. So $y \in L_{\mathfrak{v}}$ and $y \in \mathrm{Ext}(l)$.

For the bounds, each set $\{y \mid l \subseteq y \subseteq m\}$ has exactly $2^{|m|-|l|}$ elements. The union contains each such set, so its size is at least the maximum. The size of a union is at most the sum. □

Lemma 4 (Ordered recursion for weakness)**.**
Fix a finite vocabulary $\mathfrak{v} = \{p_0, \ldots, p_{n-1}\}$ under any fixed total order. For $l \in L_{\mathfrak{v}}$ and $j \in \{0, \ldots, n\}$, define

$$\mathrm{ff}_j(l) := \left|\{y \in \mathrm{Ext}(l) \mid y \setminus l \subseteq \{p_j, p_{j+1}, \ldots, p_{n-1}\}\}\right|.$$

This counts completions that only add programs with index $\geq j$. Then

$$\mathrm{ff}_j(l) = 1 + \sum_{\substack{k \in \{j, \ldots, n-1\} \\ p_k \notin l \\ l \cup \{p_k\} \in L_{\mathfrak{v}}}} \mathrm{ff}_{k+1}(l \cup \{p_k\}).$$

In particular, $\mathrm{ff}(l) = \mathrm{ff}_0(l)$.

Proof. Every y counted by $\mathrm{ff}_j(l)$ satisfies $l \subseteq y$ and $y \setminus l \subseteq \{p_j, \ldots, p_{n-1}\}$. If $y = l$, it contributes the initial 1. If $y \supsetneq l$, let p_k be the smallest-index program in $y \setminus l$. Then $k \geq j$ (by the index restriction on ff_j), $p_k \notin l$, $l \cup \{p_k\} \subseteq y$, and $l \cup \{p_k\} \in L_{\mathfrak{v}}$ (since y is a statement and $l \cup \{p_k\} \subseteq y$). The remaining programs in $y \setminus (l \cup \{p_k\})$ all have index $> k$ (because p_k was the smallest). So y is counted exactly once in $\mathrm{ff}_{k+1}(l \cup \{p_k\})$. The sets $\{y \in \mathrm{Ext}_j(l) : y \supsetneq l, \text{ min-index of } y \setminus l = k\}$ for different k are disjoint, and together with $\{l\}$ they partition the set counted by $\mathrm{ff}_j(l)$. Summing over all valid k gives the recursion. The base case $j = n$ gives $\mathrm{ff}_n(l) = 1$ (only $y = l$ survives). Setting $j = 0$ recovers $\mathrm{ff}(l) = \mathrm{ff}_0(l)$. □

4 Encoding capacity and functional sufficiency

Definition 10 (Realised encodings and capacity)**.**
For a finite vocabulary $\mathfrak{v} \subseteq \mathcal{P}$, the **realised encoding set** is the image of the encoding map

$$\mathrm{Im}(\mathrm{Enc}_{\mathfrak{v}}) := \{\mathrm{Enc}_{\mathfrak{v}}(\phi) \mid \phi \in \Phi\} \subseteq L_{\mathfrak{v}}.$$

The **encoding capacity** of a finite vocabulary $\mathfrak{v}$ is

$$\mathsf{I}(\mathfrak{v}) := \log_2 |\mathrm{Im}(\mathrm{Enc}_{\mathfrak{v}})|.$$

Definition 11 (Patch capacity ceiling)**.**
If a bounded physical region $\mathcal{R}$ can sustain at most $N_{\max}(\mathcal{R})$ reliably distinguishable internal states, its bit ceiling is

$$\mathsf{I}_{\max}(\mathcal{R}) := \log_2 N_{\max}(\mathcal{R}).$$

A vocabulary is physically realisable inside $\mathcal{R}$ only if

$$\mathsf{I}(\mathfrak{v}) \leq \mathsf{I}_{\max}(\mathcal{R}).$$

For a complete, isolated, bounded, weakly self-gravitating system with circumscribing radius $r_{\mathcal{R}}$ and total rest-frame energy E_{tot} including rest energy, the Bekenstein bound gives

$$S_{\max} \leq \frac{2\pi k_B r_{\mathcal{R}} E_{\mathrm{tot}}}{\hbar c_{\mathrm{light}}}.$$

If $2^{\mathsf{I}_{\max}(\mathcal{R})} \leq \exp(S_{\max}/k_B)$ and $\mathfrak{v}$ is physically realisable in $\mathcal{R}$, then

$$\mathsf{I}(\mathfrak{v}) \leq \mathsf{I}_{\max}(\mathcal{R}) \leq \frac{2\pi r_{\mathcal{R}} E_{\mathrm{tot}}}{\hbar c_{\mathrm{light}} \ln 2}.$$

This is the stated conditional capacity bound [Bekenstein, 1981]. For a memory reset, let n_{erase} be the decrease of its Shannon entropy in bits. If the memory is uncorrelated with retained available side information, the implementation is cyclic with no compensating stored-energy change, and a thermal environment at Θ_0 is the entropy sink, then

$$Q_{\mathrm{diss}} \geq n_{\mathrm{erase}} k_B \Theta_0 \ln 2.$$

This is Landauer's bound under the stated reset conditions [Landauer, 1961]. The quantity n_{erase} is not generally freedom, functional information, or a change in either.

Lemma 5 (Union capacity is subadditive)**.**
For finite vocabularies $\mathfrak{v}_1$ and $\mathfrak{v}_2$ over one environment,

$$\mathsf{I}(\mathfrak{v}_1 \cup \mathfrak{v}_2) = \log_2 |\{(\mathrm{Enc}_{\mathfrak{v}_1}(\phi), \mathrm{Enc}_{\mathfrak{v}_2}(\phi)) \mid \phi \in \Phi\}|,$$

so

$$\max\{\mathsf{I}(\mathfrak{v}_1), \mathsf{I}(\mathfrak{v}_2)\} \leq \mathsf{I}(\mathfrak{v}_1 \cup \mathfrak{v}_2) \leq \mathsf{I}(\mathfrak{v}_1) + \mathsf{I}(\mathfrak{v}_2),$$

with equality on the right if and only if

$$\{(\mathrm{Enc}_{\mathfrak{v}_1}(\phi), \mathrm{Enc}_{\mathfrak{v}_2}(\phi)) \mid \phi \in \Phi\} = \mathrm{Im}(\mathrm{Enc}_{\mathfrak{v}_1}) \times \mathrm{Im}(\mathrm{Enc}_{\mathfrak{v}_2}).$$

Proof. Each realised encoding $c = \mathrm{Enc}_{\mathfrak{v}_1 \cup \mathfrak{v}_2}(\phi)$ of their union determines the pair

$$(c \cap \mathfrak{v}_1, c \cap \mathfrak{v}_2) = (\mathrm{Enc}_{\mathfrak{v}_1}(\phi), \mathrm{Enc}_{\mathfrak{v}_2}(\phi)).$$

The map $c \mapsto (c \cap \mathfrak{v}_1, c \cap \mathfrak{v}_2)$ into $\mathrm{Im}(\mathrm{Enc}_{\mathfrak{v}_1}) \times \mathrm{Im}(\mathrm{Enc}_{\mathfrak{v}_2})$ is injective, because $c = (c \cap \mathfrak{v}_1) \cup (c \cap \mathfrak{v}_2)$, and its image is $\{(\mathrm{Enc}_{\mathfrak{v}_1}(\phi), \mathrm{Enc}_{\mathfrak{v}_2}(\phi)) \mid \phi \in \Phi\}$. Hence

$$|\,\mathrm{Im}(\mathrm{Enc}_{\mathfrak{v}_1 \cup \mathfrak{v}_2})| = \left|\{(\mathrm{Enc}_{\mathfrak{v}_1}(\phi), \mathrm{Enc}_{\mathfrak{v}_2}(\phi)) \mid \phi \in \Phi\}\right| \leq |\,\mathrm{Im}(\mathrm{Enc}_{\mathfrak{v}_1})|\,|\,\mathrm{Im}(\mathrm{Enc}_{\mathfrak{v}_2})|,$$

with equality if and only if the image is the whole product, both being finite. Projecting the image onto its first coordinate gives all of $\mathrm{Im}(\mathrm{Enc}_{\mathfrak{v}_1})$, and onto its second all of $\mathrm{Im}(\mathrm{Enc}_{\mathfrak{v}_2})$, so the image has at least $\max\{|\,\mathrm{Im}(\mathrm{Enc}_{\mathfrak{v}_1})|, |\,\mathrm{Im}(\mathrm{Enc}_{\mathfrak{v}_2})|\}$ elements. Applying $\log_2$, I obtain the identity and both bounds. □

Proposition 7 (Functional sufficiency)**.**
Let $\mathcal{S}$ be a set of implementations, let $A : \mathcal{S} \to \mathcal{D}$ assign a description to each implementation, and let $\Psi : \mathcal{S} \to Y$ be a functional condition with values in a set Y. There exists $F : A(\mathcal{S}) \to Y$ satisfying $\Psi = F \circ A$ if and only if

$$A(s) = A(t) \quad \Longrightarrow \quad \Psi(s) = \Psi(t) \qquad (s, t \in \mathcal{S}).$$

The descriptions are sufficient for Ψ when such an F exists. Thus one pair with equal descriptions and unequal Ψ values disproves sufficiency of those descriptions for Ψ.

Proof. If $\Psi = F \circ A$ and $A(s) = A(t)$, then by substitution $\Psi(s) = F(A(s)) = F(A(t)) = \Psi(t)$. Conversely, assume equal descriptions imply equal Ψ values. For each $d \in A(\mathcal{S})$, define $F(d)$ as that common value on $\{s : A(s) = d\}$. This set is nonempty and all its members have the same Ψ value, so F is well defined and $F(A(s)) = \Psi(s)$ for every s. □

5 Persistence order

Definition 12 (Persistence and its ordering)**.**
Fix a vocabulary $\mathfrak{v}$, a timeline $\tau : I \to \Phi$, and $u_1, u_2 \in I$ with $u_1 \leq u_2$. The persistence measure of a finite bundle $x \subseteq \mathfrak{v}$ is

$$\rho_{\tau,[u_1,u_2]}(x) := |\{u \in \{u_1, \ldots, u_2\} \mid \tau(u) \in T(x)\}|.$$

If x is inconsistent, then $T(x) = \emptyset$ and its persistence count is zero.

For finite bundles $l, l' \subseteq \mathfrak{v}$ I write

$$l \geq_{\tau,[u_1,u_2]} l' \quad \text{iff} \quad \rho_{\tau,[u_1,u_2]}(l) \geq \rho_{\tau,[u_1,u_2]}(l').$$

The strict part is written $l >_{\tau,[u_1,u_2]} l'$ and means

$$\rho_{\tau,[u_1,u_2]}(l) > \rho_{\tau,[u_1,u_2]}(l').$$

The relation $\geq_{\tau,[u_1,u_2]}$ is a total preorder on finite bundles over $\mathfrak{v}$. It is total because any two finite counts can be compared. It is a preorder because equality of counts can tie distinct bundles.

Lemma 6 (Persistence monotonicity under refinement)**.**
Fix $\mathfrak{v}$, τ, and $[u_1, u_2]$. If $l, l' \subseteq \mathfrak{v}$ are finite bundles and $T(l') \subseteq T(l)$, then

$$\rho_{\tau,[u_1,u_2]}(l') \leq \rho_{\tau,[u_1,u_2]}(l).$$

Strict inequality $\rho_{\tau,[u_1,u_2]}(l') < \rho_{\tau,[u_1,u_2]}(l)$ holds if and only if $\tau(u) \in T(l) \setminus T(l')$ for some $u \in \{u_1, \ldots, u_2\}$.

Proof. For every tick u, $\tau(u) \in T(l')$ implies $\tau(u) \in T(l)$ because $T(l') \subseteq T(l)$. Therefore the set of ticks counted by l' is a subset of the set of ticks counted by l. Taking cardinalities, I obtain the weak inequality. The difference of the two tick sets is $\{u \in \{u_1, \dots, u_2\} : \tau(u) \in T(l) \setminus T(l')\}$, so the subset is proper if and only if the timeline visits a state in $T(l) \setminus T(l')$ during the interval. □

Corollary 4 (Extra commitments cannot increase persistence)**.**
Fix a vocabulary $\mathfrak{v}$, a timeline $\tau : I \to \Phi$, and $u_1, u_2 \in I$ with $u_1 \leq u_2$. Let $l, c \subseteq \mathfrak{v}$ be finite bundles, and read $T(l \cup c) = \emptyset$ when $l \cup c$ is inconsistent. Then

$$\rho_{\tau,[u_1,u_2]}(l \cup c) \leq \rho_{\tau,[u_1,u_2]}(l),$$

with strict inequality if and only if the timeline visits a state in $T(l) \setminus T(l \cup c)$ during the interval.

Proof. Since $T(l \cup c) = T(l) \cap T(c)$, we have $T(l \cup c) \subseteq T(l)$. I apply Lemma 6. □

The Cosmic Ought. Each timeline induces the persistence ordering above, and different timelines may induce different orderings. Added commitments cannot increase persistence on any fixed fragment. The Cosmic Ought interprets this ordering as the environment's normativity preceding organisms [Bennett, 2026d].

6 Tasks, inference and learning

Definition 13 ($\mathfrak{v}$-task)**.**
Let $\mathfrak{v} \subseteq \mathcal{P}$ be a vocabulary. A **$\mathfrak{v}$-task** is a pair

$$\alpha = \langle I_\alpha, O_\alpha \rangle$$

where

$$I_\alpha \subseteq L_\mathfrak{v} \qquad \text{and} \qquad O_\alpha \subseteq \mathrm{Ext}(I_\alpha).$$

The members of I_α are **inputs**. The members of O_α are **correct outputs**. I require every input to have at least one correct output, meaning every $i \in I_\alpha$ has some $o \in O_\alpha$ with $i \subseteq o$. Set $\Gamma_\mathfrak{v}$ consists of all $\mathfrak{v}$-tasks.

For tasks α and ω, I call α a **child** of ω, written $\alpha \sqsubset \omega$, iff

$$I_\alpha \subsetneq I_\omega \qquad \text{and} \qquad O_\alpha \subseteq O_\omega.$$

I use $\alpha \sqsubseteq \omega$ when $\alpha = \omega$ or $\alpha \sqsubset \omega$. If $\mathfrak{v}$ is finite, define the **level** or **height** of a task ω by

$$\ell_\mathfrak{v}(\omega) := \max \left\{ k \in \mathbb{N} \;\middle|\; \begin{array}{l} \exists \langle \alpha_0, \dots, \alpha_k \rangle, \quad \alpha_0 = \omega, \\ \alpha_{i+1} \sqsubset \alpha_i \text{ for } 0 \leq i < k \end{array} \right\}.$$

The maximum exists because a finite vocabulary induces a finite task space. The level counts how far the task can be specialised through strict child steps. A child contains fewer input cases and no correct outputs outside those of its parent. Learning from a child to a parent therefore means generalising from fewer cases to a broader task.

Definition 14 (inference)**.**

- A $\mathfrak{v}$-task **policy** is a statement $\pi \in L_\mathfrak{v}$. It constrains how we complete inputs.
- π is a **correct policy** iff the correct outputs O_α of α are all and only the completions π' of π for which π' is also a completion of an input.

- The set of all correct policies for a task α is

$$\Pi_\alpha := \{\pi \in L_\mathfrak{v} : \mathrm{Ext}(I_\alpha) \cap \mathrm{Ext}(\pi) = O_\alpha\}.$$

Assume $\mathfrak{v}$-task ω and a policy $\pi \in L_\mathfrak{v}$. Inference proceeds as follows:

1. we are presented with an input $i \in I_\omega$, and
2. when $\mathrm{Ext}(i) \cap \mathrm{Ext}(\pi) \neq \emptyset$, we select an output $e \in \mathrm{Ext}(i) \cap \mathrm{Ext}(\pi)$. If this set is empty, inference under π is undefined on i. For a correct policy $\pi \in \Pi_\omega$, the set is nonempty for every $i \in I_\omega$, because every input has a correct output.
3. If $e \in O_\omega$, then e is correct and the task "complete". $\pi \in \Pi_\omega$ implies $e \in O_\omega$, but $e \in O_\omega$ doesn't imply $\pi \in \Pi_\omega$ (an incorrect policy can imply a correct output).

Definition 15 (λ-tasks)**.**
Fix a $\mathcal{P}$-task ρ with $I_\rho \neq \emptyset$. A **disembodied task**, or **λ-task**, associated with ρ is a function λ_ρ whose domain contains every finite vocabulary $\mathfrak{v}' \subseteq \mathcal{P}$ and may contain additional vocabularies when stated. For every vocabulary in its domain, $\lambda_\rho(\mathfrak{v}')$ is a selected $\mathfrak{v}'$-task satisfying $\lambda_\rho(\mathfrak{v}') \sqsubseteq \rho$ and

$$\ell_{\mathfrak{v}'}(\lambda_\rho(\mathfrak{v}')) = \max\Big\{\ell_{\mathfrak{v}'}(\omega) \;\Big|\; \omega \text{ is a } \mathfrak{v}'\text{-task and } \omega \sqsubseteq \rho\Big\}.$$

Thus $\lambda_\rho(\mathfrak{v}')$ is an expressible child of ρ with greatest task height in $\mathfrak{v}'$. This set of expressible children is nonempty because $I_\rho \neq \emptyset$ implies that $\langle\emptyset, \emptyset\rangle$ is a strict child of ρ. When several maximisers exist, the λ-task includes a choice among them. When $\mathfrak{v}'$ is finite, $L_{\mathfrak{v}'}$ and the induced task space are finite, so the maximum exists. To compare λ-tasks, I fix a reference vocabulary $\mathfrak{v}_{\mathrm{ref}}$ in their common domain and write

$$\lambda_1 \sqsubset \lambda_2 \quad\Longleftrightarrow\quad \lambda_1(\mathfrak{v}_{\mathrm{ref}}) \sqsubset \lambda_2(\mathfrak{v}_{\mathrm{ref}}).$$

Adjacent tasks are ordered after the higher task is grounded into the lower language. Thus the stack condition at adjacent layers is

$$g_i^\chi\Big(\lambda^{i+1}(\mathfrak{v}^{i+1})\Big) \sqsubseteq \lambda^i(\mathfrak{v}^i),$$

where $i+1$ is more abstract than i and g_i^χ is the one-step grounding of Definition 6. This compares the tasks instantiated by the stack.

Definition 16 (learning and selectors)**.**
A **selector** $<$ is a binary relation on statements. I read $l_1 < l_2$ as "l_2 is ranked ahead of l_1".

A **learner** is a system represented by an abstraction layer with vocabulary $\mathfrak{v}$. Let α be a $\mathfrak{v}$-task with $\Pi_\alpha \neq \emptyset$, let ω be a $\mathfrak{v}$-task with $\alpha \sqsubset \omega$, and let $\mathcal{F}_\alpha \subseteq \Pi_\alpha$ be the nonempty set of correct policies that the learner has found. This set is all of Π_α when the learner has found every correct policy. The learner restricts attention to $\mathcal{F}_\alpha$ and applies a selector $<$ to it. The learner **learns** ω under $<$ if it outputs a $<$-maximal policy of $\mathcal{F}_\alpha$ that is correct for ω, meaning some $\pi^\star \in \mathcal{F}_\alpha \cap \Pi_\omega$ such that

$$\forall\, \pi \in \mathcal{F}_\alpha : \quad \neg(\pi^\star < \pi).$$

The learner, and the chosen $\pi^\star$, **generalise** to ω if the learner learns ω.

Two selectors are used throughout this appendix.

- $<_{\mathrm{ff}}$ is the **freedom** selector, given by the ordering of Definition 4. Its maximal elements have maximal freedom $\mathrm{ff}(l) = |\mathrm{Ext}(l)|$. A freest correct policy was called a weakest correct policy in earlier work.

- $<_s$ is the **simplicity** selector. Writing $|l|$ for the number of programs in l, we have $l_1 <_s l_2$ iff $|l_1| > |l_2|$. Its maximal elements have minimal description length in the chosen vocabulary.

Definition 17 (Generalisation probability)**.**
Let P_α be a probability distribution over finite sets S of further demands for a task α. Let B_π be the set of further demands compatible with a policy π that is correct for α. Define

$$G_{P_\alpha}(\pi) := P_\alpha(S \subseteq B_\pi)\,.$$

A parent task generated by S includes the requirements of α and adds every member of S as an input and as a correct output. For a policy correct for α, the unseen region and buffer are $U = L_{\mathfrak{v}} \setminus \mathrm{Ext}(I_\alpha)$ and $B_\pi = \mathrm{Ext}(\pi) \cap U$. The identity $\mathrm{Ext}(I_\alpha \cup S) \cap \mathrm{Ext}(\pi) = O_\alpha \cup (\mathrm{Ext}(S) \cap B_\pi)$ shows that the policy remains correct for that parent if and only if $S \subseteq B_\pi$ and S is closed under completion inside B_π, meaning every completion of a member of S that lies in B_π is itself in S.

7 Freest-correct generalisation

Proposition 8 (Uniform generalisation equivalence)**.**
Let $\mathfrak{v}$ be finite and let α be a $\mathfrak{v}$-task with $\Pi_\alpha \neq \emptyset$. I define an unseen region $U := L_{\mathfrak{v}} \setminus \mathrm{Ext}(I_\alpha)$. A demand is a set $S \subseteq U$. A policy $\pi \in \Pi_\alpha$ generalises to S when

$$S \subseteq B_\pi, \qquad B_\pi := \mathrm{Ext}(\pi) \cap U.$$

If S is drawn uniformly from 2^U, then every correct policy has generalisation probability $G_{P_\alpha}(\pi) = 2^{|B_\pi| - |U|}$, so generalisation probability is a strictly increasing function of freedom on Π_α. Hence a correct policy maximises generalisation probability if and only if it is $<_{\mathrm{ff}}$-maximal in Π_α. Every non-$<_{\mathrm{ff}}$-maximal correct policy is strictly suboptimal.

Proof. By Lemma 9 with μ as cardinality, maximising $|\,\mathrm{Ext}(\pi)|$ over Π_α is equivalent to maximising $|B_\pi|$. Under a uniform distribution of demands, π generalises to S when S is a subset of B_π. There are $2^{|B_\pi|}$ such demands out of $2^{|U|}$ possible demands. Therefore

$$G_{P_\alpha}(\pi) = 2^{|B_\pi| - |U|},$$

which is strictly increasing in $|B_\pi|$, and hence in $|\,\mathrm{Ext}(\pi)|$ over Π_α. Therefore the policies with maximal generalisation probability are the policies with maximal freedom. □

Theorem 2 (Freest-correct is Bayes optimal under exchangeable task growth)**.**
Fix a finite vocabulary $\mathfrak{v}$ and a child task α with $\Pi_\alpha \neq \emptyset$. Let $U := L_{\mathfrak{v}} \setminus \mathrm{Ext}(I_\alpha)$ be the unseen region. For each $\pi \in \Pi_\alpha$, I write $B_\pi := \mathrm{Ext}(\pi) \cap U$ for the unseen buffer of π. Let a later task add future demands $S \subseteq U$, and assume the distribution of S is exchangeable, meaning $P(S)$ depends only on $|S|$. A policy generalises to the demand when $S \subseteq B_\pi$. Then for all $\pi, \pi' \in \Pi_\alpha$ with $|\,\mathrm{Ext}(\pi)| \leq |\,\mathrm{Ext}(\pi')|$, $P(S \subseteq B_\pi) \leq P(S \subseteq B_{\pi'})$. For any such pair with $|\,\mathrm{Ext}(\pi)| < |\,\mathrm{Ext}(\pi')|$, this inequality is strict if and only if $P(1 \leq |S| \leq |B_{\pi'}|) > 0$. In particular every freest correct policy

$$\pi^* \in \arg\max_{\pi \in \Pi_\alpha} |\,\mathrm{Ext}(\pi)|$$

maximises $P(S \subseteq B_\pi)$. Either every correct policy has generalisation probability $P(S = \emptyset)$ or the maximisers of $P(S \subseteq B_\pi)$ are the freest correct policies and no others. Under a randomised selector, a policy is drawn at random according to some probability distribution. Among all randomised selectors supported on Π_α, no selector has higher expected generalisation probability than a freest correct policy.

Proof. By Lemma 9 with μ as cardinality, maximising $|\operatorname{Ext}(\pi)|$ over Π_α is equivalent to maximising $|B_\pi|$. Let $b = |B_\pi|$. Since the distribution of S is exchangeable, there are nonnegative numbers r_m such that each subset of U with size m has probability r_m. Therefore

$$P(S \subseteq B_\pi) = \sum_{m=0}^{b} \binom{b}{m} r_m.$$

Probability $P(S \subseteq B_\pi) = \sum_{m=0}^{b} \binom{b}{m} r_m$ is nondecreasing in b, because every summand is nonnegative and $\binom{b}{m}$ is nondecreasing in b for fixed m. Hence any policy with maximal buffer, equivalently maximal freedom, maximises generalisation probability. Since $|\operatorname{Ext}(\pi)| \leq |\operatorname{Ext}(\pi')|$ if and only if $|B_\pi| \leq |B_{\pi'}|$ by Lemma 9, a correct policy with no less freedom than another has no lower generalisation probability. For $b < b'$ and $1 \leq m \leq b'$, Pascal's rule gives $\binom{b'}{m} = \binom{b'-1}{m} + \binom{b'-1}{m-1} \geq \binom{b}{m} + 1$, so

$$\sum_{m=0}^{b'} \binom{b'}{m} r_m - \sum_{m=0}^{b} \binom{b}{m} r_m = \sum_{m=1}^{b'} \Big[\binom{b'}{m} - \binom{b}{m} \Big] r_m$$

is strictly positive if and only if $r_m > 0$ for some $1 \leq m \leq b'$, that is, if and only if $P(1 \leq |S| \leq b') > 0$. Let b^* be the maximal buffer size over Π_α. If $P(1 \leq |S| \leq b^*) = 0$, then $r_m = 0$ for $1 \leq m \leq b^*$ and every correct policy has generalisation probability $r_0 = P(S = \emptyset)$. Otherwise every correct policy with a buffer smaller than b^* has strictly lower generalisation probability than a freest correct policy. Under a randomised selector, expected generalisation probability is a convex combination of the generalisation probabilities of correct policies. A convex combination cannot exceed the largest of its terms. □

Corollary 5 (Optimality does not hinge upon a uniform prior)**.**
The uniform case was the first I proved and is I think the easiest to understand. However the optimality claims do not depend on that uniform prior. Under exchangeable demands, maximising freedom is optimal. Under independent demands with unequal probabilities, generalisation probability is a strictly increasing function of weighted freedom, so maximising weighted freedom is optimal. In atomless continuous domains, suppose equally measured regions have equal survival probability. Then survival probability never falls as buffer measure grows.

Proof. The uniform case is the exchangeable case $r_m = 2^{-|U|}$ for every subset. For independent biased futures, suppose each $u \in U$ is demanded with probability $p_u \in [0, 1)$. Then

$$P(S \subseteq B_\pi) = \prod_{u \notin B_\pi} (1 - p_u),$$

so $P(S \subseteq B_\pi) = \prod_{u \in U} (1 - p_u) \exp\big(\sum_{u \in B_\pi} - \log(1 - p_u)\big)$, with $\prod_{u \in U}(1 - p_u) > 0$, is a strictly increasing function of $\sum_{u \in B_\pi} - \log(1 - p_u)$, which is weighted freedom. For continuous domains, put a reference measure μ on U and assume equally measured regions have equal survival probability. If measurable $A, B \subseteq U$ satisfy $\mu(A) \leq \mu(B)$, by atomlessness there is a measurable $B' \subseteq B$ with $\mu(B') = \mu(A)$. Therefore $P(S \subseteq A) = P(S \subseteq B') \leq P(S \subseteq B)$. □

Lemma 7 (The freest correct policy permits every completion of every correct policy)**.**
Let α be a $\mathfrak{v}$-task with $I_\alpha \neq \emptyset$ and $\Pi_\alpha \neq \emptyset$, and let $C_\alpha := \bigcap_{i \in I_\alpha} i$ be the set of programs common to every input. Then there is a policy $R_\alpha \in \Pi_\alpha$ with $R_\alpha \cap C_\alpha = \emptyset$ and

$$\Pi_\alpha = \{R_\alpha \cup X \mid X \subseteq C_\alpha\}.$$

Hence R_α is the unique freest correct policy, every other correct policy contains more programs than R_α, and $\operatorname{Ext}(\pi) \subseteq \operatorname{Ext}(R_\alpha)$ for every $\pi \in \Pi_\alpha$.

Proof. I fix $\pi \in \Pi_\alpha$ and $i \in I_\alpha$. Since $\mathrm{Ext}(i) \subseteq \mathrm{Ext}(I_\alpha)$, the correctness equation $\mathrm{Ext}(I_\alpha) \cap \mathrm{Ext}(\pi) = O_\alpha$ gives $\mathrm{Ext}(i) \cap \mathrm{Ext}(\pi) = \mathrm{Ext}(i) \cap O_\alpha$. The input i has a correct output, so this set is nonempty. Any member of it contains $i \cup \pi$, and every subset of a statement is a statement, so $i \cup \pi \in L_\mathfrak{v}$ and $\mathrm{Ext}(i \cup \pi) = \mathrm{Ext}(i) \cap O_\alpha$. The same is true of any $\pi' \in \Pi_\alpha$, so $\mathrm{Ext}(i \cup \pi) = \mathrm{Ext}(i \cup \pi')$. A statement is the smallest member of its own extension, so statements with equal extensions are equal, and $i \cup \pi = i \cup \pi'$. This is true for every input, so every program in which π and π' differ lies in every input, which is to say in C_α. Hence $R_\alpha := \pi \setminus C_\alpha$ is the same for every $\pi \in \Pi_\alpha$, and every correct policy is $R_\alpha \cup X$ with $X = \pi \cap C_\alpha$. Conversely, let $X \subseteq C_\alpha$. For every input i, $C_\alpha \subseteq i$ gives $i \cup R_\alpha \cup X = i \cup \pi$. So $R_\alpha \cup X$ is a subset of the statement $i \cup \pi$, hence a statement, and $\mathrm{Ext}(i) \cap \mathrm{Ext}(R_\alpha \cup X) = \mathrm{Ext}(i \cup \pi) = \mathrm{Ext}(i) \cap O_\alpha$. The union over the inputs gives $\mathrm{Ext}(I_\alpha) \cap \mathrm{Ext}(R_\alpha \cup X) = O_\alpha$, so $R_\alpha \cup X \in \Pi_\alpha$. Since $R_\alpha \subseteq R_\alpha \cup X$, $\mathrm{Ext}(R_\alpha \cup X) \subseteq \mathrm{Ext}(R_\alpha)$, with equality only when $X = \emptyset$, because statements with equal extensions are equal. Since $R_\alpha \cap C_\alpha = \emptyset$, $|R_\alpha \cup X| = |R_\alpha| + |X|$, which exceeds $|R_\alpha|$ whenever $X \neq \emptyset$. □

Corollary 6 (The freest correct policy survives every demand survived by any correct policy)**.** Under the hypotheses of Lemma 7, with $U := L_\mathfrak{v} \setminus \mathrm{Ext}(I_\alpha)$ and $B_\pi := \mathrm{Ext}(\pi) \cap U$,

1. $B_\pi \subseteq B_{R_\alpha}$ for every $\pi \in \Pi_\alpha$, so $G_{P_\alpha}(R_\alpha) \geq G_{P_\alpha}(\pi)$ under every probability distribution P_α of demands, exchangeable or not;

2. R_α is correct for every parent generated by a demand set, in the sense of Definition 17, for which some correct policy of α is correct;

3. if $\mathfrak{v}$ is finite, then R_α is correct for at least as many parents of α as any other correct policy.

Proof. By Lemma 7, $\mathrm{Ext}(\pi) \subseteq \mathrm{Ext}(R_\alpha)$, so $B_\pi \subseteq B_{R_\alpha}$. Every demand set inside B_π is then inside B_{R_α}, so $P_\alpha(S \subseteq B_\pi) \leq P_\alpha(S \subseteq B_{R_\alpha})$ for every distribution P_α. For the second item, let π be correct for the parent generated by S. By Definition 17, $S \subseteq B_\pi$ and S is closed under completion inside B_π. Then $S \subseteq B_{R_\alpha}$. A completion of a member of S contains a member of $\mathrm{Ext}(\pi)$, so it lies in $\mathrm{Ext}(\pi) \subseteq \mathrm{Ext}(R_\alpha)$. Such a completion lies in B_{R_α} if and only if it lies in U, and hence if and only if it lies in B_π. So S is closed under completion inside B_{R_α}, and R_α is correct for that parent. For the third item, let ω be a parent of α for which π is correct. By the correctness equation, $O_\omega = \mathrm{Ext}(I_\omega) \cap \mathrm{Ext}(\pi)$, so ω is determined by I_ω. Every input j of ω has a correct output, which contains $j \cup \pi$, so $j \cup \pi$ is a statement and so is its subset $j \cup R_\alpha$. In the task $\omega' := \langle I_\omega, \mathrm{Ext}(I_\omega) \cap \mathrm{Ext}(R_\alpha) \rangle$, every input j therefore has the correct output $j \cup R_\alpha$. The task ω' also has $I_\alpha \subsetneq I_\omega$, its correct outputs include $O_\alpha = \mathrm{Ext}(I_\alpha) \cap \mathrm{Ext}(R_\alpha)$, and R_α is one of its correct policies, so ω' is a parent of α for which R_α is correct. Distinct parents ω for which π is correct have distinct sets of inputs, so they give distinct parents ω', and R_α is correct for at least as many parents as π. □

Over the full correct-policy set, both freedom and simplicity measured by statement cardinality select R_α; their choices can diverge on a searched subset that excludes R_α. Dominance under every distribution of demands concerns parents generated by demand sets. It does not assert correctness on every arbitrary parent on which a rival succeeds. For a fixed nested sequence of children with uniform demand law at each step, freedom selection maximises generalisation probability at every step and reaches any attainable probability threshold with no more examples. With the same positive additive acquisition and processing cost per example for every selector, the corresponding cost is no greater.

8 Product vocabularies and simplicity

Definition 18 (product vocabulary)**.**
Let $B_1, \dots, B_m$ be pairwise disjoint finite sets of programs with $|B_i| \geq 2$. Assume there is a nonempty $A \subseteq \Phi$ such that each B_i partitions A, and that

$$\bigcap_{i=1}^{m} p_i \neq \varnothing \qquad \text{for every } (p_1, \dots, p_m) \in B_1 \times \cdots \times B_m.$$

The product vocabulary is

$$\mathfrak{v}_{\text{prod}} := \bigcup_{i=1}^{m} B_i.$$

For $l \in L_{\mathfrak{v}_{\text{prod}}}$, define

$$I(l) := \{i \in \{1, \dots, m\} : l \cap B_i \neq \varnothing\}.$$

The global convention is in force: statements in $L_{\mathfrak{v}_{\text{prod}}}$ are satisfiable. I use simplicity selector $<_s$ and freedom selector $<_{\text{ff}}$ from Definition 16.

Lemma 8 (product extension count)**.**
For every $l \in L_{\mathfrak{v}_{\text{prod}}}$,

$$|\operatorname{Ext}(l)| = \prod_{i \notin I(l)} (|B_i| + 1).$$

Proof. A satisfiable statement contains at most one program from each block because each B_i partitions A. Conversely, every selection of at most one program from each block is satisfiable. I select one program from each block unrepresented by the selection. The resulting full selection has nonempty intersection by assumption, so its subset does too. By definition, a completion y of l satisfies $l \subseteq y$, so it contains every program from l and therefore represents every block represented by that statement. For each block $i \notin I(l)$, a completion either adds no program from B_i or adds one of its $|B_i|$ programs. These choices are independent across blocks. Hence $|\operatorname{Ext}(l)| = \prod_{i \notin I(l)} (|B_i| + 1)$. □

Theorem 3 (when simplicity and freedom agree)**.**
On $L_{\mathfrak{v}_{\text{prod}}}$, simplicity selector $<_s$ and freedom selector $<_{\text{ff}}$ agree on all comparisons and ties iff

$$|B_1| = |B_2| = \cdots = |B_m|.$$

Proof. Assume all block sizes equal d. Since l represents exactly $|l|$ blocks, by Lemma 8,

$$|\operatorname{Ext}(l)| = (d+1)^{m-|l|}.$$

Function $(d+1)^{m-|l|}$ is strictly decreasing in $|l|$, so $l <_s l'$ iff $l <_{\text{ff}} l'$, with identical ties.

Conversely, assume two blocks have different sizes. I choose $i \neq j$ with $|B_i| \neq |B_j|$, and $p_i \in B_i$, $p_j \in B_j$. Let $l_i = \{p_i\}$ and $l_j = \{p_j\}$. Then $|l_i| = |l_j| = 1$, so simplicity selector $<_s$ ties them. By Lemma 8,

$$|\operatorname{Ext}(l_i)| = \prod_{k \neq i} (|B_k| + 1), \qquad |\operatorname{Ext}(l_j)| = \prod_{k \neq j} (|B_k| + 1).$$

Values $\prod_{k \neq i}(|B_k| + 1)$ and $\prod_{k \neq j}(|B_k| + 1)$ differ because $|B_i| \neq |B_j|$. Freedom selector $<_{\text{ff}}$ separates two statements that simplicity selector $<_s$ ties. Therefore $<_s$ and $<_{\text{ff}}$ cannot agree on all comparisons and ties unless all block sizes are equal. □

Proposition 9 (simplicity sub-optimality)**.**
Here simplicity means minimum description length, which here is measured as the cardinality of the smallest statement that implies a particular extension. When a learner chooses between two correct policies, simplicity in this sense is neither necessary nor sufficient for choosing the one with the greater generalisation probability, under the uniform distribution of demands, or under any exchangeable distribution of demands that gives positive probability to some demand of size between one and the larger of their two buffer sizes.

Proof. Let

$$\Phi = \{\phi_1, \phi_2, \phi_3, \phi_4\}$$

and let the vocabulary be $\mathfrak{v} = \{a, b, c, d, e, f\}$, where the six distinct programs are

$$\begin{aligned} a &= \{\phi_1\}, & b &= \{\phi_2\}, & c &= \{\phi_1, \phi_3\}, \\ d &= \{\phi_1, \phi_2, \phi_3\}, & e &= \{\phi_1, \phi_2, \phi_4\}, & f &= \{\phi_3, \phi_4\}. \end{aligned}$$

A subset of $\mathfrak{v}$ is a statement when its programs have nonempty intersection. Define

$$I_\alpha = \{\{b, d, e\}\}, \qquad O_\alpha = \{\{b, d, e\}\}.$$

The input $\{b, d, e\}$ is its own correct output. Its truth set is $b \cap d \cap e = \{\phi_2\}$, and no program outside $\{b, d, e\}$ is true at ϕ_2, so $\mathrm{Ext}(\{b, d, e\}) = \{\{b, d, e\}\}$ and every subset of $\{b, d, e\}$ is a correct policy. I compare two of them, $\pi_s = \{b\}$ and $\pi_{\mathrm{ff}} = \{d, e\}$. Since π_s has fewer programs, $<_s$ ranks π_s ahead of π_{ff}. The extension of π_s is

$$\mathrm{Ext}(\pi_s) = \{\{b\}, \{b, d\}, \{b, e\}, \{b, d, e\}\},$$

so $|\mathrm{Ext}(\pi_s)| = 4$. The extension of π_{ff} is

$$\mathrm{Ext}(\pi_{\mathrm{ff}}) = \{\{d, e\}, \{a, d, e\}, \{b, d, e\}, \{c, d, e\}, \{a, c, d, e\}\},$$

so $|\mathrm{Ext}(\pi_{\mathrm{ff}})| = 5$, and $<_{\mathrm{ff}}$ ranks π_{ff} ahead of π_s. By Lemma 9 their buffers have sizes $|\mathrm{Ext}(\pi_s)| - |O_\alpha| = 3$ and $|\mathrm{Ext}(\pi_{\mathrm{ff}})| - |O_\alpha| = 4$. By Proposition 8, π_{ff} then has strictly greater generalisation probability than π_s under the uniform distribution over unseen demands, and by Theorem 2 the same is true under every exchangeable distribution of demands with $P(1 \leq |S| \leq 4) > 0$. Simplicity is insufficient because $<_s$ chooses π_s. It is unnecessary because the policy with the greater generalisation probability, π_{ff}, has the longer description. Simplicity also cannot choose between π_s and $\{d\}$, since $\{d\}$ contains one program, as π_s does, while $|\mathrm{Ext}(\{d\})| = 12$. The empty statement is both the shortest and the freest correct policy, as Lemma 7 states, so the two selectors can part only when the correct policies a learner has found exclude the empty statement. □

Corollary 7 (Simplicity and freedom usually diverge on bounded machine vocabularies)**.**
In the bounded Turing configuration vocabulary of Definition 9, suppose $|\Gamma| \geq 2$ and each family

$$\{Q_q : q \in Q\}, \quad \{H_i : 1 \leq i \leq S\}, \quad \{C_{j,a} : a \in \Gamma\}\ (1 \leq j \leq S)$$

partitions $\mathcal{M}_{M,S}$. Suppose also that the implementation is configuration-complete, meaning the map

$$\phi \longmapsto (q(\phi), i(\phi), T_\phi)$$

from $\mathcal{M}_{M,S}$ onto $Q \times \{1, \ldots, S\} \times \Gamma^S$ is surjective. Then simplicity selector $<_s$ and freedom selector $<_{\mathrm{ff}}$ coincide on $L_{\mathfrak{v}^{\mathrm{conf}}_{M,S}}$ if and only if $|Q| = S = |\Gamma|$.

Proof. The partition assumptions specify one block for control, one for head position, and one for each tape cell. Every block has at least two programs. By configuration completeness, every selection of one program from each block is jointly satisfiable. If one program occurred in two blocks, select it from one and a different program from the other, then select arbitrary programs from all remaining blocks. The shared program also lies in the block from which the different program was selected. The block partitions $\mathcal{M}_{M,S}$, so these two programs have empty intersection, contradicting configuration completeness. Thus $\mathfrak{v}_{M,S}^{\mathrm{conf}}$ is a product vocabulary whose block sizes are $|Q|$, S, and $|\Gamma|$ for each tape-cell block. By Theorem 3, the simplicity and freedom selectors agree iff all block sizes are equal. □

9 Measured languages and continuous demands

Definition 19 (measured embodied language and μ-freedom)**.**
A **measured embodied language** is a triple $(\mathfrak{v}, \mathcal{A}_L, \mu)$ where $\mathfrak{v} \subseteq \mathcal{P}$ is a vocabulary, $\mathcal{A}_L$ is a σ-algebra on $L_{\mathfrak{v}}$, and μ is a σ-finite measure on $(L_{\mathfrak{v}}, \mathcal{A}_L)$. Assume $\mathrm{Ext}(l) \in \mathcal{A}_L$ for every $l \in L_{\mathfrak{v}}$. The **μ-freedom** of l is

$$\mathrm{ff}_\mu(l) := \mu(\mathrm{Ext}(l)).$$

For a task α, define the unseen region and the buffer of a correct policy by

$$U := L_{\mathfrak{v}} \setminus \mathrm{Ext}(I_\alpha), \qquad B_\pi := \mathrm{Ext}(\pi) \cap U.$$

For every continuous task considered below, assume $\mathrm{Ext}(I_\alpha)$ and O_α are measurable. Assume also that U is measurable. For the PAC-Bayes certificates of Section 10, assume in addition that $0 < \mu(U) < \infty$.

Definition 20 (μ-exchangeable distribution of demands)**.**
A probability distribution of the future demand S over the measurable subsets of U is **μ-exchangeable** when, for every measurable $B \subseteq U$, the generalisation probability

$$G(B) := P(S \subseteq B)$$

depends on B only through $\mu(B)$ and is nondecreasing in $\mu(B)$. When that dependence is strictly increasing across the buffer measures attained by the policies under comparison, a larger attained buffer has strictly larger generalisation probability.

Theorem 4 (continuous freest correct optimality)**.**
Fix a measured embodied language and a task α with $\Pi_\alpha \neq \emptyset$ and $\mu(O_\alpha) < \infty$. A correct policy's generalisation shortfall under a distribution of demands is the amount by which its generalisation probability falls below the highest generalisation probability among correct policies. Under any μ-exchangeable distribution of demands, correct policies π, π' with $\mathrm{ff}_\mu(\pi) \leq \mathrm{ff}_\mu(\pi')$ satisfy $P(S \subseteq B_\pi) \leq P(S \subseteq B_{\pi'})$, so every correct policy maximising $\mathrm{ff}_\mu(\pi)$ also maximises $P(S \subseteq B_\pi)$. If generalisation probability is strictly increasing in buffer measure across the attained buffer measures, $\mathrm{ff}_\mu(\pi) < \mathrm{ff}_\mu(\pi')$ implies $P(S \subseteq B_\pi) < P(S \subseteq B_{\pi'})$, and every correct policy with less than maximal μ-freedom is strictly suboptimal. Hence the worst-case generalisation shortfall over the μ-exchangeable distributions of demands is minimised by μ-freest correct selection.

Proof. By Lemma 9, μ-freedom and buffer measure rank correct policies identically. By Definition 20, generalisation probability is a nondecreasing function of that buffer measure. When generalisation probability is strictly increasing across the attained buffer measures, the ordering on the attained values is strict. A policy that is pointwise optimal for every distribution in the class has zero shortfall for every distribution, so its worst-case shortfall is minimal. □

Proposition 10 (measured-language reparameterisation invariance)**.**
Let $(L_{\mathfrak{v}}, \mathcal{A}_L, \mu)$ and $(L_{\mathfrak{v}'}, \mathcal{A}_{L'}, \mu')$ be measured embodied languages. Let π be a policy and U an unseen region in $L_{\mathfrak{v}}$, and let π' and U' be a policy and an unseen region in $L_{\mathfrak{v}'}$. Let $h : L_{\mathfrak{v}} \to L_{\mathfrak{v}'}$ be an injective measurable map satisfying $h_{\#}\mu = \mu'$, $h(\mathrm{Ext}(\pi)) = \mathrm{Ext}(\pi')$, and $h(U) = U'$. Then

$$\mathrm{ff}_{\mu'}(\pi') = \mathrm{ff}_{\mu}(\pi) \qquad \text{and} \qquad \mu'(B_{\pi'}) = \mu(B_{\pi}).$$

Proof. Since $h_{\#}\mu = \mu'$, $\mu'(E) = \mu(h^{-1}(E))$ for every measurable $E \subseteq L_{\mathfrak{v}'}$, and $h^{-1}(h(A)) = A$ for every $A \subseteq L_{\mathfrak{v}}$ by injectivity. Hence $\mathrm{ff}_{\mu'}(\pi') = \mu'(h(\mathrm{Ext}(\pi))) = \mu(\mathrm{Ext}(\pi)) = \mathrm{ff}_{\mu}(\pi)$, and since $B_{\pi'} = h(\mathrm{Ext}(\pi)) \cap h(U) = h(\mathrm{Ext}(\pi) \cap U)$ by injectivity, $\mu'(B_{\pi'}) = \mu(B_{\pi})$. □

Proposition 11 (function-level invariance in one language)**.**
Let $F(\theta)$ and $F(\theta')$ be the same input-output function, and fix one vocabulary $\mathfrak{v}$ whose programs describe input-output functions. Then

$$\mathrm{Enc}_{\mathfrak{v}}(F(\theta)) = \mathrm{Enc}_{\mathfrak{v}}(F(\theta')).$$

Consequently their extensions, finite freedom, and μ-freedom are equal whenever measured in that same language.

Proof. By definition $\mathrm{Enc}_{\mathfrak{v}}(F)$ is the set of programs in $\mathfrak{v}$ that hold of the function F. Since $F(\theta)$ and $F(\theta')$ are one function, a program holds of the first if and only if it holds of the second, so the two encodings are the same set of programs. A statement determines its extension, so equal encodings have equal extensions. Finite freedom is the cardinality of the extension, and μ-freedom is its measure under μ. Each is fixed by the extension alone, so each has the same value for the two encodings. □

Lemma 9 (fixed seen-region decomposition)**.**
Let α be a task. Each correct policy $\pi \in \Pi_\alpha$ satisfies

$$\mathrm{Ext}(\pi) = O_\alpha \,\dot\cup\, B_\pi$$

and

$$\mathrm{ff}_{\mu}(\pi) = \mu(O_\alpha) + \mu(B_\pi).$$

If $\mu(O_\alpha) < \infty$, μ-freedom and buffer measure rank correct policies identically. In particular, when $\mathfrak{v}$ is finite and μ is cardinality, correct policies $\pi, \pi' \in \Pi_\alpha$ satisfy $|\,\mathrm{Ext}(\pi)| \le |\,\mathrm{Ext}(\pi')|$ if and only if $|B_\pi| \le |B_{\pi'}|$.

Proof. By correctness, $\mathrm{Ext}(I_\alpha) \cap \mathrm{Ext}(\pi) = O_\alpha$. The seen region $\mathrm{Ext}(I_\alpha)$ and unseen region U partition $L_{\mathfrak{v}}$, so O_α and B_π are disjoint and satisfy $\mathrm{Ext}(\pi) = O_\alpha \,\dot\cup\, B_\pi$. By countable additivity, the measure identity holds. The finite value $\mu(O_\alpha)$ is fixed across Π_α, so adding it preserves the ordering, with cardinality as a special case. □

I use Definitions 19 and 20. Throughout this section, U is measurable, $0 < \mu(U) < \infty$, $\mu(O_\alpha) < \infty$, and $\Pi_\alpha \neq \emptyset$. I use $G(\pi, P) := P(S \subseteq B_\pi)$ for a distribution P of future demands.

Theorem 5 (Continuous sufficiency under Poisson demands)**.**
If future demands are generated by a Poisson point process on U with intensity $\lambda\mu$ with $\lambda > 0$, then for every correct policy $\pi \in \Pi_\alpha$,

$$G(\pi, P) = \exp\big(-\lambda[\mu(U) - \mu(B_\pi)]\big).$$

In particular, $G(\pi, P)$ is strictly increasing in $\mu(B_\pi)$ and therefore, by Lemma 9, strictly increasing in $\mathrm{ff}_{\mu}(\pi)$ on Π_α.

Proof. Under a Poisson point process with intensity measure $\lambda\mu$, the event $S \subseteq B_\pi$ is exactly the event that no points fall in the measurable complement $U \setminus B_\pi$. The Poisson void formula therefore gives

$$G(\pi, P) = P(S \subseteq B_\pi) = \exp\big(-\lambda\mu(U \setminus B_\pi)\big) = \exp\big(-\lambda[\mu(U) - \mu(B_\pi)]\big).$$

This is strictly increasing in $\mu(B_\pi)$. By Lemma 9, on correct policies this is equivalent to strict increase in $\mathrm{ff}_\mu(\pi)$. □

Definition 21 (Nondegeneracy)**.**
A μ-exchangeable distribution P of demands is **nondegenerate** if for every measurable $A \subseteq U$ with $\mu(A) > 0$,

$$P(\emptyset \neq S \subseteq A) > 0.$$

Theorem 6 (Continuous strict necessity)**.**
Assume that $(U, \mathcal{A}_L|_U, \mu|_U)$ is atomless. If P is μ-exchangeable and nondegenerate, and if two correct policies satisfy

$$\mu(B_{\pi_1}) > \mu(B_{\pi_2}),$$

then

$$G(\pi_1, P) > G(\pi_2, P).$$

By Lemma 9, this is equivalent to strict ordering by μ-weakness on Π_α.

Proof. Choose measurable $B' \subseteq B_{\pi_1}$ with $\mu(B') = \mu(B_{\pi_2})$. Such a set exists by atomlessness. By Definition 20,

$$P(S \subseteq B') = P(S \subseteq B_{\pi_2}).$$

Set $A := B_{\pi_1} \setminus B'$, which has positive measure. Nondegeneracy gives $P(\emptyset \neq S \subseteq A) > 0$. This event is disjoint from $\{S \subseteq B'\}$ and implies $S \subseteq B_{\pi_1}$, so

$$P(S \subseteq B_{\pi_1}) \geq P(S \subseteq B') + P(\emptyset \neq S \subseteq A) > P(S \subseteq B_{\pi_2}).$$

□

Theorem 7 (Minimax optimality of μ-weakness)**.**
Let $\mathcal{E}$ be the class of all μ-exchangeable distributions of demands on U. If a μ-freest correct policy $\pi^* \in \Pi_\alpha$ exists, then

$$\max_{\pi \in \Pi_\alpha} \inf_{P \in \mathcal{E}} G(\pi, P) \;=\; \inf_{P \in \mathcal{E}} \max_{\pi \in \Pi_\alpha} G(\pi, P),$$

and π^* attains every policy maximum in this equality.

Proof. Let π^* be a correct policy maximising ff_μ on Π_α. By Theorem 4,

$$G(\pi^*, P) \geq G(\pi, P)$$

for all $\pi \in \Pi_\alpha$ and all $P \in \mathcal{E}$. So

$$\max_{\pi \in \Pi_\alpha} G(\pi, P) = G(\pi^*, P) \qquad \text{for every } P \in \mathcal{E},$$

which gives

$$\inf_{P \in \mathcal{E}} \max_{\pi \in \Pi_\alpha} G(\pi, P) = \inf_{P \in \mathcal{E}} G(\pi^*, P).$$

Again by Theorem 4,

$$G(\pi^*, P) \geq G(\pi, P) \qquad \text{for all } \pi \in \Pi_\alpha \text{ and all } P \in \mathcal{E},$$

so

$$\inf_{P\in\mathcal{E}} G(\pi^*, P) \geq \inf_{P\in\mathcal{E}} G(\pi, P) \qquad \text{for all } \pi \in \Pi_\alpha.$$

Therefore π^* achieves $\max_{\pi\in\Pi_\alpha} \inf_{P\in\mathcal{E}} G(\pi, P)$. The minimax equality follows from the chain

$$\max_{\pi\in\Pi_\alpha} \inf_{P\in\mathcal{E}} G(\pi, P) \geq \inf_{P\in\mathcal{E}} G(\pi^*, P) = \inf_{P\in\mathcal{E}} \max_{\pi\in\Pi_\alpha} G(\pi, P) \geq \max_{\pi\in\Pi_\alpha} \inf_{P\in\mathcal{E}} G(\pi, P).$$

□

10 PAC-Bayes survival certificates

The candidate space $\mathcal{C}_0$ is measurable and fixed, and $S_1, \ldots, S_n \sim P$ are independent demands. The jointly measurable score $u(c, S) \in [0, 1]$ is fixed before sampling. For Stack Theory, $\mathcal{C}_0 = \Pi_\alpha$ and $u(\pi, S) = \mathbf{1}\{S \subseteq B_\pi\}$. The relevant expectations and divergences are

$$\widehat{G}_{n,u}(Q) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}_{c\sim Q} u(c, S_i), \qquad G_u(Q, P) = \mathbb{E}_{c\sim Q, S\sim P} u(c, S),$$

$$\mathrm{kl}(a\|b) = a \log\frac{a}{b} + (1-a)\log\frac{1-a}{1-b}, \qquad \mathrm{KL}(Q\|P_0) = \int \log\frac{dQ}{dP_0}\, dQ$$

when $Q \ll P_0$, and $\mathrm{KL}(Q\|P_0) = \infty$ otherwise. The usual endpoint conventions and natural logarithms apply. The prior may depend on the fixed child task but is fixed before the demands are drawn.

Theorem 8 (PAC-Bayes future-demand certificate)**.**
Let $n \geq 2$, let $\delta \in (0, 1)$, and let P_0 be a prior on $\mathcal{C}_0$ fixed before $S_1, \ldots, S_n$ are drawn. With probability at least $1 - \delta$, simultaneously for every posterior Q,

$$\mathrm{kl}\Big(\widehat{G}_{n,u}(Q)\Big\|G_u(Q, P)\Big) \leq \frac{\mathrm{KL}(Q\|P_0) + \log(2\sqrt{n}/\delta)}{n}.$$

Proof. I apply PAC-Bayes-kl to the bounded loss $1 - u(c, S)$ and use

$$\mathrm{kl}(1-a\|1-b) = \mathrm{kl}(a\|b)$$

[Seeger, 2002, Maurer, 2004, Chugg et al., 2023]. □

For measurable A with $0 < \mu(A) < \infty$, the normalised restriction is

$$\mu(E \mid A) := \frac{\mu(E \cap A)}{\mu(A)}.$$

Theorem 9 (The closest distribution on a buffer)**.**
Assume

$$0 < \mu(B_\pi) \leq \mu(U) < \infty.$$

For every probability measure $Q \ll \mu(\cdot \mid U)$ supported on B_π,

$$\mathrm{KL}(Q\|\mu(\cdot \mid U)) = \mathrm{KL}(Q\|\mu(\cdot \mid B_\pi)) + \log\frac{\mu(U)}{\mu(B_\pi)}.$$

Consequently,

$$\inf_{\substack{Q\ll\mu(\cdot|U)\\ Q(B_\pi)=1}} \mathrm{KL}(Q\|\mu(\cdot \mid U)) = \log\frac{\mu(U)}{\mu(B_\pi)},$$

and the unique minimiser up to null sets is $\mu(\cdot \mid B_\pi)$.

Proof. On B_π,

$$\frac{d\mu(\cdot \mid B_\pi)}{d\mu(\cdot \mid U)} = \frac{\mu(U)}{\mu(B_\pi)}.$$

For every admissible Q,

$$\log \frac{dQ}{d\mu(\cdot \mid U)} = \log \frac{dQ}{d\mu(\cdot \mid B_\pi)} + \log \frac{\mu(U)}{\mu(B_\pi)}.$$

Integrating with respect to Q, I get $\mathrm{KL}(Q\|\mu(\cdot \mid U)) = \mathrm{KL}(Q\|\mu(\cdot \mid B_\pi)) + \log(\mu(U)/\mu(B_\pi))$. Since $\mathrm{KL}(Q\|\mu(\cdot \mid B_\pi)) \geq 0$, the infimum of $\mathrm{KL}(Q\|\mu(\cdot \mid U))$ is $\log(\mu(U)/\mu(B_\pi))$, which is attained at $Q = \mu(\cdot \mid B_\pi)$, where the first term vanishes. Since $\mathrm{KL}(Q\|\mu(\cdot \mid B_\pi)) = 0$ only when $Q = \mu(\cdot \mid B_\pi)$ up to null sets, the minimiser is unique. □

Corollary 8 (completion-restriction survival certificate)**.**
Assume $n \geq 2$, $\delta \in (0,1)$, and $0 < \mu(U) < \infty$. Let $S_1, \ldots, S_n$ be independent future-demand sets from P. Assume $(y, S) \mapsto \mathbf{1}\{S \subseteq \mathrm{Ext}(y) \cap U\}$ is jointly measurable. Define

$$\widehat{C}_n(\pi) := \frac{1}{n} \sum_{i=1}^{n} \mathbb{E}_{y \sim \mu(\cdot \mid B_\pi)} \mathbf{1}\{S_i \subseteq \mathrm{Ext}(y) \cap U\},$$

$$C(\pi, P) := \mathbb{E}_{y \sim \mu(\cdot \mid B_\pi)} P(S \subseteq \mathrm{Ext}(y) \cap U).$$

With probability at least $1 - \delta$, simultaneously for every policy $\pi \in L_{\mathfrak{v}}$ with $\mu(B_\pi) > 0$, where $B_\pi := \mathrm{Ext}(\pi) \cap U$,

$$\mathrm{kl}\Big(\widehat{C}_n(\pi) \Big\| C(\pi, P)\Big) \leq \frac{\log[\mu(U)/\mu(B_\pi)] + \log(2\sqrt{n}/\delta)}{n}.$$

Moreover,

$$P(S \subseteq B_\pi) \geq C(\pi, P).$$

Proof. I apply Theorem 8 with candidate space $\mathcal{C}_0 = U$, prior $\mu(\cdot \mid U)$, posterior $\mu(\cdot \mid B_\pi)$, and survival score $u(y, S) = \mathbf{1}\{S \subseteq \mathrm{Ext}(y) \cap U\}$. Theorem 9 states the KL term. If $y \in B_\pi \subseteq \mathrm{Ext}(\pi)$, then $\mathrm{Ext}(y) \subseteq \mathrm{Ext}(\pi)$. Therefore

$$\mathbf{1}\{S \subseteq \mathrm{Ext}(y) \cap U\} \leq \mathbf{1}\{S \subseteq B_\pi\}.$$

I take expectations over y and S. □

The buffer KL identity alone gives no classification-error guarantee: a sampled completion need not be child-correct or inherit its policy's loss. The certificates concern the specified future-demand score. They do not identify arbitrary parameter-space PAC-Bayes posteriors with freedom.

Proposition 12 (Both certificates survive measure-preserving renaming)**.**
Let $h : U \to U'$ be a bimeasurable bijection[1] with $h_\# \mu = \mu'$ and $h(B_\pi) = B_{\pi'}$. Then

$$h_\# \mu(\cdot \mid U) = \mu'(\cdot \mid U'), \qquad h_\# \mu(\cdot \mid B_\pi) = \mu'(\cdot \mid B_{\pi'}),$$

$$\mu'(B_{\pi'}) = \mu(B_\pi),$$

and

$$\mathrm{KL}(\mu(\cdot \mid B_\pi)\|\mu(\cdot \mid U)) = \mathrm{KL}(\mu'(\cdot \mid B_{\pi'})\|\mu'(\cdot \mid U'))\,.$$

[1]Primed symbols U', μ', π', $B_{\pi'}$ and $\mathrm{Ext}(\cdot)'$ are the counterparts of U, μ, π, B_π and $\mathrm{Ext}(\cdot)$ in a second measured embodied language, as in Proposition 10 in Section 9. The pushforward $h_\# \mu$ gives each measurable set $E \subseteq U'$ the weight $\mu(h^{-1}(E))$, so $h_\# \mu = \mu'$ means every set weighs the same before and after renaming by h.

If $S' = h(S)$, then survival of the policy is preserved. Survival of sampled completions is also preserved when

$$h(\mathrm{Ext}(y) \cap U) = \mathrm{Ext}(h(y))' \cap U'$$

for $\mu(\cdot \mid B_\pi)$-almost every $y \in B_\pi$.

Proof. Normalisation commutes with the pushforward because $h_\# \mu = \mu'$ and h maps the buffer to the primed buffer. KL is invariant when both distributions are pushed through the same measurable bijection. Direct image under a bijection preserves containment. Under the final extension condition, h maps each completion-survival event to its primed counterpart. □

11 Description length and priors

Proposition 13 (subjectivity of description length)**.**
Suppose there is no abstraction, in the sense that the vocabulary equals the full program set ($\mathfrak{v} = \mathcal{P} = 2^\Phi$). Then every truth set of a statement other than the tautological truth set Φ has minimum description length 1, meaning that it is the truth set of some one-program statement and of no shorter statement, and Φ has minimum description length 0. More generally this holds for every vocabulary $\mathfrak{v}$ with $T(l) \in \mathfrak{v}$ for every nonempty $l \in L_\mathfrak{v}$. Consequently, description length is highly representation dependent. In the extreme case $\mathfrak{v} = \mathcal{P}$, minimising $|l|$ does not meaningfully distinguish among policies by truth set.

Proof. If $\mathfrak{v} = \mathcal{P}$, then for every $l \in L_\mathfrak{v}$ the truth set $p := T(l)$ is itself a member of $\mathfrak{v}$. If $p = \Phi$, the empty statement already has truth set p and length 0. If $p \neq \Phi$, the singleton statement $\{p\}$ has length 1 and satisfies

$$T(\{p\}) = p = T(l).$$

The empty statement is the only statement of length 0 and its truth set is Φ, so no shorter statement has truth set p. Thus every truth set expressible by any finite statement is also expressible by either the empty statement or a one-program statement in the full vocabulary. The argument uses $\mathfrak{v} = \mathcal{P}$ only to place p in $\mathfrak{v}$ when $p \neq \Phi$, where l is nonempty, so it holds for every vocabulary with $T(l) \in \mathfrak{v}$ for every nonempty $l \in L_\mathfrak{v}$. Equality $T(\{T(l)\}) = T(l)$ concerns semantic representability, not equality of extensions inside an arbitrary vocabulary. One-program representation in $\mathfrak{v} = \mathcal{P}$ shows description length can be collapsed by changing vocabulary, so description length cannot be the invariant that explains generalisation. □

Lemma 10 (Full languages collapse freedom to simplicity)**.**
Assume $\mathfrak{v}$ is finite. Then

$$\mathrm{ff}(l) \leq 2^{|\mathfrak{v}|-|l|}$$

for every $l \in L_\mathfrak{v}$. Equality holds for one $l \in L_\mathfrak{v}$, equivalently for every $l \in L_\mathfrak{v}$, if and only if $\mathfrak{v}$ is degenerate, with $T(\mathfrak{v}) \neq \emptyset$, every subset of $\mathfrak{v}$ satisfiable and $L_\mathfrak{v} = 2^\mathfrak{v}$. In this special case, freedom is a strictly decreasing function of description length (simplicity).

Proof. A completion of l is any satisfiable superset y with $l \subseteq y \subseteq \mathfrak{v}$. There are $2^{|\mathfrak{v}|-|l|}$ supersets before imposing satisfiability, one for each subset of the remaining programs $\mathfrak{v} \setminus l$, so $\mathrm{ff}(l) \leq 2^{|\mathfrak{v}|-|l|}$, with equality if and only if every superset of l in $\mathfrak{v}$ is satisfiable. A subset of a satisfiable set is satisfiable, since $y \subseteq y'$ gives $T(y') \subseteq T(y)$. The vocabulary $\mathfrak{v}$ is a superset of every l. Therefore every superset of l in $\mathfrak{v}$ is satisfiable if and only if $T(\mathfrak{v}) \neq \emptyset$, a condition independent of l. □

Corollary 9 (Weak constraints compress under usage bias)**.**
Let P be a usage distribution over the finite set $L_\mathfrak{v}$, meaning that P assigns each statement its probability of being used. Assume P satisfies the usage-bias condition that $\mathrm{ff}(l_1) > \mathrm{ff}(l_2)$ implies

$P(l_1) \geq P(l_2)$. Then there exists a prefix-free code c^* for P with minimum expected length such that

$$\mathrm{ff}(l_1) > \mathrm{ff}(l_2) \quad \Longrightarrow \quad |c^*(l_1)| \leq |c^*(l_2)|.$$

Every prefix-free code c for P with minimum expected length satisfies $P(l_1) > P(l_2) \Rightarrow |c(l_1)| \leq |c(l_2)|$. Therefore every prefix-free code with minimum expected length satisfies the displayed ordering at every pair with $P(l_1) \neq P(l_2)$. No equality claim is made for equal-freedom ties.

Proof. I take any minimum expected length prefix-free code c and fix its multiset of codewords. If $P(l_1) > P(l_2)$ but $|c(l_1)| > |c(l_2)|$, swap the two codewords. Prefix-freeness is preserved and expected length strictly decreases, a contradiction. Thus no optimal code assigns a longer codeword to a higher-probability statement than to a lower-probability statement. For each equal-probability class, reorder the codewords assigned to that class so that higher-freedom statements receive no longer codewords. Expected length is unchanged because the probabilities in the class are equal. The reordered code c^* is still a minimum expected length prefix-free code. By the usage-bias condition, a statement with strictly greater freedom never has lower probability. So $\mathrm{ff}(l_1) > \mathrm{ff}(l_2)$ implies $|c^*(l_1)| \leq |c^*(l_2)|$. □

Corollary 10 (Simple-general correlation is freedom induced)**.**
Under the conditions of Corollary 9, let $\pi_1, \pi_2 \in \Pi_\alpha$ be correct policies for a $\mathfrak{v}$-task α. If

$$\mathrm{ff}(\pi_1) > \mathrm{ff}(\pi_2),$$

then

$$|c^*(\pi_1)| \leq |c^*(\pi_2)|.$$

If the distribution P_α of further demands for α is exchangeable as in Theorem 2, then $|c^*(\pi_1)| < |c^*(\pi_2)|$ implies $G_{P_\alpha}(\pi_1) \geq G_{P_\alpha}(\pi_2)$. Code length appears to predict generalisation. Without the usage-bias condition, that prediction is no longer guaranteed.

Proof. By Theorem 2, a policy with greater freedom has no lower generalisation probability under exchangeable task growth. Its generalisation probability is strictly greater in the uniform case of Proposition 8 and in the strictly increasing case of Theorem 4. By Corollary 9 and the usage-bias condition, some optimal code assigns the greater-freedom policy a codeword no longer than the codeword of the lower-freedom policy. If $|c^*(\pi_1)| < |c^*(\pi_2)|$ then $\mathrm{ff}(\pi_1) \geq \mathrm{ff}(\pi_2)$, since $\mathrm{ff}(\pi_1) < \mathrm{ff}(\pi_2)$ would give $|c^*(\pi_2)| \leq |c^*(\pi_1)|$. Then $|B_{\pi_1}| \geq |B_{\pi_2}|$ by Lemma 9, and under exchangeable demands $G_{P_\alpha}(\pi)$ is a nondecreasing function of $|B_\pi|$ by Theorem 2, so $G_{P_\alpha}(\pi_1) \geq G_{P_\alpha}(\pi_2)$. The apparent link between code length and persistence is therefore inherited from freedom through the usage distribution. If the usage distribution stops ranking weaker policies as at least as probable, $\mathrm{ff}(\pi_1) > \mathrm{ff}(\pi_2)$ no longer implies $|c^*(\pi_1)| \leq |c^*(\pi_2)|$. □

Corollary 11 (complexity confounding)**.**
For every finite vocabulary there is a strictly positive usage distribution for which every minimum expected length prefix-free code assigns no longer codewords to weaker statements than to stronger statements, and so to weaker correct policies than to stronger correct policies. Under the assumptions of Theorem 2, code length can therefore correlate with generalisation even though the code-length ordering is induced by usage and vocabulary.

Proof. For any finite nonempty embodied language, choose

$$P(l) = \frac{\mathrm{ff}(l)}{\sum_{y \in L_\mathfrak{v}} \mathrm{ff}(y)}.$$

Every statement is a member of its own extension, so P is strictly positive and satisfies the usage-bias condition, with $P(l_1) > P(l_2)$ whenever $\mathrm{ff}(l_1) > \mathrm{ff}(l_2)$. Corollary 9 then establishes the

code-length ordering for every minimum expected length prefix-free code. Theorem 2 implies the generalisation ordering under the assumptions of Theorem 2. Changing the usage distribution can change the code-length ordering without changing any extension. □

Lemma 11 (Decoder keys are hidden entropy)**.**
Let $\mathcal{M}$ be a **finite** microstate set with $|\mathcal{M}| = 2^S$, where S is the number of bits needed to index the microstate set. Let $\mathcal{D}$ be a finite family of operationally distinct decoders.

For each $D \in \mathcal{D}$, let $\mathcal{M}^{(D)} \subseteq \mathcal{M}$ be the set of microstates on which D succeeds uniformly at some fixed accuracy. Assume a uniform coverage bound

$$|\mathcal{M}^{(D)}| \leq d_{\max} \qquad \text{for every } D \in \mathcal{D}.$$

If the protocol must succeed for every microstate, meaning $\bigcup_{D\in\mathcal{D}} \mathcal{M}^{(D)} = \mathcal{M}$, then

$$|\mathcal{D}| \geq \frac{|\mathcal{M}|}{d_{\max}} = \frac{2^S}{d_{\max}}.$$

Therefore any control register that must select among $\mathcal{D}$ at verification time must support at least $|\mathcal{D}|$ reliably distinguishable control states. Equivalently any key vocabulary $\mathfrak{v}_K$ representing that register must satisfy

$$\mathsf{I}(\mathfrak{v}_K) \geq \log_2 |\mathcal{D}| \geq S - \log_2 d_{\max}.$$

Proof. The covering condition gives

$$|\mathcal{M}| \leq \sum_{D\in\mathcal{D}} |\mathcal{M}^{(D)}| \leq |\mathcal{D}|\, d_{\max}.$$

Rearrange to get the lower bound on $|\mathcal{D}|$. A selector register that can implement any $D \in \mathcal{D}$ needs at least one reliably distinguishable control state per operationally distinct decoder. So the register has at least $|\mathcal{D}|$ internal states and any representing vocabulary satisfies $\mathsf{I}(\mathfrak{v}_K) \geq \log_2 |\mathcal{D}|$. □

Remark 1 (Weakness does not imply inclusion)**.**
Weakness compares only *cardinalities*. In general, $\mathrm{ff}(l_1) \geq \mathrm{ff}(l_2)$ does *not* imply $l_1 \subseteq l_2$, nor does it imply $\mathrm{Ext}(l_1) \supseteq \mathrm{Ext}(l_2)$. However, if $l_1 \subseteq l_2$ (i.e. l_2 is a completion of l_1), then $\mathrm{Ext}(l_2) \subseteq \mathrm{Ext}(l_1)$ and hence $\mathrm{ff}(l_1) \geq \mathrm{ff}(l_2)$.

Corollary 12 (Occam priors collapse under no abstraction)**.**
Fix a finite candidate family $H \subseteq L_{\mathcal{P}}$, where $\mathcal{P} = 2^{\Phi}$. If $\mathfrak{v} = \mathcal{P}$ (no abstraction), then every $h \in H$ has a singleton semantic representation of length 1 by Proposition 13. Therefore any prior or selector on H that depends only on this semantic length proxy is uniform on H. In particular, the semantic-length Occam prior carries no discriminatory information on a no-abstraction candidate family.

Proof. By Proposition 13, when $\mathfrak{v} = \mathcal{P}$ every candidate $h \in H$ has a singleton semantic representation $\{T(h)\}$ with description length

$$|\{T(h)\}| = 1.$$

So any prior or selector on the finite family H that depends only on this semantic length proxy assigns the same score to every candidate. After normalisation, the induced prior is therefore uniform on H. Hence the semantic-length Occam prior carries no discriminatory information on a no-abstraction candidate family. □

Proposition 14 (Every finite prior on its support is an Occam prior up to a constant)**.**
Let H be a finite hypothesis set and let Q be a probability distribution on H. The support of this distribution is

$$H_+ := \{h \in H : Q(h) > 0\}.$$

Then there exists a prefix-free code κ_Q on H_+ such that the induced Occam prior

$$P_{\kappa_Q}(h) \propto 2^{-|\kappa_Q(h)|}$$

satisfies

$$\frac{1}{2}Q(h) \le P_{\kappa_Q}(h) \le 2Q(h) \qquad \text{for all } h \in H_+.$$

Proof. Let $H_+ = \{h_1, \dots, h_n\}$ be the support of Q, so $Q(h_i) > 0$ for all i. Define integer lengths

$$\ell_i := \lceil -\log_2 Q(h_i) \rceil .$$

Then $\ell_i \ge -\log_2 Q(h_i)$ and $\ell_i < -\log_2 Q(h_i) + 1$, so

$$\frac{Q(h_i)}{2} < 2^{-\ell_i} \le Q(h_i).$$

Also

$$\sum_{i=1}^{n} 2^{-\ell_i} \le \sum_{i=1}^{n} Q(h_i) = 1,$$

so by the Kraft-McMillan theorem there exists a prefix-free code κ_Q on H_+ with $|\kappa_Q(h_i)| = \ell_i$ for all i. Let

$$Z := \sum_{j=1}^{n} 2^{-|\kappa_Q(h_j)|} = \sum_{j=1}^{n} 2^{-\ell_j}.$$

From the bounds above,

$$\frac{1}{2} = \sum_{j=1}^{n} \frac{Q(h_j)}{2} < Z \le 1.$$

The induced Occam prior is

$$P_{\kappa_Q}(h_i) = \frac{2^{-\ell_i}}{Z}.$$

Using $2^{-\ell_i} > Q(h_i)/2$ and $Z \le 1$ gives

$$P_{\kappa_Q}(h_i) \ge \frac{Q(h_i)/2}{1} = \frac{Q(h_i)}{2}.$$

Using $2^{-\ell_i} \le Q(h_i)$ and $Z > 1/2$ gives

$$P_{\kappa_Q}(h_i) \le \frac{Q(h_i)}{1/2} = 2Q(h_i).$$

Therefore

$$\frac{1}{2}Q(h_i) \le P_{\kappa_Q}(h_i) \le 2Q(h_i) \qquad \text{for all } h_i \in H_+.$$

□

12 Stacks and delegation

Definition 22 (unseen-buffer utility of intelligence)**.**
For any task $\gamma = \langle I_\gamma, O_\gamma \rangle$ whose vocabulary $\mathfrak{v}$ is finite, define

$$U_\gamma := L_\mathfrak{v} \setminus \mathrm{Ext}(I_\gamma), \qquad B_\pi^\gamma := \mathrm{Ext}(\pi) \cap U_\gamma.$$

For each correct policy $\pi \in \Pi_\gamma$, define its unseen-buffer value by

$$\beta_\gamma(\pi) := |B_\pi^\gamma|.$$

The **utility of intelligence** of γ is the largest unseen-buffer value achievable by a correct policy:

$$\epsilon(\gamma) = \begin{cases} 0, & \Pi_\gamma = \emptyset, \\ \max_{\pi \in \Pi_\gamma} \beta_\gamma(\pi), & \Pi_\gamma \neq \emptyset. \end{cases}$$

For $\pi \in \Pi_\gamma$, by Lemma 9 with μ as cardinality, $\beta_\gamma(\pi) = |\,\mathrm{Ext}(\pi)| - |O_\gamma| = |\,\mathrm{Ext}(\pi) \setminus O_\gamma|$. The maximum exists because the vocabulary is finite.

Definition 23 (stack)**.**
A **stack** [Bennett, 2024] of depth n is specified by the following data.

- A sequence of disembodied tasks $\langle \lambda^0, \lambda^1, \ldots, \lambda^n \rangle$ such that $\lambda^{i+1} \sqsubset \lambda^i$.
- A sequence of vocabularies $\langle \mathfrak{v}^0, \mathfrak{v}^1, \ldots, \mathfrak{v}^n \rangle$.
- A sequence of policies $\langle \pi^0, \pi^1, \ldots, \pi^n \rangle$ with $\pi^i \in L_{\mathfrak{v}^i}$ for each i.

For each $i = 0, \ldots, n-1$, $\mathfrak{v}^{i+1} \subseteq \mathfrak{f}(\mathfrak{v}^i, \pi^i)$, so each higher layer embodies some or all of the truth sets of the completions of the policy beneath it. I call a stack **over-constrained** if there exists some layer i with $\Pi_{\lambda^i(\mathfrak{v}^i)} = \varnothing$.

Theorem 10 (Time in a stack)**.**
Let $\langle \mathfrak{v}^0, \ldots, \mathfrak{v}^n \rangle$ be the finite vocabularies of a stack, and let $\tau : I \to \Phi$ be a timeline. For every adjacent pair of layers and all states $\phi, \phi' \in \Phi$, $\mathrm{Enc}_{\mathfrak{v}^i}(\phi) = \mathrm{Enc}_{\mathfrak{v}^i}(\phi')$ implies $\mathrm{Enc}_{\mathfrak{v}^{i+1}}(\phi) = \mathrm{Enc}_{\mathfrak{v}^{i+1}}(\phi')$. Hence for every finite timeline fragment h of τ,

$$\kappa_{\mathfrak{v}^{i+1}}(h) \leq \kappa_{\mathfrak{v}^i}(h).$$

Every tick of layer $i{+}1$ is a tick of layer i, so for $u \leq u'$ in I,

$$\kappa^\tau_{\mathfrak{v}^i}(u) = \kappa^\tau_{\mathfrak{v}^i}(u') \quad \Longrightarrow \quad \kappa^\tau_{\mathfrak{v}^{i+1}}(u) = \kappa^\tau_{\mathfrak{v}^{i+1}}(u'),$$

and every uninterrupted block of constant lower encoding lies within one block of constant higher encoding. Consequently, for every $t \in I$,

$$\kappa^\tau_{\mathfrak{v}^{i+1}}(t) \leq \kappa^\tau_{\mathfrak{v}^i}(t),$$

and hence

$$0 \leq \kappa^\tau_{\mathfrak{v}^n}(t) \leq \cdots \leq \kappa^\tau_{\mathfrak{v}^0}(t) \leq t.$$

For any vocabulary $\mathfrak{v}$, equality $\kappa^\tau_\mathfrak{v}(t) = t$ holds if and only if its encoding changes at every adjacent transition from 0 through t.

Proof. By the stack and abstractor definitions, every $q \in \mathfrak{v}^{i+1}$ is $T(o)$ for some $o \in \mathrm{Ext}(\pi^i)$ with $o \subseteq \mathfrak{v}^i$. Therefore $\phi \in q$ if and only if $o \subseteq \mathrm{Enc}_{\mathfrak{v}^i}(\phi)$. Equal lower encodings at two states therefore give equal higher encodings. Suppose layer $i{+}1$ records a tick between objective ticks $u-1$ and u. Then some program $q \in \mathfrak{v}^{i+1}$ changes truth value between $\tau(u-1)$ and $\tau(u)$. By the stack and abstractor definitions, $q = T(o)$ for some $o \in \mathrm{Ext}(\pi^i)$. If every program $p \in o$ had the same truth value at both environmental states, then their intersection $T(o) = q$ would also have the same truth value. Therefore some $p \in o \subseteq \mathfrak{v}^i$ changes truth value. The lower encoding changes, so layer i records a layer tick at the same objective tick. Every layer tick counted at layer $i{+}1$ is therefore counted at layer i. Thus $\kappa_{\mathfrak{v}^{i+1}}(h) \le \kappa_{\mathfrak{v}^i}(h)$. Taking $h = \tau_{\le t}$, I obtain $\kappa^{\tau}_{\mathfrak{v}^{i+1}}(t) \le \kappa^{\tau}_{\mathfrak{v}^i}(t)$. Applying this adjacent-layer inequality for $i = 0, \dots, n-1$ shows that $\kappa^{\tau}_{\mathfrak{v}^n}(t) \le \cdots \le \kappa^{\tau}_{\mathfrak{v}^0}(t)$. A layer clock increments by one at each change of its encoding and otherwise stays constant, so if a lower clock is constant between u and u', its higher clock is constant there too. Each of t objective ticks advances any layer clock by at most one, so its upper bound is t. Equality holds if and only if all t increments are one. □

Proposition 15 (The Law of the Stack)**.**
In a stack with adjacent layers i and $i{+}1$, suppose $\mathfrak{v}^{i+1} \subseteq \mathfrak{f}(\mathfrak{v}^i, \pi^i) = \{T(o) \mid o \in \mathrm{Ext}(\pi^i)\}$. Then the achievable behavioural flexibility at layer $i{+}1$ is bottlenecked by the freedom of the realised policy at layer i in the following cardinal sense:

$$\sup_{\pi \in L_{\mathfrak{v}^{i+1}}} |\mathrm{Ext}(\pi)| \;\le\; 2^{|\,\mathrm{Ext}(\pi^i)|}.$$

In the finite case, let γ^{i+1} be any $\mathfrak{v}^{i+1}$-task with $\Pi_{\gamma^{i+1}} \neq \emptyset$, such as the induced layer-$(i{+}1)$ task $\lambda^{i+1}(\mathfrak{v}^{i+1})$ when it has a correct policy. Then $\epsilon(\gamma^{i+1}) + |O_{\gamma^{i+1}}| \ge 1$, and the familiar bottleneck form holds:

$$|\,\mathrm{Ext}(\pi^i)| \;\ge\; \log_2\!\big(\epsilon(\gamma^{i+1}) + |O_{\gamma^{i+1}}|\big).$$

Higher targets for the utility of intelligence at layer $i{+}1$ imply larger lower bounds on the freedom of the layer-i policy. A target above the bound set by the present freedom of the layer-i policy cannot be reached without increasing that freedom.

Proof. A layer i policy π^i has a set of completions $\mathrm{Ext}(\pi^i)$. Every program of the next layer vocabulary $\mathfrak{v}^{i+1}$ is the truth set of one of those completions, so $|\mathfrak{v}^{i+1}| \le |\,\mathrm{Ext}(\pi^i)|$. Any layer $i{+}1$ statement is a subset of $\mathfrak{v}^{i+1}$, so the entire layer-$i{+}1$ language has size at most $2^{|\mathfrak{v}^{i+1}|}$, which is at most $2^{|\,\mathrm{Ext}(\pi^i)|}$. Every extension set $\mathrm{Ext}(\pi)$ at layer $i{+}1$ is a subset of that language, so $|\,\mathrm{Ext}(\pi)| \le 2^{|\,\mathrm{Ext}(\pi^i)|}$. In the finite case, by the definition of ϵ there is some correct layer-$(i{+}1)$ policy with $|\,\mathrm{Ext}(\pi)| = \epsilon(\gamma^{i+1}) + |O_{\gamma^{i+1}}|$, because $\Pi_{\gamma^{i+1}} \neq \emptyset$ and the set of correct policies is finite. This cardinality is at least one, because a correct policy is a statement and is a member of its own extension. Applying $|\,\mathrm{Ext}(\pi)| \le 2^{|\,\mathrm{Ext}(\pi^i)|}$ to that policy and taking base-2 logarithms, I get $|\,\mathrm{Ext}(\pi^i)| \ge \log_2(\epsilon(\gamma^{i+1}) + |O_{\gamma^{i+1}}|)$. □

Corollary 13 (Delegation is necessary for lowering F_2)**.**
Let $\mathfrak{v}^i$ be finite and $\mathfrak{v}^{i+1} \subseteq \mathfrak{f}(\mathfrak{v}^i, \pi^i)$ as in Proposition 15. Fix a layer-$(i{+}1)$ viability statement $\mu \in L_{\mathfrak{v}^{i+1}}$ whose extension $\mathrm{Ext}(\mu)$ represents viable completions. For any policy $\pi \in L_{\mathfrak{v}^{i+1}}$ with $\Omega_\pi := \mathrm{Ext}(\pi) \cap \mathrm{Ext}(\mu) \neq \emptyset$, let p_π be uniform on Ω_π and p_μ uniform on $\mathrm{Ext}(\mu)$. I use $D_{\mathrm{KL},2}$ for Kullback–Leibler divergence with base-2 logarithms. Define

$$F_2(\pi) := D_{\mathrm{KL},2}(p_\pi \| p_\mu) = \log_2 |\,\mathrm{Ext}(\mu)| - \log_2 |\Omega_\pi|.$$

This is a divergence in bits, not thermodynamic free energy. The physical identification follows below. Then

$$|\Omega_\pi| \le |\,\mathrm{Ext}(\pi)| \le 2^{|\,\mathrm{Ext}(\pi^i)|}.$$

Consequently, if at least N viable completions must remain open to the layer-$(i{+}1)$ policy, then

$$|\operatorname{Ext}(\pi^i)| \geq \log_2 N.$$

Lower F_2 targets therefore imply larger lower bounds on the freedom of the layer-i policy [Bennett, 2024].

Proof. The vocabulary $\mathfrak{v}^{i+1} \subseteq \{T(o) \mid o \in \operatorname{Ext}(\pi^i)\}$ has at most $|\operatorname{Ext}(\pi^i)|$ programs. Every statement of $L_{\mathfrak{v}^{i+1}}$ is a subset of this vocabulary, and every completion of π is such a statement. Therefore $|\operatorname{Ext}(\pi)| \leq 2^{|\operatorname{Ext}(\pi^i)|}$, which is the bound of Proposition 15. Since $\Omega_\pi \subseteq \operatorname{Ext}(\pi)$, $|\Omega_\pi| \leq 2^{|\operatorname{Ext}(\pi^i)|}$. If $N \leq |\Omega_\pi|$ then $N \leq 2^{|\operatorname{Ext}(\pi^i)|}$, and taking base-2 logarithms gives $|\operatorname{Ext}(\pi^i)| \geq \log_2 N$. □

The F_2 ordering uses the stated uniform p_π and p_μ. Lower-layer restrictions may make the viability statement μ inexpressible or leave no correct upper-layer policy. The premise $\mu \in L_{\mathfrak{v}^{i+1}}$ already presupposes its representability; optimisation confined to the upper layer cannot remove either obstruction without changing lower constraints. Under a bijection from completions in $\operatorname{Ext}(\mu)$ to the full physical microstate space, with one fixed Hamiltonian and p_μ its canonical equilibrium distribution at temperature Θ_0, the nonequilibrium identity gives

$$F_{\text{phys}}[p_\pi] - F_{\text{phys}}[p_\mu] = k_B \Theta_0 \ln 2 \, F_2(\pi).$$

The uniform p_μ assumed in Corollary 13 can be canonical only when the mapped microstates have equal Boltzmann weights. Without these identifications, F_2 is an information quantity in bits.

13 Regret and weighted forgetting

Theorem 11 (Regret is freedom lost)**.**
Fix a finite list Z of history-task pairs $z = (h, Q)$, where h is an interaction record and Q is a binary prediction task. Here an interaction record is any finite record used by the prediction setup. It need not be a timeline or a finite timeline fragment. Each z has evaluation weight $\rho_z \geq 0$ with $\sum_{z \in Z} \rho_z = 1$. Let $p_z \in [0, 1]$ be its conditional probability that test Q returns 1, given record h. The optimal answer is $a^*(z) = 1$ when $p_z > 1/2$, $a^*(z) = 0$ when $p_z < 1/2$, and either answer at a tie. I define the normalised decision margin for this history-task pair

$$m_z := \frac{|2p_z - 1|}{\max\{p_z, 1 - p_z\}} \in [0, 1].$$

This is the normalised-regret weight $\rho_i = 2|p_i - \frac{1}{2}|/(\frac{1}{2} + |p_i - \frac{1}{2}|)$ in Bennett [2026b], where $|p_i - \frac{1}{2}|$ is called the margin. Here m_z denotes its normalised-regret weight, rather than its unnormalised margin. The weight is zero at a tie. A representation M maps each interaction record to a memory state. This map partitions Z into groups

$$C_{s,Q_0} = \{(h, Q) \in Z \mid M(h) = s,\ Q = Q_0\}.$$

An M-based policy gives one answer $a_C \in \{0, 1\}$ on each group. I define the group weights

$$W_C(a) := \sum_{z \in C \,:\, a^*(z) = a} \rho_z m_z$$

Normalised regret at z is one minus the ratio of the policy's actual success probability to the optimal success probability $\max\{p_z, 1 - p_z\}$. Its average weighted by W_C is

$$R_\rho^M(a_C) := \sum_C W_C(1 - a_C).$$

Finally define the weighted forgetting cost

$$K_\rho(M) := \sum_C \min\{W_C(0), W_C(1)\}.$$

Then

$$\min_{M\text{-based policies}} R_\rho^M(a_C) = K_\rho(M).$$

Moreover, the regret of every M-based policy decomposes as

$$R_\rho^M(a_C) = K_\rho(M) + \sum_C \Big(W_C(1-a_C) - \min\{W_C(0), W_C(1)\}\Big).$$

Term $K_\rho(M)$ is the forgetting unavoidable under memory M. Sum $\sum_C(W_C(1-a_C) - \min\{W_C(0), W_C(1)\})$ is avoidable prediction error given representation M. An M-based policy attains $K_\rho(M)$ if and only if $a_C \in \arg\max_{a\in\{0,1\}} W_C(a)$ for every group C.

Proof. Choosing a worse answer at z reduces success probability by $|2p_z - 1|$. Normalising by $\max\{p_z, 1-p_z\}$, I recover m_z. Groups are disjoint, and an M-based policy chooses one answer per group. If it answers 0 on group C, it incurs cost equal to the weighted mass of points whose optimal answer is 1, namely $W_C(1)$. If it answers 1, it pays $W_C(0)$. The smallest possible cost on that group is therefore $\min\{W_C(0), W_C(1)\}$. Summing over groups, I get $K_\rho(M)$. For a policy with answers a_C, add and subtract this groupwise minimum:

$$\begin{aligned} R_\rho^M(a_C) = &\sum_C \min\{W_C(0), W_C(1)\} \\ &+ \sum_C \Big(W_C(1-a_C) - \min\{W_C(0), W_C(1)\}\Big). \end{aligned}$$

The first sum is the forgetting cost, since $\sum_C \min\{W_C(0), W_C(1)\} = K_\rho(M)$. The second sum, $\sum_C(W_C(1-a_C) - \min\{W_C(0), W_C(1)\})$, is nonnegative. A sum of nonnegative terms vanishes if and only if every term does, and the term for C vanishes if and only if $W_C(1-a_C) = \min\{W_C(0), W_C(1)\}$, meaning $a_C \in \arg\max_{a\in\{0,1\}} W_C(a)$. □

Weighted freedom in the degenerate vocabulary. The degenerate vocabulary contains two distinct programs $p_{z,0}, p_{z,1}$ for each $z \in Z$, all true at one common state, so every bundle is satisfiable. With no task inputs every policy is correct and every statement unseen. For an M-based decision rule $a = (a_C)$, put

$$\pi_a = \{p_{z,a_{C(z)}} : z \in Z\}, \qquad o_z = \{p_{z,a^*(z)}\} \cup \{p_{z',0}, p_{z',1} : z' \neq z\}.$$

The weight of o_z is $\rho_z m_z$, and every other statement has weight zero. Then $\pi_a \subseteq o_z$ iff $a_{C(z)} = a^*(z)$. The weighted freedom of π_a is $W_Z - R_\rho^M(a)$, where $W_Z := \sum_z \rho_z m_z$. With $W^*(M)$ the maximum weighted freedom among M-based policies,

$$W_Z - W^*(M) = \min_a R_\rho^M(a) = K_\rho(M).$$

If the demands o_z occur independently with probabilities $1 - \exp(-\rho_z m_z)$, optimal M-based generalisation is $\exp[-K_\rho(M)]$, since the product of probabilities of avoiding excluded demands is the exponential of their negative total weight. For any nonempty admissible policy family permitting outputs from at most one answer per group,

$$W_Z - W^*(M) \geq K_\rho(M),$$

with equality iff a single admissible policy attains the larger of $W_C(0)$ and $W_C(1)$ in every group simultaneously. Within each test, the coarsest zero-forgetting grouping of positive-weight

points identifies points with the same optimal answer. Representation convergence up to renaming additionally requires this grouping to be realisable by one memory code $M(h)$ across all tests. Weighted freedom uses a fixed demand prior; cross-vocabulary comparisons require a correspondence of weights, as well as outputs.

I use indexed notation for these extensions of Theorem 11. Evaluation points are $x_i = (h_i, T_i)$ with masses $\mu_i > 0$ summing to 1, conditional probabilities p_i, and optimal predictions $y_i \in \{0, 1\}$. Here μ_i corresponds to ρ_{z_i} in that theorem. I set

$$d_i := |p_i - \tfrac{1}{2}|, \qquad r_i := \frac{2d_i}{\frac{1}{2} + d_i},$$

so r_i corresponds to its m_{z_i}. A representation M groups points with equal $M(h_i)$ and equal T_i. I denote this partition by $\mathcal{C}_M$. On each group C, a randomised policy predicts 1 with probability $q_C \in [0, 1]$. Normalised regret at x_i is $\delta_i(\pi)$.

For this section I abbreviate $K_\rho(M)$ as $K(M)$. With $A_C := \sum_{i \in C,\, y_i = 1} \mu_i r_i$ and $B_C := \sum_{i \in C,\, y_i = 0} \mu_i r_i$, its equivalent deletion form is

$$K(M) = \sum_C \min\{A_C, B_C\} = \min\left\{ \sum_{i \in B} \mu_i r_i \;\middle|\; y \text{ is constant on } C \setminus B \text{ for every } C \in \mathcal{C}_M \right\}.$$

Each group becomes constant after deleting one prediction class. Deleting its lighter class minimises cost.

Lemma 12 (Pointwise regret decomposition)**.**
Let π be any M-based policy with report probability q_C on cell C. For $i \in C$,

$$\delta_i(\pi) = r_i\big((1 - q_C)\mathbf{1}\{y_i = 1\} + q_C \mathbf{1}\{y_i = 0\}\big).$$

Proof. If $y_i = 1$, then $p_i \geq \frac{1}{2}$ and $V_i^\star = p_i = \frac{1}{2} + d_i$. The success probability under report q_C is $V_i^\pi = q_C p_i + (1 - q_C)(1 - p_i)$. So $\delta_i = 1 - V_i^\pi / V_i^\star = 2d_i(1 - q_C)/(\frac{1}{2} + d_i) = r_i(1 - q_C)$. The $y_i = 0$ case is symmetric, giving $\delta_i = r_i q_C$. □

Corollary 14 (Policy-wise decomposition)**.**
For each cell $C \in \mathcal{C}_M$, let $A_C := \sum_{i \in C,\, y_i = 1} \mu_i r_i$, $B_C := \sum_{i \in C,\, y_i = 0} \mu_i r_i$, and choose any

$$q_C^\star \in \arg\min_{q \in [0,1]} \big(A_C(1 - q) + B_C q\big).$$

Then every M-based policy π with report probabilities q_C satisfies

$$\sum_{i=1}^n \mu_i \delta_i(\pi) = K(M) + \sum_{C \in \mathcal{C}_M} |A_C - B_C|\, |q_C - q_C^\star|.$$

In particular,

$$K(M) \leq \sum_{i=1}^n \mu_i \delta_i(\pi),$$

with equality exactly when the policy is cellwise optimal on every non-tied cell.

Proof. On each cell C, the policy contributes $A_C(1 - q_C) + B_C q_C$. The minimum is $\min\{A_C, B_C\}$, achieved at $q_C^\star$. If $A_C > B_C$, then $q_C^\star = 1$ and

$$A_C(1 - q_C) + B_C q_C = B_C + (A_C - B_C)(1 - q_C) = \min\{A_C, B_C\} + |A_C - B_C|\, |q_C - q_C^\star|.$$

If $A_C < B_C$, then $q_C^\star = 0$ and

$$A_C(1 - q_C) + B_C q_C = A_C + (B_C - A_C) q_C = \min\{A_C, B_C\} + |A_C - B_C|\, |q_C - q_C^\star|.$$

If $A_C = B_C$, both sides reduce to A_C for every q_C. Summing over cells and using Theorem 11 gives the decomposition. The equality criterion is immediate from the absolute-value term. □

Theorem 12 (Multi-class exact reduction)**.**
Assume now a K-class setting in which each support point has an optimal label $y_i \in \{1, \dots, K\}$ and, for $i \in C$, pointwise regret takes the weighted misreport form

$$\delta_i(\pi) = r_i(1 - q_C^{(y_i)}),$$

where $q_C \in \Delta^{K-1}$ is the class-distribution reported on cell C. Define

$$A_C^{(k)} := \sum_{i \in C,\, y_i = k} \mu_i r_i$$

for each cell C and class k. Then

$$\inf_{\pi \text{ is } M\text{-based}} \sum_{i=1}^{n} \mu_i \delta_i(\pi) = \sum_{C \in \mathcal{C}_M} \Big(\sum_{k=1}^{K} A_C^{(k)} - \max_k A_C^{(k)} \Big).$$

Proof. An M-based policy assigns a distribution $q_C \in \Delta^{K-1}$ over the K classes for each cell C. Under the pointwise assumption in the theorem statement, its expected regret on cell C is

$$\sum_{k=1}^{K} A_C^{(k)}(1 - q_C^{(k)}) = \Big(\sum_k A_C^{(k)} \Big) - \sum_k A_C^{(k)} q_C^{(k)}.$$

To minimise this over $q_C \in \Delta^{K-1}$, maximise the inner product $\sum_k A_C^{(k)} q_C^{(k)}$. This is a linear function on the simplex, so its maximum is achieved at a vertex, namely all mass on the class $k^\star = \arg\max_k A_C^{(k)}$. The minimum per-cell regret is therefore $(\sum_k A_C^{(k)}) - \max_k A_C^{(k)}$. To make y single-valued on C, one must delete every point not in the dominant class, costing the same quantity. Summing over cells gives both sides. □

Proposition 16 (Lipschitz stability of K)**.**
For fixed optimal labels, if $\hat{r}_i$ are perturbed margin weights, then $|\widehat{K}(M) - K(M)| \le \sum_{i=1}^{n} \mu_i |\hat{r}_i - r_i|$. Since r is Lipschitz with constant 4 on $(0,1)$ (the one-sided derivatives satisfy $|dr/dp| \le 4$ on $(0, \frac{1}{2}) \cup (\frac{1}{2}, 1)$), estimation error in conditional probabilities, including a change of optimal label, satisfies $|\widehat{K}(M) - K(M)| \le 4 \sum_{i=1}^{n} \mu_i |\hat{p}_i - p_i|$.

Proof. For each cell C, $|\min\{\hat{A}_C, \hat{B}_C\} - \min\{A_C, B_C\}| \le |\hat{A}_C - A_C| + |\hat{B}_C - B_C| \le \sum_{i \in C} \mu_i |\hat{r}_i - r_i|$. Sum over cells. For the second bound, on $(0, \frac{1}{2})$ we have $r = (1-2p)/(1-p)$ with $|dr/dp| = 1/(1-p)^2 \le 4$, and on $(\frac{1}{2}, 1)$ we have $r = (2p-1)/p$ with $|dr/dp| = 1/p^2 \le 4$. At $p = \frac{1}{2}$, r has a kink but both one-sided derivatives have magnitude 4. So r is Lipschitz with constant 4 on $(0,1)$, giving $|\hat{r}_i - r_i| \le 4|\hat{p}_i - p_i|$ when optimal labels stay fixed. If an estimated probability crosses $1/2$, each fixed prediction's pointwise regret remains 4-Lipschitz. For prediction 0 it is 0 below $1/2$ and $(2p-1)/p$ above $1/2$. For prediction 1 it is $(1-2p)/(1-p)$ below $1/2$ and 0 above $1/2$. Both are continuous at $1/2$. Every fixed policy's expected regret therefore changes by at most $4 \sum_i \mu_i |\hat{p}_i - p_i|$, and taking minima over the same policies preserves this bound. □

Proposition 17 (Measurable exact reduction)**.**
Let $(\mathcal{X}, \Sigma, \mu)$ be a probability space. Let $(\mathcal{Z}, \mathcal{B})$ be a standard Borel space and let $\zeta : \mathcal{X} \to \mathcal{Z}$, $y : \mathcal{X} \to \{0,1\}$, and $r : \mathcal{X} \to [0,1]$ be measurable.

For a measurable decision rule $q : \mathcal{Z} \to [0,1]$, define

$$R(q) := \mathbb{E}\Big[r\Big((1 - q(\zeta)) \mathbf{1}\{y = 1\} + q(\zeta) \mathbf{1}\{y = 0\} \Big) \Big].$$

Let $a, b : \mathcal{Z} \to [0,1]$ be measurable versions satisfying

$$a(\zeta) = \mathbb{E}[r \mathbf{1}\{y = 1\} \mid \sigma(\zeta)] \quad \text{and} \quad b(\zeta) = \mathbb{E}[r \mathbf{1}\{y = 0\} \mid \sigma(\zeta)] \qquad \text{a.s.}$$

Then

$$\inf_{q:\mathcal{Z}\to[0,1] \text{ measurable}} R(q) = \mathbb{E}[\min\{a(\zeta), b(\zeta)\}].$$

Moreover any measurable selector

$$q^\star(z) \in \arg\min_{q\in[0,1]} \big(a(z)(1-q) + b(z)q\big)$$

attains the infimum; for example $q^\star(z) = \mathbf{1}\{a(z) > b(z)\}$.

Proof. Because $(\mathcal{Z}, \mathcal{B})$ is standard Borel, the conditional expectations with respect to $\sigma(\zeta)$ admit measurable versions as functions of ζ; those are the displayed functions a and b. For any measurable $q : \mathcal{Z} \to [0, 1]$, the random variable $q(\zeta)$ is $\sigma(\zeta)$-measurable, so the tower property gives

$$R(q) = \mathbb{E}[a(\zeta)(1 - q(\zeta)) + b(\zeta)q(\zeta)]\,.$$

For each fixed z, the integrand

$$a(z)(1-q) + b(z)q$$

is affine in $q \in [0, 1]$, so its minimum is $\min\{a(z), b(z)\}$, attained by any selector $q^\star(z) \in \arg\min_{q\in[0,1]}(a(z)(1-q) + b(z)q)$. The explicit choice $q^\star(z) = \mathbf{1}\{a(z) > b(z)\}$ is measurable because a and b are measurable.

Substituting this pointwise minimiser into the previous display yields

$$\inf_{q:\mathcal{Z}\to[0,1] \text{ measurable}} R(q) = \mathbb{E}[\min\{a(\zeta), b(\zeta)\}],$$

which is finite because $0 \le a, b \le 1$. □

Theorem 13 (Representation convergence)**.**
Let $I_+ = \{i : \mu_i r_i > 0\}$ be the informative support. Define the coarsest pure diagnostic partition $\mathcal{P}^\star_+$ on I_+ by grouping points that share the same test value T_i and optimal label y_i. Then $K(M) = 0$ iff $\mathcal{C}_M|_{I_+}$ refines $\mathcal{P}^\star_+$.

Assume that an admissible representation $M(h)$ induces $\mathcal{P}^\star_+$ on I_+. This requires a single code for each history, used across every test. Call a representation M **informative-minimal zero-regret** if $K(M) = 0$ and there is no representation M' with $K(M') = 0$ such that $\mathcal{C}_{M'}|_{I_+}$ is strictly coarser than $\mathcal{C}_M|_{I_+}$. Then M is informative-minimal zero-regret iff

$$\mathcal{C}_M|_{I_+} = \mathcal{P}^\star_+.$$

Consequently, any two informative-minimal zero-regret representations induce the same diagnostic partition on I_+, up to labels for its groups.

Proof. By Theorem 11, $K(M) = 0$ iff $\min\{A_C, B_C\} = 0$ for every cell $C \in \mathcal{C}_M$. On the informative support I_+, this is equivalent to saying that each cell contains only one optimal label at its fixed test value. That is exactly the statement that $\mathcal{C}_M|_{I_+}$ refines $\mathcal{P}^\star_+$. This proves the first claim.

For the minimality claim, note that by assumption the partition $\mathcal{P}^\star_+$ itself is representable, and it is zero-regret because every one of its cells is pure by construction. So any zero-regret representation whose induced partition on I_+ is strictly finer than $\mathcal{P}^\star_+$ cannot be informative-minimal: coarsening it down to $\mathcal{P}^\star_+$ preserves zero regret. Conversely, if $\mathcal{C}_M|_{I_+} = \mathcal{P}^\star_+$, then no strictly coarser zero-regret partition exists, because any strict coarsening would merge two groups with different optimal predictions at one test and would therefore create an impure informative group. Hence M is informative-minimal zero-regret iff $\mathcal{C}_M|_{I_+} = \mathcal{P}^\star_+$.

Finally, if M_1 and M_2 are both informative-minimal zero-regret, then they induce the same partition $\mathcal{P}^\star_+$ on I_+. Their diagnostic group labels may differ, but matching equal groups gives a bijection between those labels. □

Proposition 18 (Coordination cost of shared representations)**.**
Let N subsystems each face a diagnostic family on the same informative support I_+ with the same single test T. Let $\mathcal{P}_j^\star$ be the coarsest pure partition for subsystem j on I_+, with $|\mathcal{P}_j^\star| = k_j$ cells. Then every shared representation that achieves zero regret for all N subsystems simultaneously must have at least

$$|\mathcal{P}_1^\star \vee \cdots \vee \mathcal{P}_N^\star|$$

cells on I_+, where $\vee$ denotes the common refinement. This lower bound is tight in the unrestricted class of shared representations. More generally, it is tight whenever the class contains a representation whose informative quotient is exactly the common refinement. Hence in that case the minimum number of cells in a shared zero-regret representation is exactly

$$|\mathcal{P}_1^\star \vee \cdots \vee \mathcal{P}_N^\star|.$$

Moreover,

$$|\mathcal{P}_1^\star \vee \cdots \vee \mathcal{P}_N^\star| \;\le\; \min\Big\{|I_+|,\; \prod_{j=1}^{N} k_j\Big\}.$$

The product bound is achievable when $|I_+| \ge \prod_{j=1}^N k_j$ and every tuple of one cell from each partition has nonempty intersection.

Proof. By Theorem 13, zero regret for subsystem j requires the shared partition to refine $\mathcal{P}_j^\star$. The coarsest partition refining all N partitions is the common refinement $\mathcal{P}_1^\star \vee \cdots \vee \mathcal{P}_N^\star$, so any shared zero-regret representation needs at least that many cells.

This lower bound is tight whenever the shared representation class contains a representation whose informative quotient is the common refinement. In particular, this is automatic in the unrestricted quotient class. Such a representation has induced partition exactly $\mathcal{P}_1^\star \vee \cdots \vee \mathcal{P}_N^\star$. Because each refinement cell lies inside a pure cell of every subsystem partition, it has zero regret for every subsystem simultaneously. So in that case the minimum possible cell count is exactly the size of the common refinement.

Each refinement cell is an intersection of one cell from each subsystem partition. There are at most $\prod_{j=1}^N k_j$ such tuples, and of course no partition of I_+ can have more than $|I_+|$ nonempty cells. Hence

$$|\mathcal{P}_1^\star \vee \cdots \vee \mathcal{P}_N^\star| \le \min\Big\{|I_+|,\; \prod_{j=1}^{N} k_j\Big\}.$$

If $|I_+| \ge \prod_{j=1}^N k_j$ and every tuple of subsystem cells has nonempty intersection, then every possible intersection occurs and the refinement has exactly $\prod_j k_j$ cells. □

Remark 2 (Splintering on the diagnostic quotient)**.**
Take three support points x_1, x_2, x_3 under one common test T.

Subsystem A has optimal labels $y^A(x_1) = 1$, $y^A(x_2) = 0$, $y^A(x_3) = 0$. Its coarsest pure quotient is $\mathcal{P}_{A,+}^\star = \{\{x_1\}, \{x_2, x_3\}\}$.

Subsystem B has labels $y^B(x_1) = 0$, $y^B(x_2) = 1$, $y^B(x_3) = 0$. Its coarsest pure quotient is $\mathcal{P}_{B,+}^\star = \{\{x_2\}, \{x_1, x_3\}\}$.

These minimal zero-cost quotients do not coincide. If each subsystem separately minimises its own weighted forgetting cost, they converge to different partitions (Theorem 13). If they are forced to share one common zero-cost quotient, they must use the finer common refinement $\{\{x_1\}, \{x_2\}, \{x_3\}\}$.

Proposition 19 (Regret ordering)**.**
Define $M_1 \preceq_\rho M_2$ iff $K(M_1) \ge K(M_2)$. This is a total preorder on representations, and it

becomes a partial order on the equivalence classes under

$$M_1 \sim_\rho M_2 \iff K(M_1) = K(M_2).$$

It is consistent with the partition refinement lattice. If $\mathcal{C}_{M_2}|_{I_+}$ refines $\mathcal{C}_{M_1}|_{I_+}$, then $K(M_2) \leq K(M_1)$. More sharply, let a parent cell C_1 split into subcells $C_2^{(j)}$ and define $\Delta_j := A_{C_2^{(j)}} - B_{C_2^{(j)}}$. Then

$$\min\{A_{C_1}, B_{C_1}\} - \sum_j \min\{A_{C_2^{(j)}}, B_{C_2^{(j)}}\} = \frac{1}{2}\left(\sum_j |\Delta_j| - \left|\sum_j \Delta_j\right|\right).$$

So the inequality is strict iff at least one mixed parent cell is split into subcells with opposite-signed imbalances. Equivalently, strict improvement occurs iff some mixed parent cell is split into subcells with different dominant labels.

Proof. If $\mathcal{C}_{M_2}$ refines $\mathcal{C}_{M_1}$, every cell $C_1 \in \mathcal{C}_{M_1}$ is partitioned into subcells $C_2^{(j)} \in \mathcal{C}_{M_2}$. The total weights therefore satisfy

$$A_{C_1} = \sum_j A_{C_2^{(j)}}, \qquad B_{C_1} = \sum_j B_{C_2^{(j)}}.$$

Using $\min\{x, y\} = \frac{x+y-|x-y|}{2}$, I get

$$\min\{A_{C_1}, B_{C_1}\} = \frac{\sum_j (A_{C_2^{(j)}} + B_{C_2^{(j)}}) - \left|\sum_j (A_{C_2^{(j)}} - B_{C_2^{(j)}})\right|}{2},$$

while

$$\sum_j \min\{A_{C_2^{(j)}}, B_{C_2^{(j)}}\} = \frac{\sum_j (A_{C_2^{(j)}} + B_{C_2^{(j)}}) - \sum_j |A_{C_2^{(j)}} - B_{C_2^{(j)}}|}{2}.$$

Subtracting gives the displayed identity

$$\min\{A_{C_1}, B_{C_1}\} - \sum_j \min\{A_{C_2^{(j)}}, B_{C_2^{(j)}}\} = \frac{1}{2}\left(\sum_j |\Delta_j| - \left|\sum_j \Delta_j\right|\right) \geq 0,$$

by the triangle inequality. Summing over parent cells yields $K(M_1) \geq K(M_2)$.

The relation $\preceq_\rho$ is reflexive, transitive, and total because it is induced by the scalar score K. It need not be antisymmetric on representations themselves, since distinct representations can have the same regret floor. Quotienting by $\sim_\rho$ restores antisymmetry, so the induced relation on equivalence classes is a partial order.

For strictness, the triangle inequality is strict exactly when the summands are not all of the same sign. So the improvement on a given parent cell is strict iff the subcell imbalances Δ_j take both positive and negative values. That is equivalent to saying the refined mixed parent cell contains subcells with different dominant labels. Therefore the global inequality is strict iff this happens for at least one refined mixed parent cell. □

Corollary 15 (Unweighted forgetting on the informative region)**.**
Fix $\gamma \in (0, \frac{1}{2}]$ and let $\mathcal{X}_\gamma := \{i : d_i \geq \gamma\}$. The unweighted forgetting cost on the informative region, $K_\gamma(M)$, is defined by replacing $\mu_i r_i$ with μ_i in Theorem 11 and restricting to $\mathcal{X}_\gamma$. Then with $c(\gamma) = 4\gamma/(1+2\gamma)$,

$$c(\gamma) K_\gamma(M) \;\leq\; \inf_{\pi \text{ is } M\text{-based}} \sum_{i \in \mathcal{X}_\gamma} \mu_i \delta_i(\pi) \;\leq\; K_\gamma(M).$$

Proof. Let

$$s_\gamma := \sum_{i \in \mathcal{X}_\gamma} \mu_i.$$

If $s_\gamma = 0$, then both $K_\gamma(M)$ and the restricted regret are zero, so the claim is trivial. Assume $s_\gamma > 0$ and define normalised masses

$$\tilde{\mu}_i := \frac{\mu_i}{s_\gamma} \qquad \text{for } i \in \mathcal{X}_\gamma.$$

Apply Theorem 11 to the restricted support $\mathcal{X}_\gamma$ with masses $\tilde{\mu}_i$. This gives

$$\inf_{\pi \text{ is } M\text{-based}} \sum_{i \in \mathcal{X}_\gamma} \tilde{\mu}_i \delta_i(\pi) = \widetilde{K}_{r,\gamma}(M),$$

where $\widetilde{K}_{r,\gamma}(M)$ is the corresponding weighted deletion cost on the restricted support. Multiplying by s_γ yields the identity for the original masses on the restricted support

$$\inf_{\pi \text{ is } M\text{-based}} \sum_{i \in \mathcal{X}_\gamma} \mu_i \delta_i(\pi) = K_{r,\gamma}(M),$$

with

$$K_{r,\gamma}(M) := \min\Big\{ \sum_{i \in B \cap \mathcal{X}_\gamma} \mu_i r_i \;\Big|\; B \subseteq \{1, \dots, n\},\ y \text{ is constant on } (C \cap \mathcal{X}_\gamma) \setminus B \text{ for every } C \in \mathcal{C}_M \Big\}.$$

Now on $\mathcal{X}_\gamma$ I have $r_i \in [c(\gamma), 1]$. Therefore, for every admissible deletion set B,

$$c(\gamma) \sum_{i \in B \cap \mathcal{X}_\gamma} \mu_i \le \sum_{i \in B \cap \mathcal{X}_\gamma} \mu_i r_i \le \sum_{i \in B \cap \mathcal{X}_\gamma} \mu_i.$$

Taking minima over the same admissible family gives

$$c(\gamma) K_\gamma(M) \le K_{r,\gamma}(M) \le K_\gamma(M).$$

Combining this with the restricted exact reduction proves the claim. □

I use the independent binary prediction construction associated with Theorem 11. Each evaluation point corresponds to an unseen output u_i with weight $w_i := \mu_i r_i$. For an admissible policy ϑ,

$$W(\vartheta) := \sum_{u_i \in \mathrm{Ext}(\vartheta) \cap U} w_i, \qquad Z := \sum_i w_i, \qquad W^*(M) := \max_\vartheta W(\vartheta).$$

Independent choices across prediction groups give $W^*(M) = Z - K(M)$. Under these assumptions, the following corollary relates regret to a weighted KL floor.

Corollary 16 (Free-energy floor from regret)**.**
Under the independent prediction construction above, assume $Z = \sum_{i=1}^n w_i > 0$. Define the finite measure

$$\nu(A) := \sum_{u_i \in A} w_i \qquad \text{on} \qquad U = \{u_1, \dots, u_n\}.$$

For any admissible policy ϑ with $W(\vartheta) > 0$, define the weighted prior and posterior

$$p_U(u_i) := \frac{w_i}{Z}, \qquad q_\vartheta(u_i) := \frac{w_i \, \mathbf{1}\{u_i \in \mathrm{Ext}(\vartheta) \cap U\}}{W(\vartheta)}.$$

Let

$$F_2^\nu(\vartheta) := D_{\mathrm{KL},2}(q_\vartheta \| p_U).$$

Then

$$F_2^\nu(\vartheta) = \log_2 \frac{Z}{W(\vartheta)}.$$

If $W^\star(M) > 0$ (equivalently $K(M) < Z$), then the minimum weighted free-energy achievable by an admissible policy is

$$F_{2,\min}^\nu(M) = \log_2 \frac{Z}{W^\star(M)} = \log_2 \frac{Z}{Z - K(M)}.$$

If $W^\star(M) = 0$ (equivalently $K(M) = Z$), then no admissible policy retains positive weighted unseen mass, so there is no admissible policy with finite weighted free-energy relative to p_U. In that case the attainable floor is $+\infty$. In particular, whenever $0 < K(M) < Z$, the weighted free-energy floor is strictly above the zero-regret value 0 and is strictly increasing in $K(M)$.

Proof. By construction, q_ϑ is the normalised restriction of the measure ν to the retained unseen region $\mathrm{Ext}(\vartheta) \cap U$, while p_U is the normalised version of ν on all of U. Therefore, for q_ϑ-almost every u_i,

$$\frac{q_\vartheta(u_i)}{p_U(u_i)} = \frac{w_i/W(\vartheta)}{w_i/Z} = \frac{Z}{W(\vartheta)}.$$

The ratio is constant on the support of q_ϑ. Hence

$$F_2^\nu(\vartheta) = D_{\mathrm{KL},2}(q_\vartheta \| p_U) = \sum_{u_i} q_\vartheta(u_i) \log_2 \frac{q_\vartheta(u_i)}{p_U(u_i)} = \log_2 \frac{Z}{W(\vartheta)}.$$

This is strictly decreasing in $W(\vartheta)$. Therefore, if $W^\star(M) > 0$, the smallest attainable weighted free-energy is achieved by maximising $W(\vartheta)$, i.e. by $W^\star(M)$. Theorem 11 gives $W^\star(M) = Z - K(M)$, so

$$F_{2,\min}^\nu(M) = \log_2 \frac{Z}{W^\star(M)} = \log_2 \frac{Z}{Z - K(M)}.$$

If instead $W^\star(M) = 0$, then there is no admissible policy with $W(\vartheta) > 0$, so the domain on which F_2^ν is finite is empty. In the extended reals the attainable floor is therefore $+\infty$. If $K(M) = 0$, the finite formula gives 0. If $0 < K(M) < Z$, the denominator decreases and the logarithm increases strictly. □

14 Bagi and Unagi

For $\Pi_\alpha \neq \emptyset$, Bagi selects

$$\mathrm{Bagi}_{\mathfrak{v}}(\alpha) \in \arg\max_{\pi \in \Pi_\alpha} |\mathrm{Ext}(\pi)|.$$

Operational Unagi first maximises the specified utility of intelligence over admissible embodiments, then minimises K_ρ among utility ties, then applies Bagi within the selected embodiment. Semantic Unagi instead compares the distinct unseen truth sets specified below. These optimisations presume that the required optima are attained.

Definition 24 (Semantic Unagi score for a disembodied task)**.**
Fix a disembodied task λ_ρ and a nonempty class V of finite vocabularies in its domain. I define the set of admissible pairs

$$\mathcal{A}(\lambda_\rho, V) := \{(\mathfrak{v}, \pi) \mid \mathfrak{v} \in V,\ \pi \in \Pi_{\lambda_\rho(\mathfrak{v})}\}.$$

Assume $\mathcal{A}(\lambda_\rho, V) \neq \emptyset$. For an admissible pair, define

$$\beta_{\lambda_\rho, V}(\mathfrak{v}, \pi) := \Big| \{T(y) \mid y \in \mathrm{Ext}(\pi)\} \setminus \{T(o) \mid o \in O_{\lambda_\rho(\mathfrak{v})}\} \Big|,$$

where the extension is computed in the paired vocabulary $\mathfrak{v}$. Statements with the same truth set are tallied once. I call $\beta_{\lambda_\rho,V}(\mathfrak{v},\pi)$ the **semantic Unagi score** of the pair. Semantic Unagi ranks admissible pairs by

$$(\mathfrak{v}_1,\pi_1) \preceq_{\lambda_\rho,V} (\mathfrak{v}_2,\pi_2) \quad\Longleftrightarrow\quad \beta_{\lambda_\rho,V}(\mathfrak{v}_1,\pi_1) \le \beta_{\lambda_\rho,V}(\mathfrak{v}_2,\pi_2).$$

Proposition 20 (Semantic Unagi optimisation)**.**
Fix λ_ρ and V as above. Let

$$V_{\mathrm{ad}} := \{\mathfrak{v} \in V \mid \Pi_{\lambda_\rho(\mathfrak{v})} \neq \emptyset\},$$

and assume the following maximum is attained. Define

$$V^\star := \arg\max_{\mathfrak{v}\in V_{\mathrm{ad}}} \max_{\pi\in\Pi_{\lambda_\rho(\mathfrak{v})}} \beta_{\lambda_\rho,V}(\mathfrak{v},\pi).$$

An admissible pair $(\mathfrak{v}^\star,\pi^\star)$ maximises $\preceq_{\lambda_\rho,V}$ iff

$$\mathfrak{v}^\star \in V^\star$$

and

$$\beta_{\lambda_\rho,V}(\mathfrak{v}^\star,\pi^\star) = \max_{\pi\in\Pi_{\lambda_\rho(\mathfrak{v}^\star)}} \beta_{\lambda_\rho,V}(\mathfrak{v}^\star,\pi).$$

Equivalently, its score is the maximum over all admissible pairs.

Proof. For each admissible pair, its policy belongs to the class associated with its vocabulary in the definition of $V^\star$. The pair's score cannot exceed the maximum over that class, and hence cannot exceed the largest of those maxima. A pair whose vocabulary lies in $V^\star$ and whose policy attains the maximum for that vocabulary attains the global maximum. Conversely, any global maximiser must attain the maximum for its vocabulary, and that vocabulary must lie in $V^\star$. □

For the finite set $\mathcal{T}$ of distinct truth sets of unseen test demands withheld during learning,

$$\widehat{\beta}_{\mathcal{T}}(\mathfrak{v},\pi) := \Big|\mathcal{T} \cap \Big(\{T(y) : y \in \mathrm{Ext}(\pi)\} \setminus \{T(o) : o \in O_{\lambda_\rho(\mathfrak{v})}\}\Big)\Big| \le \beta_{\lambda_\rho,V}(\mathfrak{v},\pi).$$

The inequality follows by inclusion. Equal positive weights give the corresponding weighted bound; equality requires exhaustion of the distinct unseen truth sets. Among policies attaining a maximal semantic score in a tied embodiment, Bagi may break ties. In the full-vocabulary limit, infinite cardinalities may require a measure or finite restriction before comparison.

Corollary 17 (Semantic and operational Unagi)**.**
Fix one list of history-task pairs Z with total weight $W_Z := \sum_{z\in Z} \rho_z m_z$, and let each admissible vocabulary $\mathfrak{v}$ induce a representation $M_{\mathfrak{v}}$. For each representation, assume a nonempty admissible policy family whose members permit outputs from at most one prediction per group. In this degenerate vocabulary,

$$W^*(M_{\mathfrak{v}}) = W_Z - K_\rho(M_{\mathfrak{v}}).$$

Hence minimising K_ρ maximises weighted freedom $W^*(M_{\mathfrak{v}})$ for the task in this degenerate vocabulary. For a task built in the same way in any embodied language,

$$W^*(M_{\mathfrak{v}}) \le W_Z - K_\rho(M_{\mathfrak{v}}).$$

Thus K_ρ lower-bounds the weighted freedom lost, $W_Z - W^*(M_{\mathfrak{v}})$. Forgetting cost K_ρ gives no lower bound on semantic Unagi's score itself. Suppose each $M_{\mathfrak{v}}$-based policy is paired with a correct policy π of $\lambda_\rho(\mathfrak{v})$. The pairing is assumed to match the unseen outputs in the buffer of the $M_{\mathfrak{v}}$-based policy one to one with the distinct unseen truth sets of π, all with equal weight. Then the weighted freedom of the $M_{\mathfrak{v}}$-based policy is proportional to semantic Unagi's score $\beta_{\lambda_\rho,V}(\mathfrak{v},\pi)$ in Definition 24, so the two scores give one ranking.

Proof. The task in this degenerate vocabulary has independent predictions for each group. So I can use Section 13's identity $\min R_\rho^M = W_Z - W^*(M)$ over M-based policies. Theorem 11 states that $\min R_\rho^M = K_\rho(M)$. Equating the two expressions for minimum regret, I get $W_Z - W^*(M_\mathfrak{v}) = K_\rho(M_\mathfrak{v})$. Since W_Z is fixed across the compared representations, minimising K_ρ is equivalent to maximising W^*. Each admissible policy holds weight at most $\max\{W_C(0), W_C(1)\}$ in each group C. Summing this upper bound and subtracting it from total weight, I get $W_Z - W^*(M_\mathfrak{v}) \geq K_\rho(M_\mathfrak{v})$, which rearranges to $W^*(M_\mathfrak{v}) \leq W_Z - K_\rho(M_\mathfrak{v})$. Under equal weights, the weighted mass of a policy's unseen buffer is proportional to the number of unseen outputs in that buffer. Under the pairing, these outputs correspond one to one with the distinct unseen truth sets of the paired policy. Therefore the two scores give one ranking. □

15 Novelty, damage and autopoiesis

Definition 25 (Viable policy buffers)**.**
Fix a finite vocabulary $\mathfrak{v}$ and a child task $\alpha = \langle I_\alpha, O_\alpha \rangle$ with a finite nonempty subset $\mathcal{V} \subseteq \Pi_\alpha$ of correct policies under comparison. The unseen region is

$$U := L_\mathfrak{v} \setminus \mathrm{Ext}(I_\alpha).$$

This represents the demands an unknown future might present us with. For each policy $\pi \in \mathcal{V}$, I define its unseen buffer as

$$B_\pi := \mathrm{Ext}(\pi) \cap U.$$

This is the part of the unseen region covered by the policy.

Definition 26 (Timeline continuations and demand sets)**.**
Fix a timeline $\tau : I \to \Phi$ and $t \in I$. Fix its prefix $\tau_{\leq t}$ and a finite future test interval $J_t = \{t+1, \ldots, t+r\}$ with $r \in \mathbb{N}$ and $r \geq 1$. For each state $\phi \in \Phi$, the maximal statement seen through $\mathfrak{v}$ is the encoding $\mathrm{Enc}_\mathfrak{v}(\phi)$ from Definition 7. Since $\phi \in \bigcap \mathrm{Enc}_\mathfrak{v}(\phi)$, this is an element of $L_\mathfrak{v}$. An infinite timeline $c : \mathbb{N} \to \Phi$ is admissible for α on J_t when $c(s) = \tau(s)$ for every $s \leq t$, and

$$\mathrm{Enc}_\mathfrak{v}(c(s)) \in \mathrm{Ext}(I_\alpha) \quad \Longrightarrow \quad \mathrm{Enc}_\mathfrak{v}(c(s)) \in O_\alpha$$

for every $s \in J_t$. I call c a timeline continuation, because it continues a particular past. Let $\mathcal{C}_{\tau,t}$ be a finite nonempty set of such admissible timeline continuations, equipped with probability measure $P_\mathcal{C}$. In a physical application, $\mathcal{C}_{\tau,t}$ is restricted to timeline continuations permitted by the chosen physical model. For $c \in \mathcal{C}_{\tau,t}$ and a finite bundle $x \subseteq \mathfrak{v}$, $\rho_{c,J_t}(x)$ is the number of ticks for which x persists in the timeline c, defined as

$$\rho_{c,J_t}(x) := |\{s \in J_t \mid c(s) \in T(x)\}|.$$

For $c \in \mathcal{C}_{\tau,t}$, define the demand set

$$\Delta(c) := \{\, \mathrm{Enc}_\mathfrak{v}(c(s)) \mid s \in J_t,\ \mathrm{Enc}_\mathfrak{v}(c(s)) \in U \,\}.$$

Thus $\Delta(c)$ is the set of unseen maximal outputs that are true during the future test interval of timeline c. Note that every member of $\Delta(c)$ is in the set of unseen demands that could intersect with an unseen buffer. Now remember how I defined a probability measure $P_\mathcal{C}$ over these timeline continuations. Each continuation determines a demand set, so $P_\mathcal{C}$ determines a probability measure over demand sets. The induced novelty distribution on 2^U is

$$P_\tau(S) := P_\mathcal{C}(\{c \in \mathcal{C}_{\tau,t} : \Delta(c) = S\}).$$

The prior P_τ is exchangeable when $P_\tau(S) = P_\tau(S')$ whenever S and S' are sets of unseen encodings with $|S| = |S'|$.[2]

Lemma 13 (A policy survives a future when, and only when, that future's demands are inside its buffer)**.**
Under Definition 26, for every $\pi \in \mathcal{V}$ and every $c \in \mathcal{C}_{\tau,t}$,

$$\rho_{c,J_t}(\pi) = |J_t| \quad \Longleftrightarrow \quad \Delta(c) \subseteq B_\pi.$$

Moreover,

$$\rho_{c,J_t}(\pi) = |J_t| - \big|\{s \in J_t : \mathrm{Enc}_{\mathfrak{v}}(c(s)) \in U \setminus B_\pi\}\big|.$$

Consequently,

$$P_{\mathcal{C}}\big(\rho_{c,J_t}(\pi) = |J_t|\big) = P_\tau(\{S \subseteq U : S \subseteq B_\pi\}).$$

Proof. For any $s \in J_t$, the statement π is true at $c(s)$ when

$$\pi \subseteq \mathrm{Enc}_{\mathfrak{v}}(c(s)),$$

which is when $\mathrm{Enc}_{\mathfrak{v}}(c(s)) \in \mathrm{Ext}(\pi)$.

If $\mathrm{Enc}_{\mathfrak{v}}(c(s)) \in U$, then π is true at $c(s)$ if and only if that encoding lies in $B_\pi = \mathrm{Ext}(\pi) \cap U$. If $\mathrm{Enc}_{\mathfrak{v}}(c(s)) \in \mathrm{Ext}(I_\alpha)$, by admissibility it lies in O_α. Since $\pi \in \Pi_\alpha$, by correctness $O_\alpha \subseteq \mathrm{Ext}(\pi)$. The sets U and $\mathrm{Ext}(I_\alpha)$ partition $L_{\mathfrak{v}}$, so π is false at $c(s)$ iff $\mathrm{Enc}_{\mathfrak{v}}(c(s)) \in U \setminus B_\pi$. Counting the ticks of J_t gives the second display, and π is true at every tick in J_t iff every unseen encoding in the continuation lies in B_π, meaning $\Delta(c) \subseteq B_\pi$.

The probability equality follows because P_τ is the pushforward of $P_{\mathcal{C}}$ under Δ. □

Proposition 21 (Survival under novelty)**.**
Let $S \subseteq U$ be drawn from a probability distribution P on 2^U that is exchangeable, meaning $P(S)$ depends only on $|S|$. Then $P(\{S \subseteq U : S \subseteq B_\pi\})$ is a nondecreasing function of $|B_\pi|$, and when $|B_{\pi'}| > |B_\pi|$, $P(\{S \subseteq U : S \subseteq B_{\pi'}\}) > P(\{S \subseteq U : S \subseteq B_\pi\})$ if and only if $P(1 \le |S| \le |B_{\pi'}|) > 0$. Under the uniform P where every $S \subseteq U$ is equally likely,

$$P(\{S \subseteq U : S \subseteq B_\pi\}) = 2^{|B_\pi| - |U|}.$$

Proof. Let $b = |B_\pi|$ and $N = |U|$. Exchangeability means there are nonnegative numbers r_m such that every subset of U with size m has probability r_m. Hence

$$P(\{S \subseteq U : S \subseteq B_\pi\}) = \sum_{m=0}^{b} \binom{b}{m} r_m.$$

For fixed m, the term $\binom{b}{m}$ is nondecreasing in b, and every coefficient r_m is nonnegative. The whole sum is nondecreasing in b. For $b < b'$, the difference between the sums at b' and at b is $\sum_{m=1}^{b'} \big[\binom{b'}{m} - \binom{b}{m}\big] r_m$ with $\binom{b}{m} = 0$ for $m > b$. Every bracket is positive, since $\binom{b'}{m} > \binom{b}{m}$ for $1 \le m \le b'$, so the difference is positive if and only if $r_m > 0$ for some $1 \le m \le b'$, which is $P(1 \le |S| \le b') > 0$. In the uniform case, $r_m = 2^{-N}$ for every subset of size m, so

$$P(\{S \subseteq U : S \subseteq B_\pi\}) = \sum_{m=0}^{b} \binom{b}{m} 2^{-N} = 2^{b-N}.$$

□

[2] Only encodings occur in $\Delta(c)$, so P_τ is concentrated on sets of unseen encodings and exchangeability is a condition on those sets. Imposed on every subset of U it would force $P_\tau(\emptyset) = 1$ whenever U contains a statement that is no state's encoding, as it does in any product vocabulary over Φ, where every encoding has one program per block and a nonempty U contains the empty statement.

For the finite viable-policy set $\mathcal{V}$, put

$$F(\pi) := P_\tau\{S \subseteq U : S \subseteq B_\pi\}, \qquad M_\pi := \{\pi' \in \mathcal{V} : F(\pi') \geq F(\pi)\}, \qquad I(\pi) := -\log_2 \frac{|M_\pi|}{|\mathcal{V}|}.$$

Thus functional information is the survival-rank statistic on this fixed $\mathcal{V}$, in bits [Hazen et al., 2007, Szostak, 2003].

Theorem 14 (The Law of Increasing Functional Information)**.**
Fix the objects in Definitions 25 and 26. Let π be drawn from $\mathcal{V}$ by a probability distribution w, uniform or otherwise, and a continuation c from $P_\mathcal{C}$ independently of π. If the full persistence event $\rho_{c,J_t}(\pi) = |J_t|$ has positive probability, then

$$\mathbb{E}[I(\pi) \mid \rho_{c,J_t}(\pi) = |J_t|] \geq \mathbb{E}[I(\pi)]\,,$$

with strict inequality if and only if $F(\pi)$ is not constant on the policies to which w gives positive probability.

Proof. By Lemma 13, conditioning on full persistence through the continuation is conditioning on the event $S \subseteq B_\pi$. For each $\pi \in \mathcal{V}$, this event has probability $F(\pi)$.

I list the policies in $\mathcal{V}$ as $\pi_1, \ldots, \pi_k$ in nondecreasing order of F.

$$F(\pi_1) \leq F(\pi_2) \leq \cdots \leq F(\pi_k).$$

Let $q_j := F(\pi_j)$ and $I_j := I(\pi_j)$. From the definition of M_π, the sequence (I_j) is nondecreasing in the same order as (q_j). If $q_i < q_j$, then $M_{\pi_j} \subsetneq M_{\pi_i}$, so $I_j > I_i$.

Before selection, π is π_j with probability $w_j := w(\pi_j)$, so

$$\mathbb{E}[I(\pi)] = \sum_{j=1}^{k} w_j I_j.$$

After conditioning on survival, policy π_j is weighted by $w_j q_j$. Since the survival event has positive probability, $\sum_j w_j q_j > 0$, and

$$\mathbb{E}[I(\pi) \mid \rho_{c,J_t}(\pi) = |J_t|] = \frac{\sum_{j=1}^{k} w_j q_j I_j}{\sum_{j=1}^{k} w_j q_j}.$$

By the covariance identity, since $\sum_j w_j = 1$,

$$\sum_{j=1}^{k} w_j q_j I_j - \left(\sum_{j=1}^{k} w_j q_j\right)\left(\sum_{j=1}^{k} w_j I_j\right) = \frac{1}{2}\sum_{i,j} w_i w_j (q_i - q_j)(I_i - I_j).$$

Every term on the right is nonnegative because (q_j) and (I_j) are ordered the same way. Therefore

$$\frac{\sum_{j=1}^{k} w_j q_j I_j}{\sum_{j=1}^{k} w_j q_j} \geq \sum_{j=1}^{k} w_j I_j.$$

If F is not constant on the policies with $w_j > 0$, then some i, j with $w_i, w_j > 0$ satisfy $q_i < q_j$ and $I_i < I_j$, so $w_i w_j (q_i - q_j)(I_i - I_j) > 0$. At least one summand in the covariance identity is then positive, so the inequality is strict. If F is constant on those policies, then every q_j with $w_j > 0$ equals one positive number, so the two expectations coincide. The uniform draw, $w_j = 1/k$, is one case. □

For repeated independent continuations under one novelty law, selection updates w_j to $w'_j = w_j F(\pi_j)/\sum_i w_i F(\pi_i)$. Keeping $\mathcal{V}$ and its rank statistic I fixed, Theorem 14 applies at every round; expected functional information is nondecreasing.

Definition 27 (Damage history, mend, and closure)**.**
Fix a finite embodied language $L_{\mathfrak{v}}$, a viability statement $\mu \in L_{\mathfrak{v}}$, a policy $\pi \in L_{\mathfrak{v}}$, a timeline prefix $\tau_{\le t}$, a horizon $H \in \mathbb{N}$, and transition constraints that determine which timelines are **permitted**. The prefix $\tau_{\le t}$ is the history of timeline τ from tick 0 through the present tick t (Definition 2). In a physical application, transition constraints include conservation laws and thermodynamic requirements on energy and material flows (Section 27). Damage can occur at ticks $t+1, \dots, t+H$. A **damage history** is a set

$$\delta \subseteq \{1, \dots, H\} \times \mathfrak{v}.$$

A pair $(j, r) \in \delta$ records a **disruption** of program r at tick $t+j$. A timeline c **continues** $\tau_{\le t}$ when $c_{\le t} = \tau_{\le t}$. A timeline c that continues $\tau_{\le t}$ **realises** δ when every disruption in δ occurs on c. Disruption (j, r) occurs on c when program r is false at tick $t+j$, meaning $c(t+j) \notin r$. Basically, a disruption (j, r) occurs when r is not true at time j after t. All damage histories form a finite set $\mathbb{D}_H := 2^{\{1,\dots,H\}\times\mathfrak{v}}$. A **damage distribution** is a probability measure Q_H on $\mathbb{D}_H$.

A program $r \in \pi$ is **damage-sensitive** when Q_H gives positive probability to damage histories that contain a disruption of r. It is **essential** for μ when viability ends if r is never restored. Formally, every permitted timeline on which r is false at every tick from some tick s onward is outside $T(\mu)$ at some tick $u \ge s$. A program that is both damage-sensitive and essential is a **point of failure**. $k(\pi)$ denotes the number of points of failure for π.

A damage history δ is **repairable** from $\tau_{\le t}$ when some permitted timeline c continues $\tau_{\le t}$, realises δ, and has $c(u) \in T(\mu) \cap T(\pi)$ at some tick $u > t+H$. I call such c a **repair timeline** for δ, and such u a **repair tick**. Damage histories repairable from $\tau_{\le t}$ form π's **mend** over horizon H,

$$R_\pi(H) := \{\delta \in \mathbb{D}_H \mid \delta \text{ is repairable from } \tau_{\le t}\}.$$

Damage drawn from damage distribution Q_H lies in this mend with probability

$$q_\pi(H) := Q_H(R_\pi(H)),$$

which I call π's **mend probability**. It is the chance that π can recover from the damage it meets over horizon H.

A policy π has **finite repair capacity** $m_\pi \in \mathbb{N}$ when, for every horizon H, each damage history in $R_\pi(H)$ contains at most m_π disruptions of points of failure in π.

The same definition of repairability applies from any timeline prefix h in place of $\tau_{\le t}$, with the final tick of h as the present, and I then say δ is **repairable from** h. A policy π is **organisationally closed** relative to (μ, Q_H, H) when after each repair the system can be ready for the next. Formally, there is a set $\mathcal{E}$ of timeline prefixes that contains $\tau_{\le t}$. From each $h \in \mathcal{E}$, each $\delta \in R_\pi(H)$ with $Q_H(\{\delta\}) > 0$ is repairable from h along a repair timeline c. The repair timeline can be chosen so that its history $c_{\le u}$ up to a repair tick u lies in $\mathcal{E}$. I call $\mathcal{E}$ the prefix set. I call π **organisationally closed** when it is organisationally closed relative to (μ, Q_H, H) for every $H \in \mathbb{N}$.

Definition 28 (Joint persistence)**.**
Let U be an unseen region and let $B_\pi = \mathrm{Ext}(\pi) \cap U$ be the unseen buffer of π. Let an unseen demand $S \subseteq U$ be drawn from P_N and a damage history $\delta \in \mathbb{D}_H$ from Q_H, independently. Policy π survives the joint test when $S \subseteq B_\pi$ and $\delta \in R_\pi(H)$. Its persistence probability is

$$P_H(\pi) = P_N(S \subseteq B_\pi)\, Q_H(\delta \in R_\pi(H)).$$

Under the uniform novelty distribution on U, define log persistence by

$$\Lambda_H(\pi) := \log_2 P_H(\pi) = |B_\pi| - |U| + \log_2 q_\pi(H),$$

whenever $q_\pi(H) > 0$, and $\Lambda_H(\pi) := -\infty$ when $q_\pi(H) = 0$.

In a physical application, a damage history belongs to $R_\pi(H)$ only if its repair continuation satisfies the balances and resource bounds, restores viability and restores the conditions holding π in force. Physical feasibility therefore determines part of $q_\pi(H)$ and $P_H(\pi)$, rather than supplying an extra persistence factor. Acquisition, maintenance, repair and reproduction share the stipulated resource budget.

Lemma 14 (Complex passivity must die)**.**
Assume a non-repairing policy π has $k(\pi) = k > 0$ points of failure. In the independent damage model, each such program is disrupted at each tick with probability $p \in (0, 1)$, independently across programs and ticks. Suppose the policy survives if and only if no such program is disrupted. Then

$$q_\pi(H) = (1 - p)^{kH},$$

and therefore

$$\lim_{H\to\infty} q_\pi(H) = 0.$$

Under uniform novelty,

$$\Lambda_H(\pi) = |B_\pi| - |U| + kH \log_2(1 - p),$$

which tends to $-\infty$ as $H \to \infty$.

Proof. There are kH independent opportunities for a point of failure to be disrupted. The probability that none of them is disrupted is $(1 - p)^{kH}$. Since $0 < 1 - p < 1$ and $k > 0$, this probability tends to zero with H. Substituting $q_\pi(H) = (1 - p)^{kH}$ into Definition 28, I obtain $\Lambda_H(\pi) = |B_\pi| - |U| + kH \log_2(1 - p)$. Because $\log_2(1 - p) < 0$, the final term tends to $-\infty$. □

Lemma 15 (Repair without closure is murdered by damage)**.**
Assume a policy π has $k(\pi) = k > 0$ points of failure and finite repair capacity $m_\pi < \infty$. In the independent damage model of Lemma 14, let $N_H \sim \text{Binomial}(kH, p)$ count essential disruptions over horizon H. Then

$$q_\pi(H) \le \Pr[N_H \le m_\pi],$$

and hence

$$\lim_{H\to\infty} q_\pi(H) = 0.$$

Proof. Under the independent damage model, N_H is the number of essential disruptions among the kH independent opportunities. A policy with finite repair capacity m_π cannot survive histories with $N_H > m_\pi$, so $q_\pi(H) \le \Pr[N_H \le m_\pi]$. Since $p > 0$ and $k > 0$, the mean of N_H is $kHp \to \infty$ while m_π is fixed. Therefore $\Pr[N_H \le m_\pi] \to 0$, for example by the law of large numbers. Thus $q_\pi(H) \to 0$. □

Proposition 22 (Closure provideth an abiding chance)**.**
Let π be organisationally closed relative to (μ, Q_W, W) with prefix set $\mathcal{E}$. Let its repairs be **prompt**. This means that, from each $h \in \mathcal{E}$ with final tick s, each $\delta \in R_\pi(W)$ with $Q_W(\{\delta\}) > 0$ has a repair timeline with repair tick $s + W + 1$. That repair timeline can be chosen so that its history up to this tick lies in $\mathcal{E}$. The horizon is divided into **periods**. For $i \ge 1$, period i occupies positions $(i-1)(W+1)+1$ to $(i-1)(W+1)+W$ of a damage history, and position $i(W+1)$ has no damage. The damage in period i is read as a set $\delta_i \subseteq \{1, \dots, W\} \times \mathfrak{v}$, which is drawn from a distribution $Q_W^{(i)}$ that gives positive probability only to histories to which Q_W gives positive

probability. The draws for different periods are independent. For each horizon H, the damage distribution Q_H is the distribution of the resulting history restricted to positions $1, \ldots, H$. Let

$$\varepsilon_i := Q_W^{(i)}(\mathbb{D}_W \setminus R_\pi(W))$$

be the chance that the damage in period i lies outside the mend. Then for every horizon H, with $n := \lfloor H/(W+1) \rfloor + 1$,

$$q_\pi(H) \;\geq\; \prod_{i=1}^{n}(1-\varepsilon_i).$$

If $\varepsilon_i < 1$ for every i and $\sum_{i\geq 1} \varepsilon_i < \infty$, then for every horizon H

$$q_\pi(H) \;\geq\; q_c := \prod_{i\geq 1}(1-\varepsilon_i) \;>\; 0,$$

so q_c is an abiding chance for π.

Proof. Fix a horizon H and set $n := \lfloor H/(W+1) \rfloor + 1$, so that periods $1, \ldots, n$ cover every position up to H and $n(W+1) > H$. I draw the period histories $\delta_1, \ldots, \delta_n$ and let δ be the horizon-H history they form. Suppose every δ_i lies in $R_\pi(W)$. Each drawn δ_i has positive probability under $Q_W^{(i)}$, hence under Q_W, so closure with prompt repair applies to it from every prefix in $\mathcal{E}$. I set $h_0 := \tau_{\leq t}$, which lies in $\mathcal{E}$ and has final tick $s_0 := t$. For $i = 1, \ldots, n$ in turn, I write $s_i := s_{i-1} + W + 1$. By closure with prompt repair there is a permitted timeline c^i that continues h_{i-1} and realises δ_i at ticks $s_{i-1}+1, \ldots, s_{i-1}+W$. This timeline satisfies $c^i(s_i) \in T(\mu) \cap T(\pi)$ and has $h_i := c^i_{\leq s_i} \in \mathcal{E}$. Then $s_i = t + i(W+1)$ for every i.

The final timeline c^n continues $\tau_{\leq t}$, because c^n agrees with h_{n-1} up to tick s_{n-1}, h_{n-1} agrees with h_{n-2} up to tick s_{n-2}, and so on down to $h_0 = \tau_{\leq t}$. The timeline c^n realises every disruption of δ. A disruption at position $(i-1)(W+1)+j$ with $1 \leq j \leq W$ is a disruption of δ_i at tick $s_{i-1}+j \leq s_i$, realised by c^i. The timeline c^n agrees with c^i up to tick s_i, so c^n preserves this disruption. Positions $i(W+1)$ have no damage. Finally $c^n(s_n) \in T(\mu) \cap T(\pi)$ with $s_n = t + n(W+1) > t + H$. So c^n is a repair timeline for δ from $\tau_{\leq t}$, and $\delta \in R_\pi(H)$.

Hence $\delta \in R_\pi(H)$ whenever $\delta_1, \ldots, \delta_n \in R_\pi(W)$, and by independence of the periods,

$$\begin{aligned} q_\pi(H) = Q_H(R_\pi(H)) &\geq \prod_{i=1}^{n} Q_W^{(i)}(R_\pi(W)) = \prod_{i=1}^{n}(1-\varepsilon_i) \\ &\geq \prod_{i\geq 1}(1-\varepsilon_i) = q_c. \end{aligned}$$

Because every $\varepsilon_i < 1$ and $\sum_i \varepsilon_i < \infty$, only finitely many ε_i exceed $1/2$. Also, $\log(1-\varepsilon) \geq -2\varepsilon$ for $0 \leq \varepsilon \leq 1/2$. So $\sum_i \log(1-\varepsilon_i)$ is finite and $q_c > 0$. □

Lemma 16 (An abiding chance means probability of persistence is greater than zero)**.** Let π_c be a policy satisfying $\liminf_{H\to\infty} q_{\pi_c}(H) \geq q_c > 0$. Then

$$\liminf_{H\to\infty} P_H(\pi_c) \geq P_N(S \subseteq B_{\pi_c}) q_c.$$

Under the uniform novelty distribution this becomes

$$\liminf_{H\to\infty} P_H(\pi_c) \geq 2^{|B_{\pi_c}|-|U|} q_c.$$

Let π_o be another policy and set $\Delta b := |B_{\pi_o}| - |B_{\pi_c}|$, which may be negative. Suppose both mend probabilities are positive at horizon H. Then π_c has greater log persistence than π_o if and only if

$$\log_2 \frac{q_{\pi_c}(H)}{q_{\pi_o}(H)} > \Delta b.$$

If $q_{\pi_o}(H) = 0 < q_{\pi_c}(H)$, then π_c wins automatically.

Proof. By independence, $P_H(\pi_c) = P_N(S \subseteq B_{\pi_c}) q_{\pi_c}(H)$. Taking $\liminf$ and using $\liminf q_{\pi_c}(H) \geq q_c$, I obtain $\liminf_{H\to\infty} P_H(\pi_c) \geq P_N(S \subseteq B_{\pi_c}) q_c$. Under uniform novelty, $P_N(S \subseteq B_{\pi_c}) = 2^{|B_{\pi_c}|-|U|}$ by Proposition 21. For the comparison, subtract log persistences:

$$\begin{aligned}\Lambda_H(\pi_c) - \Lambda_H(\pi_o) &= (|B_{\pi_c}| - |B_{\pi_o}|) + \log_2 \frac{q_{\pi_c}(H)}{q_{\pi_o}(H)} \\ &= -\Delta b + \log_2 \frac{q_{\pi_c}(H)}{q_{\pi_o}(H)}.\end{aligned}$$

Difference $-\Delta b + \log_2(q_{\pi_c}(H)/q_{\pi_o}(H))$ is positive if and only if $\log_2(q_{\pi_c}(H)/q_{\pi_o}(H)) > \Delta b$, when both mend probabilities are positive. If $q_{\pi_o}(H) = 0 < q_{\pi_c}(H)$, then $\Lambda_H(\pi_o) = -\infty$ while $\Lambda_H(\pi_c)$ is finite, so π_c wins automatically. □

Relative to bounds b_s and k_s on static policies, a policy is **non-static** when $|B_\pi| > b_s$ and **complex** when $k(\pi) > k_s$.

Theorem 15 (Closure Necessity Theorem)**.**
Fix a finite embodied language, an unseen region U, a novelty distribution P_N on 2^U, and the independent damage model of Lemma 14 with $p \in (0, 1)$. Let $\mathcal{S}$ be a static policy class whose unseen buffers are bounded by b_s and whose points of failure number at most k_s, so $|B_\pi| \leq b_s$ and $k(\pi) \leq k_s$ for all $\pi \in \mathcal{S}$. Suppose $P_N(|S| > b_s) > 0$, so static policies face positive-probability demands they cannot contain. Let $\mathcal{N}$ be a class of complex policies with finite repair capacity. Then:

1. no policy in $\mathcal{S}$ survives any demand S with $|S| > b_s$, so every $\pi \in \mathcal{S}$ has $P_H(\pi) \leq 1 - P_N(|S| > b_s) < 1$ for every H;
2. every policy in $\mathcal{N}$ has asymptotic persistence probability zero;
3. any policy π_c with $\liminf_H q_{\pi_c}(H) > 0$ and $P_N(S \subseteq B_{\pi_c}) > 0$ has positive long-horizon persistence.

Thus, once novelty with positive probability exceeds the static bound b_s, and non-closed complex policies have only finite repair capacity, closure is how complex policies remain viable under ongoing damage over long horizons.

Proof. If $|S| > b_s$ and $|B_\pi| \leq b_s$, then $S \not\subseteq B_\pi$, so no static policy with buffer at most b_s survives that demand. Hence for every $\pi \in \mathcal{S}$, $P_N(S \subseteq B_\pi) \leq P_N(|S| \leq b_s) = 1 - P_N(|S| > b_s) < 1$, and $P_H(\pi) = P_N(S \subseteq B_\pi)\, q_\pi(H) \leq 1 - P_N(|S| > b_s)$ for every H. Each $\pi \in \mathcal{N}$ has $k(\pi) > k_s \geq 0$ and finite repair capacity, so by Lemma 15, $q_\pi(H) \to 0$ and hence $P_H(\pi) \leq q_\pi(H) \to 0$. Let $q_c := \liminf_H q_{\pi_c}(H) > 0$. Lemma 16 shows that $\liminf_H P_H(\pi_c) \geq P_N(S \subseteq B_{\pi_c}) q_c$, and positive $P_N(S \subseteq B_{\pi_c})$ then implies $\liminf_H P_H(\pi_c) > 0$. Buffer inequality proves item 1 of Theorem 15, limit $P_H(\pi) \to 0$ proves item 2, and $\liminf_H P_H(\pi_c) > 0$ proves item 3. □

Theorem 16 (Void of the Unviable)**.**
Assume a finite embodied language, an unseen region U, uniform novelty, and the independent damage model of Lemma 14 with $p \in (0, 1)$, and fix bounds b_s and k_s. Let π_c be a policy with a buffer of size b_c and an abiding chance q_c, meaning $\liminf_{H\to\infty} q_{\pi_c}(H) \geq q_c > 0$. Let $\mathcal{M}$ be any finite set of complex policies with finite repair capacity. Let π_s be a static policy with a buffer of size b_s and at most k_s points of failure. Suppose π_s satisfies one of the following.

1. $q_{\pi_s}(H) = q_s > 0$ for every H, and $b_c - b_s + \log_2(q_c/q_s) > 0$.
2. π_s is non-repairing and survives if and only if none of its $k(\pi_s) \geq 1$ points of failure is disrupted.

Then there exists H_0 such that for every $H \geq H_0$,

$$\Lambda_H(\pi_c) > \Lambda_H(\pi_s) > \max_{\pi \in \mathcal{M}} \Lambda_H(\pi).$$

Proof. For $\varepsilon \in (0, q_c)$, the bound $\liminf_{H\to\infty} q_{\pi_c}(H) \geq q_c$ gives H_1 with $q_{\pi_c}(H) \geq q_c - \varepsilon$ for all $H \geq H_1$. It follows that $\Lambda_H(\pi_c) \geq b_c - |U| + \log_2(q_c - \varepsilon)$ for all $H \geq H_1$. In case 1, premise $b_c - b_s + \log_2(q_c/q_s) > 0$ is strict, so I choose ε with $b_c - b_s + \log_2((q_c - \varepsilon)/q_s) > 0$. Then for all $H \geq H_1$,

$$\Lambda_H(\pi_c) \geq (b_c - |U|) + \log_2(q_c - \varepsilon) > (b_s - |U|) + \log_2 q_s = \Lambda_H(\pi_s).$$

For every $\pi \in \mathcal{M}$, Lemma 15 shows that $q_\pi(H) \to 0$. Because $L_{\mathfrak{v}}$ is finite and $\mathcal{M}$ is finite, the buffers $|B_\pi|$ are bounded above on $\mathcal{M}$. Therefore

$$\Lambda_H(\pi) = |B_\pi| - |U| + \log_2 q_\pi(H)$$

tends to $-\infty$ uniformly over $\mathcal{M}$. Hence for sufficiently large H, every policy in $\mathcal{M}$ has lower log persistence than π_s, while $b_c - b_s + \log_2(q_c/q_s) > 0$ implies higher log persistence for π_c than for π_s. Hence $\Lambda_H(\pi_c) > \Lambda_H(\pi_s) > \max_{\pi\in\mathcal{M}} \Lambda_H(\pi)$ for all sufficiently large H.

In case 2, by the argument of Lemma 14,

$$\Lambda_H(\pi_s) = b_s - |U| + k(\pi_s)H \log_2(1-p),$$

which tends to $-\infty$ because $k(\pi_s) \geq 1$, while $\Lambda_H(\pi_c) \geq b_c - |U| + \log_2(q_c - \varepsilon)$ for all $H \geq H_1$. Hence $\Lambda_H(\pi_c) > \Lambda_H(\pi_s)$ for all sufficiently large H. Now I take $\pi \in \mathcal{M}$ with $k := k(\pi)$ and $m := m_\pi$, and write $n := kH$. Whenever $n \geq \max\{1, m\}$, by Lemma 15,

$$q_\pi(H) \leq \sum_{i=0}^{m} \binom{n}{i} p^i (1-p)^{n-i} \leq (m+1)\, n^m (1-p)^{n-m},$$

because $\binom{n}{i} \leq n^m$, $p^i \leq 1$ and $(1-p)^{n-i} \leq (1-p)^{n-m}$ for every $i \leq m$. Therefore

$$\begin{aligned}\Lambda_H(\pi_s) - \Lambda_H(\pi) \geq b_s - |B_\pi| - \log_2(m+1) \\ - m\log_2(kH) + m\log_2(1-p) \\ + (k - k(\pi_s))H\,\big|\log_2(1-p)\big|.\end{aligned}$$

Since $k(\pi_s) \leq k_s < k$ implies $k - k(\pi_s) \geq 1$, the final term grows linearly in H while $m\log_2(kH)$ grows logarithmically, so $\Lambda_H(\pi_s) - \Lambda_H(\pi) \to \infty$. Buffers are bounded because $L_{\mathfrak{v}}$ is finite, and $\mathcal{M}$ is finite, so one H_0 serves every $\pi \in \mathcal{M}$. □

Full-support exploration means that each unseen completion in the current buffer has positive probability of being expressed over repeated operation; it can occur in a fixed finite buffer. For a fixed countable set $\mathcal{P}_*$ of higher-layer policies, local finiteness means that only finitely many policies have been realised at each finite time. Open-ended policy generation means that the realised policies are not contained in any fixed finite subset. These are distinct conditions.

Theorem 17 (Autopoietic Theorem)**.**
At every finite time t, let the system have a finite embodied vocabulary $\mathfrak{v}_t$ and a finite set of realised policies drawn from one fixed countable set $\mathcal{P}_*$ of possible higher-layer policies. The vocabulary may change with time. Suppose the system is exposed indefinitely to damage and to viability-relevant demands, meaning demands that end viability when unmet. Damage and demand opportunities may overlap, and no independence is assumed. Assume the following.

1. Along every timeline that remains viable, there are infinitely many successive designated damage opportunities. At each opportunity, conditional upon the full preceding timeline prefix and viability to that point, a damaging event has probability at least $p_0 > 0$.

2. For every viable finite timeline prefix h, there are finite capacities $m_0(h), m_1(h) \in \mathbb{N}$, determined by h. On every continuation of h with no further response to damage, viability cannot continue through its $(m_0(h)+1)$st subsequent damaging event. A renewal restores the lower-layer conditions under which the same response policy or process is available again. On every continuation of h with no further renewal, viability cannot continue through its $(m_1(h)+1)$st subsequent damaging event. These capacities may grow across successive prefixes.

3. For every fixed finite class $\mathcal{K} \subseteq \mathcal{P}_*$ of higher-layer policies, there is $\varepsilon_{\mathcal{K}} > 0$ and an infinite sequence of designated demand opportunities along every timeline that remains viable while confined to $\mathcal{K}$. At each opportunity, conditional upon the full preceding timeline prefix and viability to that point, a viability-relevant demand that no policy in $\mathcal{K}$ can meet has probability at least $\varepsilon_{\mathcal{K}}$.

4. The system has positive probability of remaining viable indefinitely.

Then, under Assumptions 1 to 3, the probability is zero that a timeline remains viable indefinitely with only finitely many responses to damage, or only finitely many renewals of the conditions for those responses, or with every realised higher-layer policy inside one fixed finite subset of $\mathcal{P}_*$. In particular, under Assumption 4, conditional on indefinite viability, almost every timeline contains infinitely many responses to damage, infinitely many renewals of the conditions for those responses, and open-ended policy generation. By almost every timeline I mean all but a set of probability zero. Those renewals constitute organisational closure. If they are generated through processes internal to the system's specified organisational boundary, they also meet autopoiesis' self-production condition.

Proof. Let D_j indicate damage at the jth designated damage opportunity and let V_j be the event of viability through that opportunity. For $r \geq 0$, set $N_{r,n} := \sum_{j=r+1}^{r+n} D_j$. By Assumption 1 and induction over successive opportunities,

$$\Pr(V_{r+n} \cap \{N_{r,n} \leq m\}) \leq \sum_{i=0}^{m} \binom{n}{i} p_0^i (1-p_0)^{n-i} \longrightarrow 0$$

for every r and m. For $r, m \geq 0$, let $E_{r,m}$ denote indefinite viability with no response to damage after opportunity r and $N_{r,n} \leq m$ for every $n \geq 1$. At $r = 0$, use the initial prefix. For every n,

$$\Pr(E_{r,m}) \leq \Pr(V_{r+n} \cap \{N_{r,n} \leq m\}) \longrightarrow 0,$$

so $\Pr(E_{r,m}) = 0$. On an indefinitely viable timeline with finitely many responses, I can choose an opportunity r after responses have ceased, using $r = 0$ if none occur. Let h be the timeline's prefix at opportunity r. Assumption 2 then implies that this timeline lies in $E_{r,m}$ with $m = m_0(h) < \infty$. Thus $\bigcup_{r,m \geq 0} E_{r,m}$ contains every indefinitely viable timeline with finitely many responses, and this countable union has probability zero.

Replacing response with renewal and $m_0(h)$ with $m_1(h)$ proves that indefinitely viable timelines with finitely many renewals also have probability zero. Each renewal restores lower-layer conditions for the response policy, so these renewals constitute organisational closure.

Fix any finite class $\mathcal{K}$ of higher-layer policies. By Assumption 3 and successive conditioning, the probability of remaining viable through n designated demand opportunities while confined to $\mathcal{K}$ is at most $(1-\varepsilon_{\mathcal{K}})^n$. This bound tends to zero. Thus indefinite viability while confined to $\mathcal{K}$ has probability zero. The set $\mathcal{P}_*$ is countable, so its finite subsets form a countable family. Indefinite viability while confined to some fixed finite class is therefore a countable union of probability-zero events. By Assumption 4, conditioning on indefinite viability is well defined. Consequently, assuming indefinite viability, almost every timeline has infinitely many damage responses, infinitely many renewals, and policies outside every fixed finite class. The vocabulary and the set of realised policies at each time are finite, while their union need not be. □

Corollary 18 (A widening mend provideth an abiding chance)**.**
With the periods and the period distributions $Q_W^{(i)}$ of Proposition 22, let $B_1, B_2, \ldots \subseteq \mathbb{D}_W$ be sets of period damage histories. Each B_i is a set of damage histories the system can repair in period i. Let $\mathcal{E}_0, \mathcal{E}_1, \ldots$ be sets of timeline prefixes with $\tau_{\leq t} \in \mathcal{E}_0$[3]. Each $\mathcal{E}_i$ holds histories the system can have once period i has been repaired. Suppose that for every period $i \geq 1$ and every prefix $h \in \mathcal{E}_{i-1}$ with final tick s, each damage history $\delta \in B_i$ with $Q_W^{(i)}(\{\delta\}) > 0$ has a repair timeline for π with repair tick $s + W + 1$ whose history up to that tick lies in $\mathcal{E}_i$. Let $\varepsilon_i := Q_W^{(i)}(\mathbb{D}_W \setminus B_i)$ be the chance that period i's damage lies outside B_i. If $\varepsilon_i < 1$ for every i and $\sum_{i \geq 1} \varepsilon_i < \infty$, then

$$q_\pi(H) \;\geq\; \prod_{i \geq 1} (1 - \varepsilon_i) \;>\; 0$$

for every horizon H. Now take the independent damage model of Lemma 14, in which each of π's k points of failure is disrupted at each tick with probability p. So a period holds kW opportunities for such a disruption. Suppose that in period i the system can repair any damage that disrupts at most m_i of those points. So B_i contains every such period history. Then

$$\varepsilon_i \leq \Pr\big[\text{Binomial}(kW, p) > m_i\big].$$

If $m_i \geq kW$ from some period i^* onward, no later period can be fatal, so $\varepsilon_i = 0$ for every $i \geq i^*$ and

$$q_c = \prod_{i < i^*} (1 - \varepsilon_i)$$

is a positive abiding chance.

Proof. At step i, the construction in the proof of Proposition 22 rests only on δ_i being promptly repairable from a prefix in one set into a prefix in the next. So it applies with $\mathcal{E}_{i-1}$ and B_i in place of $\mathcal{E}$ and $R_\pi(W)$, starting from $h_0 = \tau_{\leq t} \in \mathcal{E}_0$. Hence $\delta \in R_\pi(H)$ whenever $\delta_1 \in B_1, \ldots, \delta_n \in B_n$, so $q_\pi(H) \geq \prod_{i \leq n}(1 - \varepsilon_i) \geq \prod_{i \geq 1}(1 - \varepsilon_i)$, and the product is positive as before. Under the independent damage model, a period history outside B_i disrupts more than m_i of the k points of failure, and the number of such disruptions in a period is Binomial(kW, p), so $\varepsilon_i \leq \Pr[\text{Binomial}(kW, p) > m_i]$. This bound is zero when $m_i \geq kW$ and at most $1 - (1-p)^{kW} < 1$ otherwise. So when $m_i \geq kW$ for every $i \geq i^*$, the sum $\sum_i \varepsilon_i$ has finitely many nonzero terms, each below one. Hence $\prod_{i < i^*}(1 - \varepsilon_i)$ is positive. □

Proposition 23 (Surplus sustains repair)**.**
Fix a finite embodied language, viability statement μ, and policy π as in Definition 27. Let $\mathcal{E}$ be a set of permitted timeline prefixes ending in $T(\mu) \cap T(\pi)$. I fix a process that generates permitted timelines by acquiring resource and repairing damage, using events observed so far. The process checks viability after repair at cycle boundaries.

For a fixed positive integer W, suppose that from every prefix in $\mathcal{E}$ with at least $a > 0$ units of resource, this process completes one cycle of W ticks and ends at another prefix in $\mathcal{E}$ for every cycle outcome of positive probability. The operating requirement a covers every resource requirement within a cycle, including costs incurred before acquisition succeeds. Storage permits every finite reserve.

From each such prefix, resource gain and cost of acquisition and repair have a common bounded joint distribution (G, C), with $G, C \geq 0$. Cycle outcomes are independent of earlier cycles. Gain and cost within one cycle may be dependent. Reserve changes by $G - C$ at cycle completion.

[3] $\tau_{\leq t}$ is the present, meaning the history of the timeline from tick 0 to the current tick t, as in Definition 27. Repairs are only assumed from prefixes in the sets $\mathcal{E}_i$. So the chain of period repairs can only begin from a prefix in $\mathcal{E}_0$. The mend probability $q_\pi(H)$ measures repairability from the present, so the present $\tau_{\leq t}$ must lie in $\mathcal{E}_0$.

Let an initial prefix $\tau_{\leq t} \in \mathcal{E}$ have reserve $A_0 = a + r$, where $r > 0$. For a comparison, I draw independent cycle outcomes (G_n, C_n), each with the distribution of (G, C), for all $n \geq 1$, and set

$$X_n := G_n - C_n, \qquad \gamma := \mathbb{E}[X_1] > 0,$$

$$A_n := a + r + \sum_{j=1}^{n} X_j, \qquad E_\infty := \{A_n \geq a \text{ for every } n \geq 0\}.$$

Then some $\theta > 0$ has $\mathbb{E}[e^{-\theta X_1}] < 1$, and every $\theta > 0$ with $\mathbb{E}[e^{-\theta X_1}] \leq 1$ satisfies

$$\Pr(E_\infty) \geq 1 - e^{-\theta r} > 0.$$

On E_∞, the process reaches these reserves, every cycle completes, and μ and π hold at every cycle boundary. For damage distributions Q_H induced by the process,

$$q_\pi(H) \geq \Pr(E_\infty) \geq 1 - e^{-\theta r} \qquad \text{for every such } \theta \text{ and every } H \in \mathbb{N}.$$

Suppose a permitted finite initial phase has probability $p_{\text{start}} > 0$ of meeting its viability checks and reaching $\mathcal{E}$ with reserve at least $a+r$. Then that initial phase followed by perpetual operation has probability at least $p_{\text{start}}(1 - e^{-\theta r})$. Mend probabilities from the initial prefix of that phase have this lower bound as well.

Proof. Boundedness permits differentiation at zero, giving

$$\frac{d}{d\theta}\mathbb{E}[e^{-\theta X_1}]\Big|_{\theta=0} = -\gamma < 0.$$

Since this expectation equals one at zero, some $\theta > 0$ has $\mathbb{E}[e^{-\theta X_1}] < 1$. Fix any $\theta > 0$ with $\mathbb{E}[e^{-\theta X_1}] \leq 1$.

Let $S_n := A_n - a$ and let $\sigma := \inf\{n \geq 1 : S_n < 0\}$, with $\sigma = \infty$ if this set is empty. Independence of these cycle outcomes gives, by conditioning successively,

$$\mathbb{E}[e^{-\theta S_{\min\{\sigma,n\}}}] \leq e^{-\theta r}.$$

Indeed, after σ the quantity inside the expectation remains fixed, while before σ its conditional expectation at the next step is multiplied by $\mathbb{E}[e^{-\theta X_1}] \leq 1$. On $\{\sigma \leq n\}$ that quantity exceeds one, so

$$\Pr(\sigma \leq n) \leq e^{-\theta r}.$$

Taking $n \to \infty$ proves $\Pr(E_\infty) \geq 1 - e^{-\theta r} > 0$.

I use identical random cycle outcomes for this comparison and for the process itself until reserve falls below a. Initially, $\tau_{\leq t} \in \mathcal{E}$ and $A_0 \geq a$. Whenever a completed cycle ends in $\mathcal{E}$ with reserve at least a, the specified process executes another cycle, whose gain and cost are those of its fresh cycle outcome. So reserve changes by X_n, and the cycle's final prefix is again in $\mathcal{E}$. Induction establishes actual completion of every cycle on E_∞.

Fix a horizon H. Every realised history on E_∞ continues $\tau_{\leq t}$, realises its damage history through H, and reaches a cycle boundary $u > t + H$ at which μ and π hold, so the actual continuation is a repair timeline under Definition 27. Hence on E_∞ the realised horizon-H damage history lies in $R_\pi(H)$. This proves $q_\pi(H) \geq \Pr(E_\infty)$. Conditioning on each prefix in which the initial phase succeeds proves the final claim. □

Corollary 19 (Surplus accumulates)**.**
Under Proposition 23, conditional on E_∞,

$$\frac{A_n}{n} \longrightarrow \gamma > 0 \qquad \text{almost surely.}$$

Consequently, every fixed finite reserve is eventually exceeded at every later cycle boundary.

Proof. Before conditioning, the bounded independent increments obey the strong law of large numbers,

$$\frac{1}{n}\sum_{j=1}^{n} X_j \longrightarrow \gamma \qquad \text{almost surely.}$$

This probability-one event also has conditional probability one given E_∞, because $\Pr(E_\infty) > 0$. On E_∞, actual reserve equals $a + r + \sum_{j=1}^{n} X_j$, so $A_n/n \to \gamma$. Positive γ implies $A_n \to \infty$. □

Proposition 24 (Surplus funds learner improvement)**.**
Assume the cycle process, operating requirement a, and bounded independent and identically distributed net changes X_i of Proposition 23, with $\gamma := \mathbb{E}[X_1] > 0$. Assume the distribution of the X_i, the cycle timing and the operating requirement a all continue to apply during every trial and installation described below. Let the operating reserve at the start be $A_0 = a + r$, where $r > 0$. I define $X_i^+ := \max\{X_i, 0\}$ and choose

$$0 < \lambda < \frac{\gamma}{\mathbb{E}[X_1^+]}.$$

After cycle i, λX_i^+ units of resource are allocated to trials. The operating reserve then changes by $Y_i := X_i - \lambda X_i^+$. Since $\gamma \le \mathbb{E}[X_1^+]$, this choice has $\lambda < 1$. Deductions occur after repair and reduce only positive net gains, so the operating requirement a remains sufficient. If K_n is the total expenditure on trials through cycle n, the resource available for trials is

$$J_n := \lambda \sum_{i=1}^{n} X_i^+ - K_n \ge 0.$$

Every trial, including evaluation and installation, has a total cost of at most $c > 0$ and terminates after finitely many cycles. Trials run sequentially, another starting after a finite delay whenever c units are available. All additional operating costs are included in that trial cost.

Fix a finite vocabulary $\mathfrak{v}$, a learning task α, a finite nonempty set $\mathcal{V} \subseteq \Pi_\alpha$ of learning procedures, and a novelty distribution P_N on 2^U, where $U := L_\mathfrak{v} \setminus \mathrm{Ext}(I_\alpha)$. I use the existing buffers and performance

$$B_\ell := \mathrm{Ext}(\ell) \cap U, \qquad F(\ell) := P_N(S \subseteq B_\ell).$$

The current procedure $\ell \in \mathcal{V}$ performs each trial. A proposed successor $\ell' \in \mathcal{V}$ is adopted only when $F(\ell') > F(\ell)$, with every comparison correct. Installation preserves the cycle process of Proposition 23 and replaces the procedure that performs subsequent learning and trials. Whenever the current performance is below $\max_{\eta\in\mathcal{V}} F(\eta)$, assume each funded trial has probability at least $p \in (0, 1]$ of producing and installing an F-maximiser, conditional upon the full viable timeline prefix at which that trial starts. One fixed p applies to every such prefix.

Let E_∞ be continued operation with $A_n \ge a$ for every n under this allocation. Then some $\theta > 0$ has $\mathbb{E}[e^{-\theta Y_1}] \le 1$. The bound $\Pr(E_\infty) \ge 1 - e^{-\theta r} > 0$ holds for every such θ. Moreover, almost every timeline in E_∞ reaches $\max_{\eta\in\mathcal{V}} F(\eta)$ after finitely many trials and remains at it thereafter. For every k, the probability of remaining viable through k trials without reaching that maximum is at most $(1-p)^k$. If P_N is uniform on 2^U, the procedure reached also maximises freedom among procedures in $\mathcal{V}$.

Proof. The operating changes Y_i are bounded, independent and identically distributed, and

$$\mathbb{E}[Y_1] = \gamma - \lambda\mathbb{E}[X_1^+] > 0.$$

Proposition 23 applied to Y_i gives $\Pr(E_\infty) \ge 1 - e^{-\theta r} > 0$ for every $\theta > 0$ with $\mathbb{E}[e^{-\theta Y_1}] \le 1$. By the strong law of large numbers, almost surely on E_∞,

$$\frac{A_n}{n} \longrightarrow \gamma - \lambda\mathbb{E}[X_1^+] > 0, \qquad \frac{\lambda\sum_{i=1}^{n} X_i^+}{n} \longrightarrow \lambda\mathbb{E}[X_1^+] > 0.$$

The resource allocated to trials therefore grows without bound. If only finitely many trials occurred, their total expenditure would be finite, so J_n would eventually exceed c and another trial would start. Hence trials continue indefinitely on E_∞. Successive conditioning gives

$$\Pr\left(\begin{array}{c}\text{viable through } k \text{ trials and}\\ \text{maximal performance not yet reached}\end{array}\right) \le (1-p)^k \longrightarrow 0.$$

Thus timelines in E_∞ that never reach maximal performance have probability zero. The comparisons preserve maximal performance after it is reached. Under uniform P_N, Proposition 21 gives $F(\ell) = 2^{|B_\ell|-|U|}$, and correctness for the common task α gives $|\operatorname{Ext}(\ell)| = |O_\alpha| + |B_\ell|$. Maximising F therefore maximises freedom on $\mathcal{V}$. □

For a mass stream crossing a gravitational potential drop $\Delta\phi_g > 0$,

$$P_{\text{grav}} \le \dot{m}\Delta\phi_g = \dot{m} g_{\text{grav}} \Delta h$$

when gravitational acceleration is approximately constant. A static conservative field gives zero net drop around a closed cycle; resetting the mass costs at least the recovered work in the reversible limit. Sustained output requires a supplied stream or a powered reset. In the engineered niche, the lower-layer statement g remains true only while these flows, waste export and physical balances are maintained.

Corollary 20 (Artificial worlds can grow life)**.**
Let g be a lower-layer statement that remains true along every viable timeline fragment under consideration. Suppose the higher-layer systems conditional on g satisfy the hypotheses of Theorem 17. Then, conditional on indefinite viability, almost every timeline of each such higher-layer system contains infinitely many responses to damage, infinitely many renewals of the conditions for those responses, and open-ended policy generation. Those renewals constitute organisational closure. Response to damage without end, renewal, and policy generation beyond every fixed finite class follow whether g arose without design or is maintained by engineered machinery.

Proof. I restrict the higher-layer process to the part of the environment $T(g)$ in which the lower-layer condition holds. By assumption, the damage, restoration, novelty, and indefinite-viability premises of Theorem 17 hold on that restriction. Theorem 17 therefore implies response to damage without end, organisational closure, and generation beyond every fixed finite policy class. Only the persistence of g is utilised in this proof, and not the process that produced g. Response to damage without end, closure, and open-ended policy generation therefore follow whether geology, prior life, or engineering produced that lower layer. □

16 Causality and organisms

Definition 29 (intervention)**.**
If *int* and *obs* are "events" which have happened, then we say that *int* has **caused** *obs* if *obs* happens whenever *int* does. In our formalism, an **event** is a statement in $L_{\mathfrak{v}}$, and an event **happens** or is **observed** iff it is a true statement. If $obs \in L_{\mathfrak{v}}$ is sensorimotor activity we interpret as an "observed event", and $int \in L_{\mathfrak{v}}$ is an **intervention** (by an acting system or other agency, in the sense described by Pearl [Pearl and Mackenzie, 2018]) that causes that event, then the intervention entails the observation:

$$T(int) \subseteq T(obs).$$

The syntactic condition $obs \subseteq int$ is always sufficient for $T(int) \subseteq T(obs)$. The converse requires a vocabulary in which semantic entailment is reflected by inclusion of statements. I use inclusion

of truth sets here because the causal definition should not depend on redundant vocabulary items.

Definition 30 (Causal identities and intent)**.**
Fix an embodied language $L_{\mathfrak{v}}$. Let $INT \subseteq L_{\mathfrak{v}}$ be a set of (putatively) *intervention statements* and $OBS \subseteq L_{\mathfrak{v}}$ a set of matched *non-intervention observations*, meaning observations of the same outcomes occurring without the intervention. I define the set of causal identities

$$\mathcal{C}(INT, OBS) := \left\{ c \in L_{\mathfrak{v}} \setminus \{\emptyset\} \;\middle|\; \begin{array}{l} (\forall\, int \in INT)\ T(int) \subseteq T(c) \text{ and} \\ (\forall\, obs \in OBS)\ T(obs) \nsubseteq T(c) \end{array} \right\}.$$

Each $c \in \mathcal{C}(INT, OBS)$ is a **causal identity** for whatever is common to the interventions but not present in the observations. When $L_{\mathfrak{v}}$ is finite and $\mathcal{C}(INT, OBS) \neq \emptyset$, a **freedom maximising causal identity** is any $c^\star \in \mathcal{C}(INT, OBS)$ that maximises freedom over $\mathcal{C}(INT, OBS)$: $|\mathrm{Ext}(c^\star)| = \max\{|\mathrm{Ext}(c)| \mid c \in \mathcal{C}(INT, OBS)\}$. If a policy π is a causal identity, then the associated task is to classify interventions: given an event $e \in L_{\mathfrak{v}}$, decide "source present" iff $T(e) \subseteq T(\pi)$.

A causal identity c corresponding to INT and OBS is the **intent**, **purpose** or **goal** ascribed to those interventions. Correct intent inference requires a causal identity that implies subsequent interventions.

Theorem 18 (The Psychophysical Principle of Causality)**.**
Fix a finite vocabulary. Let μ be a viability statement encoding the most basic valence partition, viable versus not viable. For any policy π, define the set of viable completions

$$\Omega_\pi := \mathrm{Ext}(\pi) \cap \mathrm{Ext}(\mu).$$

I use the viability divergence F_2 of Corollary 13,

$$F_2(\pi) := \log_2 |\mathrm{Ext}(\mu)| - \log_2 |\Omega_\pi|.$$

Let π be any policy with $\Omega_\pi \neq \emptyset$. Let the added commitment $c \subseteq \mathfrak{v}$ be a finite bundle. Commitment c is **not required** by viability inside π when there exists some $o \in \Omega_\pi$ with $T(o) \nsubseteq T(c)$. If c is not required, then either $\Omega_{\pi \cup c} = \emptyset$ or $F_2(\pi \cup c) > F_2(\pi)$, where $\Omega_{\pi \cup c}$ is taken to be empty when $\pi \cup c \notin L_{\mathfrak{v}}$. In fact $\Omega_{\pi \cup c} = \{o \in \Omega_\pi : c \subseteq o\}$, so $\Omega_{\pi \cup c} = \emptyset$ or $F_2(\pi \cup c) > F_2(\pi)$ if and only if some $o \in \Omega_\pi$ omits a program of c, which holds for every commitment not required by viability.

Proof. If $\pi \cup c \notin L_{\mathfrak{v}}$, then $\mathrm{Ext}(\pi \cup c) = \emptyset$ and hence $\Omega_{\pi \cup c} = \emptyset$. In this case, no $o \in \Omega_\pi$ contains c, since $\pi \cup c \subseteq o$ would give $\emptyset \neq T(o) \subseteq T(\pi \cup c) = \emptyset$. Otherwise $o \in \Omega_{\pi \cup c}$ iff $o \in L_{\mathfrak{v}}$ contains π, c and μ, iff $o \in \Omega_\pi$ and $c \subseteq o$. In either case $\Omega_{\pi \cup c} = \{o \in \Omega_\pi : c \subseteq o\} \subseteq \Omega_\pi$. If every $o \in \Omega_\pi$ contains c, then $\Omega_{\pi \cup c} = \Omega_\pi$ and $F_2(\pi \cup c) = F_2(\pi)$. If some $o \in \Omega_\pi$ omits a program of c, then $o \notin \Omega_{\pi \cup c}$, so $\Omega_{\pi \cup c} \subsetneq \Omega_\pi$. If $\Omega_{\pi \cup c}$ is empty, the first disjunct holds. Otherwise $|\Omega_{\pi \cup c}| < |\Omega_\pi|$, which implies $F_2(\pi \cup c) > F_2(\pi)$. Finally, if c is not required, choose $o \in \Omega_\pi$ with $T(o) \nsubseteq T(c)$. If $c \subseteq o$, then $T(o) \subseteq T(c)$. This contradicts the choice of o. Hence o omits a program of c. □

Theorem 19 (Bottom-up Psychophysical Principle from persistence)**.**
Fix a timeline $\tau : I \to \Phi$, an interval $[u_1, u_2]$ with $u_1, u_2 \in I$, and finite bundles $\pi, \mu, c \subseteq \mathfrak{v}$, where an inconsistent bundle has persistence count zero by Definition 12. Here μ is viability and π the current policy. Then

$$\rho_{\tau,[u_1,u_2]}(\pi \cup \mu \cup c) \leq \rho_{\tau,[u_1,u_2]}(\pi \cup \mu).$$

Strict inequality $\rho_{\tau,[u_1,u_2]}(\pi \cup \mu \cup c) < \rho_{\tau,[u_1,u_2]}(\pi \cup \mu)$ holds if and only if $\pi \cup \mu$ is true and c is false at $\tau(u)$ for some $u \in \{u_1, \ldots, u_2\}$. Thus any commitment not required by viability reduces persistence if and only if the timeline visits a viable completion that the commitment excludes.

Proof. I apply Corollary 4 with $l = \pi \cup \mu$. The extra commitment c refines l to $l \cup c$, so persistence cannot increase. The two values differ by $|\{u \in \{u_1, \ldots, u_2\} : \tau(u) \in T(l) \setminus T(c)\}|$, the number of ticks at which l is true and c is false, so $\rho_{\tau,[u_1,u_2]}(l \cup c) < \rho_{\tau,[u_1,u_2]}(l)$ if and only if this set is nonempty. □

Corollary 21 (Representation-evaluation collapse)**.**
In the setting of Theorems 18 and 19, an added commitment c is **zero-cost inside** π when it does not empty the set of viable completions, does not raise F_2 and does not lower persistence on the timelines being considered. If c is not entailed by the viability partition inside π, then c is not zero-cost inside π. Indeed c is zero-cost inside π if and only if $c \subseteq \pi \cup \mu$. A primitive commitment is one that the organism has from the start rather than builds out of others. Therefore any primitive commitment that is guaranteed zero-cost across all viable timelines inside π must be required by viability inside π, meaning true wherever $\pi \cup \mu$ is true. For the primitive policy $\pi_0 = \emptyset$, this means that the commitment is true wherever μ is true. Since viability is an evaluative partition, every such guaranteed primitive representation is evaluative in this formal sense.

Proof. If c is not required by viability inside π, then by Theorem 18 adding c either empties the set of viable completions or raises F_2, and neither depends on the timeline. If the bottom-up timeline visits a viable state excluded by c, then by Theorem 19 adding c strictly lowers persistence. Hence a commitment that is not entailed by viability is not zero-cost on any timeline. Thus any primitive commitment with that guarantee is required by viability inside π, so $T(o) \subseteq T(c)$ for every $o \in \Omega_\pi$. Since $\Omega_\pi \neq \emptyset$, some statement contains $\pi \cup \mu$, so $\pi \cup \mu \in L_{\mathfrak{v}}$ and $\pi \cup \mu \in \Omega_\pi$, which gives $T(\pi \cup \mu) \subseteq T(c)$. For the equivalence, if $c \subseteq \pi \cup \mu$ then $\pi \cup \mu \cup c = \pi \cup \mu$, so persistence is unchanged on every timeline, and every $o \in \Omega_\pi$ contains c, so $\Omega_{\pi \cup c} = \Omega_\pi$ by Theorem 18 and F_2 is unchanged. Conversely, if $\Omega_{\pi \cup c} \neq \emptyset$ and F_2 is not raised, then by Theorem 18 every $o \in \Omega_\pi$ contains c, and $\pi \cup \mu \in \Omega_\pi$, so $c \subseteq \pi \cup \mu$. □

Corollary 22 (Value-first construction order)**.**
Let μ be a viability statement and let $\pi_0 := \emptyset$ be the primitive current policy. A primitive commitment c is **viability-entailing** when $T(\mu) \subseteq T(c)$ at the level being considered. Let $\mathcal{Q}$ be a primitive representational partition of states, meaning a block in the sense of Definition 18, so every state satisfies one and only one program in $\mathcal{Q}$. Committing to $\mathcal{Q}$ at a state means adding to the current policy the program of $\mathcal{Q}$ true at that state. Partition $\mathcal{Q}$ is **zero-cost** when committing to it at any viable state is zero-cost inside π_0 in the sense of Corollary 21. If $\mathcal{Q}$ separates two viable states, then $\mathcal{Q}$ is not zero-cost. Therefore any guaranteed zero-cost primitive partition must be evaluative. Its distinctions must follow from attraction or repulsion relative to viability. When the primitive distinctions are all evaluative, value neutral object partitions can appear at later layers of a stack only as patterns over those evaluative distinctions. In other words, the system will be blind to any objects and properties to which valence is indifferent.

Proof. Suppose $\mathcal{Q}$ separates two viable states ϕ and ϕ'. Committing to $\mathcal{Q}$ at ϕ adds the program $q \in \mathcal{Q}$ true at ϕ, and q is false at ϕ', so $T(\mu) \not\subseteq q$ and q is not viability-entailing. Then q is an added commitment beyond viability inside π_0, since $\mu \in \Omega_{\pi_0}$ and $T(\mu) \not\subseteq q$. By Corollary 21 with $\pi = \pi_0$, q is not zero-cost. Hence $\mathcal{Q}$ is not zero-cost. By contraposition, any partition that is guaranteed zero-cost at the primitive stage separates no two viable states, so the program it adds at a viable state is entailed by viability. Those distinctions are evaluative by definition. For the last two claims, suppose every program of the primitive layer is evaluative, and each later layer is built on the one beneath as in Definition 23. Every program of a later layer is the truth set $T(o)$ of a statement o of the layer beneath, so two states that satisfy the same programs

beneath satisfy the same programs above. By induction over the layers, if no evaluative program separates two states, then no program of any later layer separates them. Every later partition is therefore a pattern over evaluative distinctions, and states between which valence is indifferent remain indistinguishable at every layer. □

Definition 31 (organism)**.**
I describe the circumstances of an organism $\mathfrak{o}$ by the four-tuple

$$\mathfrak{o} = \langle \mathfrak{v}_\mathfrak{o}, \mu_\mathfrak{o}, \mathfrak{p}_\mathfrak{o}, <_\mathfrak{o} \rangle,$$

where

- $\mathfrak{v}_\mathfrak{o}$ is the organism's embodied vocabulary.
- $\mu_\mathfrak{o}$ is its viability task, and $O_{\mu_\mathfrak{o}}$ contains the outputs counted as fit.
- $\mathfrak{p}_\mathfrak{o}$ is the set of policies currently retained by the organism.
- $<_\mathfrak{o}$ is a binary relation over $\Gamma_{\mathfrak{v}_\mathfrak{o}}$ called the organism's **preferences**.

An organism stores only so many policies, and there are two ways of coming by them. Let $\mathfrak{p}_{n.s.} \subseteq L_{\mathfrak{v}_\mathfrak{o}}$ be the reflexes fixed by natural selection, which an organism has without having learnt them. Let $\mathcal{H}_{<t_\mathfrak{o}} \subseteq \Gamma_{\mathfrak{v}_\mathfrak{o}}$ be a finite nonempty family of history tasks recording what it has undergone, and I write

$$\mathfrak{p}_{\mathcal{H}_{<t_\mathfrak{o}}} := \bigcup_{\zeta \in \mathcal{H}_{<t_\mathfrak{o}}} \Pi_\zeta$$

for the policies correct on at least one of those tasks. Then

$$\mathfrak{p}_\mathfrak{o} \subseteq \mathfrak{p}_{n.s.} \cup \mathfrak{p}_{\mathcal{H}_{<t_\mathfrak{o}}}.$$

Inheritance and experience together bound what an organism can have at its disposal. Where a later argument needs a history task, $\mathfrak{h}_{<t_\mathfrak{o}} \in \mathcal{H}_{<t_\mathfrak{o}}$ denotes the one I've chosen. It is shorthand, and the organism remains a four-tuple. For $\zeta \in \mathcal{H}_{<t_\mathfrak{o}}$, I write $\hat{\zeta}$ for the task holding every interaction the organism underwent in the stretch of history that ζ summarises. The organism has **selective memory** when some ζ omits inputs of $\hat{\zeta}$, so that $\zeta \sqsubset \hat{\zeta}$ with $O_\zeta = O_{\hat{\zeta}} \cap \mathrm{Ext}(I_\zeta)$. Dropping an input removes a demand, so $\Pi_{\hat{\zeta}} \subseteq \Pi_\zeta$ and the set of correct policies grows. One inconsistent interaction can leave $\Pi_{\hat{\zeta}}$ empty, and discarding that interaction leaves a task with correct policies [Bennett, 2023b].

Definition 32 (preconditions)**.**
If $\mathfrak{o}$ is an organism, and c is a causal identity:

- the **representation** precondition is met iff $c \in L_{\mathfrak{v}_\mathfrak{o}}$, and
- the **incentive** precondition is met if $\mathfrak{o}$ must learn c to remain "fit".

Proposition 25 (Causal identities are minimal commitments)**.**
Fix a finite vocabulary, a viability statement μ, a policy π, and sets INT, OBS with $\mathcal{C}(INT, OBS) \neq \emptyset$. Let $c \subsetneq c'$ be members of $\mathcal{C}(INT, OBS)$ with $\Omega_{\pi \cup c} \neq \emptyset$, and read $\Omega_{\pi \cup c'} = \emptyset$ when $\pi \cup c' \notin L_\mathfrak{v}$. Then

1. $\rho_{\tau,[u_1,u_2]}(\pi \cup \mu \cup c') \leq \rho_{\tau,[u_1,u_2]}(\pi \cup \mu \cup c)$ for every timeline τ and interval $[u_1, u_2]$, with strict inequality if and only if the timeline visits a state at which $\pi \cup \mu \cup c$ is true and c' is false;

2. $\Omega_{\pi\cup c'} \subseteq \Omega_{\pi\cup c}$, so $\Omega_{\pi\cup c'} = \emptyset$ or $F_2(\pi\cup c') \geq F_2(\pi\cup c)$, and if $c' \setminus c$ is not required by viability inside $\pi \cup c$, then $\Omega_{\pi\cup c'} = \emptyset$ or $F_2(\pi\cup c') > F_2(\pi\cup c)$;

3. $\mathrm{Ext}(\pi\cup c') \subseteq \mathrm{Ext}(\pi\cup c)$, strictly if and only if $c' \nsubseteq \pi\cup c$, and $\mathrm{Ext}(c') \subsetneq \mathrm{Ext}(c)$.

Hence a $\subseteq$-minimal member of $\mathcal{C}(INT, OBS)$ is never ranked below a member that properly contains it when rankings are based on persistence or on F_2. The minimal member is ranked strictly above that member by freedom. Therefore the freedom maximising causal identity of Definition 30 is $\subseteq$-minimal.

Proof. For the first part apply Theorem 19 with policy $\pi \cup c$ and added commitment $c' \setminus c$, so that $(\pi\cup c)\cup\mu\cup(c'\setminus c) = \pi\cup\mu\cup c'$. At a tick at which $\pi\cup\mu\cup c$ is true, c' is false if and only if $c'\setminus c$ is false, so the theorem's strict case holds if and only if the timeline visits a state at which $\pi\cup\mu\cup c$ is true and c' is false. For the second part, every completion of $\pi\cup c'$ completes $\pi\cup c$, so $\Omega_{\pi\cup c'} \subseteq \Omega_{\pi\cup c}$. This gives $F_2(\pi\cup c') \geq F_2(\pi\cup c)$ when $\Omega_{\pi\cup c'} \neq \emptyset$. The strict case follows from Theorem 18 with policy $\pi\cup c$ and added commitment $c'\setminus c$. For the third part, $\pi\cup c \subseteq \pi\cup c'$, so every completion of $\pi\cup c'$ completes $\pi\cup c$. When $c' \nsubseteq \pi\cup c$ the statement $\pi\cup c$ completes $\pi\cup c$ but not $\pi\cup c'$, and when $c' \subseteq \pi\cup c$ the two extensions coincide since $\pi\cup c' = \pi\cup c$. Likewise $c \in \mathrm{Ext}(c)\setminus\mathrm{Ext}(c')$. A freedom maximising causal identity therefore has no proper subset in $\mathcal{C}(INT, OBS)$, since such a subset would have strictly more completions. □

Proposition 26 (Do-operator as conditioning on an intervention indicator)**.**
Let X and Y be random variables and let $do(X = x)$ denote the intervention that sets X to the value x. Introduce an auxiliary binary variable $A \in \{0, 1\}$ indicating the regime, where $A = 1$ means "intervene and set $X = x$" and $A = 0$ means "observe" (no intervention). Consider an augmented model that preserves each regime's mechanisms and whose regime choice is independent of its exogenous variables. Assume $\mathbb{P}(A = 1) > 0$. In this model, X is generated by

$$X := \begin{cases} x, & A = 1, \\ (\text{its original mechanism}), & A = 0. \end{cases}$$

Then the interventional distribution is an ordinary conditional in the augmented model:

$$\mathbb{P}(Y \in B \mid do(X = x)) \;=\; \mathbb{P}(Y \in B \mid A = 1) \qquad \text{for all measurable } B.$$

Moreover, whenever $\mathbb{P}(X = x) > 0$,

$$\begin{aligned}\mathbb{P}(Y \in B \mid X = x) = \mathbb{P}(Y \in B \mid A = 1)\,\mathbb{P}(A = 1 \mid X = x) \\ + \mathbb{P}(Y \in B \mid A = 0, X = x)\,\mathbb{P}(A = 0 \mid X = x).\end{aligned}$$

Proof. Fix x. Let $P_{\mathrm{obs}}(X, Y)$ denote the observational joint distribution and let $P_{\mathrm{do}(x)}(Y)$ denote the interventional distribution of Y under $do(X = x)$. I can always construct a joint distribution on (A, X, Y) as follows:

- Sample $A \in \{0, 1\}$ according to any distribution with $\mathbb{P}(A = 1) > 0$.
- If $A = 0$, sample (X, Y) from the observational distribution $P_{\mathrm{obs}}(X, Y)$.
- If $A = 1$, set $X := x$ deterministically and sample Y from $P_{\mathrm{do}(x)}(Y)$.

By construction, the conditional law of Y given $A = 1$ is exactly the interventional law of Y under $do(X = x)$, so $\mathbb{P}(Y \in B \mid A = 1) = \mathbb{P}(Y \in B \mid do(X = x))$ for all measurable B. Because $A = 1$ implies $X = x$, I also have $\mathbb{P}(Y \in B \mid A = 1, X = x) = \mathbb{P}(Y \in B \mid A = 1)$.

Finally, assuming $\mathbb{P}(X = x) > 0$, apply the law of total probability conditioned on $X = x$:

$$\mathbb{P}(Y \in B \mid X = x) = \mathbb{P}(Y \in B \mid A = 1, X = x)\,\mathbb{P}(A = 1 \mid X = x) + \mathbb{P}(Y \in B \mid A = 0, X = x)\,\mathbb{P}(A = 0 \mid X = x).$$

and substitute $\mathbb{P}(Y \in B \mid A = 1, X = x) = \mathbb{P}(Y \in B \mid A = 1)$ to obtain the stated decomposition. □

17 Orders of self

Definition 33 (1ST-order-self)**.**
Let $INT_{\mathfrak{o}} \subseteq L_{\mathfrak{v}_{\mathfrak{o}}}$ denote idealised self-intervention statements for an organism $\mathfrak{o}$, and let $OBS_{\mathfrak{o}} \subseteq L_{\mathfrak{v}_{\mathfrak{o}}}$ denote matched non-self observations. Any freedom maximising causal identity

$$\mathfrak{o}^1 \in \arg \max_{c \in \mathcal{C}(INT_{\mathfrak{o}}, OBS_{\mathfrak{o}})} |\operatorname{Ext}(c)|$$

is called a **1ST-order-self** for $\mathfrak{o}$. Existence requires the representation precondition $\mathcal{C}(INT_{\mathfrak{o}}, OBS_{\mathfrak{o}}) \neq \emptyset$ and the incentive precondition that viability depends on distinguishing self-intervention from matched observation.

An intervention includes the context needed for its specified consequence; an action label alone need not determine an invariant effect. Incentive together with representation is a sufficient precondition for learning the identity, not a necessary one: an organism may construct an identity that viability does not require.

Theorem 20 (Intervention control requires a represented distinction)**.**
Let a finite decision problem have cases that are pairs (e, r) of a statement $e \in L_{\mathfrak{v}}$ and a value $r \in \{\text{int}, \text{obs}\}$ recording whether the organism brought e about or only observed it. A policy chooses its action from an internal state that represents the case. Suppose the cases (e, int) and (e, obs) have no action in common that maximises the utility of intelligence. Then no policy whose internal state omits r is optimal on both, so any optimal policy encodes a distinction between self-intervention and matched observation on which its action depends.

Proof. If the internal state omits r, the two cases are represented identically, so any policy reading that state chooses the same action in both. No action is optimal in both, so the policy is suboptimal in at least one. □

Definition 34 (Chain notation and higher-order selves)**.**
Suppose I have two organisms, $\mathfrak{a}$ (Alice) and $\mathfrak{b}$ (Bob). A causal identity $c_{\mathfrak{a}}^{\mathfrak{b}}$ denotes the causal identity $\mathfrak{a}$ constructs for $\mathfrak{b}$: informally, what Alice takes Bob to intend or cause.

More generally, for a finite chain of agents $\mathfrak{x}_1, \ldots, \mathfrak{x}_k$ I write

$$c_{\mathfrak{a}}^{\mathfrak{x}_1\mathfrak{x}_2\cdots\mathfrak{x}_k}$$

for Alice's nested causal identity in which $\mathfrak{x}_1$ models $\mathfrak{x}_2$ models … models $\mathfrak{x}_k$. For example, $c_{\mathfrak{a}}^{\mathfrak{b}\mathfrak{a}}$ denotes Alice's model of Bob's model of Alice, what Alice thinks Bob thinks Alice intends. The maximal chain depth that $\mathfrak{a}$ can represent is constrained by $\mathfrak{a}$'s vocabulary $\mathfrak{v}_{\mathfrak{a}}$. Finally, Bob need not be a biological organism. Bob can be anything for which Alice constructs a causal identity.

Fix an agent $\mathfrak{a}$. An $n^{\mathbf{th}}$-order self of $\mathfrak{a}$ is any causal identity whose chain ends at $\mathfrak{a}$ after interleaving $\mathfrak{a}$ with $n - 1$ models of other agents.

Formally, let $c^{\mathfrak{a}}_{\mathfrak{a}}$ denote the 1ST-order-self from Definition 33, and set

$$\mathfrak{a}^1 := c^{\mathfrak{a}}_{\mathfrak{a}}.$$

For $n \geq 2$, choose agents $\mathfrak{x}_1, \ldots, \mathfrak{x}_{n-1}$ (not necessarily distinct) and set

$$\mathfrak{a}^n := c^{\mathfrak{x}_1\mathfrak{a}\mathfrak{x}_2\mathfrak{a}\cdots\mathfrak{x}_{n-1}\mathfrak{a}}_{\mathfrak{a}}.$$

For example, relative to another agent $\mathfrak{b}$, I have

$$\mathfrak{a}^1 = c^{\mathfrak{a}}_{\mathfrak{a}}, \qquad \mathfrak{a}^2 = c^{\mathfrak{b}\mathfrak{a}}_{\mathfrak{a}}, \qquad \mathfrak{a}^3 = c^{\mathfrak{b}\mathfrak{a}\mathfrak{b}\mathfrak{a}}_{\mathfrak{a}}.$$

I use $\mathfrak{a}^n$ to denote an arbitrary $n^{\mathbf{th}}$-order self, and chain notation to denote a specific one, for example $c^{\mathfrak{b}\mathfrak{a}}_{\mathfrak{a}}$.

A 3RD-order-self predicts how another agent reads present commitments. The term also denotes the policy that the causal identity licenses to constrain future policies on that prediction, represented by Bind in Definition 39.

Proposition 27 (n$^{\mathbf{th}}$ order self convergence)**.**
An organism $\mathfrak{o}$ has current history task $\mathfrak{h}_{<t_\mathfrak{o}}$ and viability task $\mu_\mathfrak{o}$ with

$$\mathfrak{h}_{<t_\mathfrak{o}} \sqsubset \mu_\mathfrak{o}.$$

Successful learning of $\mu_\mathfrak{o}$ from $\mathfrak{h}_{<t_\mathfrak{o}}$ yields a retained correct policy,

$$\pi^\star \in \mathfrak{p}_\mathfrak{o} \cap \Pi_{\mu_\mathfrak{o}}.$$

An $n^{\mathbf{th}}$-order self $\mathfrak{o}^n$ is expressible in $L_{\mathfrak{v}_\mathfrak{o}}$. The incentive precondition is that every correct policy for $\mu_\mathfrak{o}$ learnable from $\mathfrak{h}_{<t_\mathfrak{o}}$ entails that self:

$$T(\pi) \subseteq T(\mathfrak{o}^n) \qquad \text{for every such policy } \pi.$$

The learned policy therefore entails the self,

$$T(\pi^\star) \subseteq T(\mathfrak{o}^n).$$

If $\mathfrak{p}_\mathfrak{o}$ is closed under entailed commitments, meaning

$$\pi \in \mathfrak{p}_\mathfrak{o}, \quad c \in L_{\mathfrak{v}_\mathfrak{o}}, \quad T(\pi) \subseteq T(c) \quad \Longrightarrow \quad c \in \mathfrak{p}_\mathfrak{o},$$

then $\mathfrak{o}^n \in \mathfrak{p}_\mathfrak{o}$.

Proof. The learned policy $\pi^\star$ is one of the correct policies learnable from $\mathfrak{h}_{<t_\mathfrak{o}}$. The incentive precondition therefore gives $T(\pi^\star) \subseteq T(\mathfrak{o}^n)$. Since $\pi^\star \in \mathfrak{p}_\mathfrak{o}$ and $\mathfrak{o}^n \in L_{\mathfrak{v}_\mathfrak{o}}$, closure under entailed commitments gives $\mathfrak{o}^n \in \mathfrak{p}_\mathfrak{o}$. This implication holds for any such policy selected by freedom maximisation. □

Proposition 28 (Agreement of 1ST-order-selves)**.**
Let $c, c' \in \mathcal{C}(INT_\mathfrak{o}, OBS_\mathfrak{o})$. Then for every statement $e \in INT_\mathfrak{o} \cup OBS_\mathfrak{o}$,

$$T(e) \subseteq T(c) \quad \Longleftrightarrow \quad T(e) \subseteq T(c').$$

Thus every 1ST-order-self classifies the organism's interventions and its matched observations the same way.

Proof. If $e \in INT_\mathfrak{o}$, both inclusions hold by Definition 30. If $e \in OBS_\mathfrak{o}$, both inclusions fail. The two statements therefore agree on every statement that defines the causal identity. □

Decoder mismatch (hypothesis). When every available mind hypothesis gives an observed trace a much smaller likelihood than the noise model, Bayes' rule lowers the posterior odds of each available mind hypothesis against noise. The suggested application to the Fermi paradox is a hypothesis about recognition across incompatible stacks [Bennett, 2026a].

18 Communication, meaning and commitment

Theorem 21 (A speaker needs a model of the listener)**.**
Let U be a set of utterances and M a set of meanings. A listener is characterised by a decoder $d : U \to M$, the function taking what it hears to what it understands. Let D be the set of decoders the speaker admits as possible. The speaker wants the listener to arrive at $m \in M$, and an utterance u works for decoder d when $d(u) = m$. For a speaker whose utterance is fixed before d is known, every utterance fails for some $d \in D$ if and only if no $u \in U$ has $d(u) = m$ for every $d \in D$. A speaker whose utterance can depend on d has, for every $d \in D$, an utterance that works for d if and only if every $d \in D$ has some u_d with $d(u_d) = m$.

Now take the binary case. Let $m \in \{0, 1\}$ be drawn from any distribution, and let $D = \{d_0, d_1\}$ where d_0 returns the bit it was sent and d_1 flips it. The listener's decoder is drawn uniformly from D, independently of m, and hidden from the speaker. The speaker sends one probe and observes the listener's response before its final utterance, and the response is the bit the listener's decoder returns for the probe. A speaker whose final utterance is independent of that response succeeds with probability $\frac{1}{2}$, which is chance. A speaker whose final utterance depends on that response succeeds with probability 1 [Bennett, 2026a].

The state separating the two speakers represents how this listener maps utterances to meanings, and that is the functional role of a 2ND-order-self.

Proof. A speaker whose utterance is fixed before d is known sends the same u to every listener. If no member of U works for every $d \in D$, then $d(u) \neq m$ for some $d \in D$, so that utterance fails for at least one member of D. Conversely, if some member of U works for every $d \in D$, a speaker that fixes it fails for no member of D. If every $d \in D$ has some u_d with $d(u_d) = m$, a speaker whose utterance depends on d sends u_d, and $d(u_d) = m$ holds by construction, so it works for every member of D. Conversely, if no utterance works for some $d \in D$, then no speaker has an utterance that works for that d.

For the binary case, take a speaker whose final utterance s depends on m alone. Since $d_1(s) = 1 \oplus d_0(s)$, the occasions on which $d_0(s) = m$ are those on which $d_1(s) \neq m$. The decoder is drawn uniformly and independently of m, so, averaged over d_0 and d_1, the success probability is $\frac{1}{2}$ whatever distribution the speaker uses. Now let the speaker send probe $p = 0$ and read the response, which is 0 under d_0 and 1 under d_1, so the speaker identifies the decoder. The speaker sends m when the decoder is d_0 and $1 \oplus m$ when it is d_1, and the listener arrives at m in both cases. Telling the members of D apart is therefore what separates chance from certainty, and a state that does so represents how the listener will take the speaker's utterance. □

Corollary 23 (Deception has the same self-order cost)**.**
In either case of Theorem 21, suppose its conditions hold for a meaning m' that the speaker holds to be false, in place of m. Causing the listener to arrive at m' then requires the state that separates the two speakers in Theorem 21. Deception and report differ in which meaning the speaker wants the listener to arrive at, and call for the same model of the listener.

Proof. I replace m by m' throughout. A speaker whose utterance is fixed before d is known fails for some $d \in D$ when no utterance works for every member of D, while a speaker that tells the members of D apart chooses u_d with $d(u_d) = m'$. In the binary case a speaker that ignores the probe response plants m' with probability $\frac{1}{2}$, and one that reads it plants m' with probability 1. □

Definition 35 (protosymbol system)**.**
Assume an organism $\mathfrak{o}$. For each policy $p \in \mathfrak{p}_\mathfrak{o}$ there exists a set $\mathfrak{s}_p = \{\alpha \in \Gamma_{\mathfrak{v}_\mathfrak{o}} : p \in \Pi_\alpha\}$ of all

tasks for which p is a correct policy. The union of all such sets is

$$\mathfrak{s}_{\mathfrak{o}} = \bigcup_{p \in \mathfrak{p}_{\mathfrak{o}}} \{\alpha \in \Gamma_{\mathfrak{v}_{\mathfrak{o}}} : p \in \Pi_\alpha\}$$

We call $\mathfrak{s}_{\mathfrak{o}}$ a "protosymbol system". A $\mathfrak{v}$-task $\alpha \in \mathfrak{s}_{\mathfrak{o}}$ is called a "protosymbol".

Definition 36 (interpretation)**.**
For each input $i \in L_{\mathfrak{v}_{\mathfrak{o}}}$, define the protosymbols signified by i as

$$\mathfrak{S}_{\mathfrak{o}}(i) := \{\, \alpha \in \mathfrak{s}_{\mathfrak{o}} \mid i \in I_\alpha \,\}.$$

Whenever $\mathfrak{S}_{\mathfrak{o}}(i)$ is finite and nonempty, restrict $<_{\mathfrak{o}}$ to a total preorder on this set and fix one tie-breaker for its maximal elements. The organism selects

$$\alpha^\star \in \max_{<_{\mathfrak{o}}} \mathfrak{S}_{\mathfrak{o}}(i),$$

chooses a policy known to be correct, $p_{\alpha^\star} \in \mathfrak{p}_{\mathfrak{o}} \cap \Pi_{\alpha^\star}$, and infers an output

$$o \in \mathrm{Ext}(i) \cap \mathrm{Ext}(p_{\alpha^\star}).$$

If $\mathfrak{S}_{\mathfrak{o}}(i) = \emptyset$, no retained correct policy exists, or $\mathrm{Ext}(i) \cap \mathrm{Ext}(p_{\alpha^\star}) = \emptyset$, interpretation fails on input i. Otherwise $\alpha^\star$ is the **interpretation** of i and o is the interpreted output.

Definition 37 (affect)**.**
Suppose organisms $\mathfrak{a}$ and $\mathfrak{b}$ inhabit the same environment. Suppose $\mathfrak{a}$ interprets input $i \in L_{\mathfrak{v}_{\mathfrak{a}}}$ as output o. For a part $v \subseteq i$ of the input, I write $i \setminus v$ for the input obtained by deleting the programs in v. If $\mathfrak{b}$ makes output k, let $k^{\mathfrak{a}} \in L_{\mathfrak{v}_{\mathfrak{a}}}$ be the part of k registered in $\mathfrak{a}$'s vocabulary.

- A **statement** $v \subseteq i$ affects $\mathfrak{a}$ on i when interpreting $i \setminus v$ produces an output different from o.
- The **organism** $\mathfrak{b}$ affects $\mathfrak{a}$ on i by making k when some $v \subseteq i \cap k^{\mathfrak{a}}$ affects $\mathfrak{a}$ on i.

Definition 38 (meaning)**.**
The **meaning** an organism $\mathfrak{o}$ ascribes to an input i is the protosymbol $\alpha^\star \in \mathfrak{s}_{\mathfrak{o}}$ selected in Definition 36.

Assume organism $\mathfrak{a}$ chooses output u on input $i_{\mathfrak{a}}$, and organism $\mathfrak{b}$ subsequently interprets input $i_{\mathfrak{b}}$ and produces output $o_{\mathfrak{b}}$. Let $o_{\mathfrak{b}}^{\mathfrak{a}} \in L_{\mathfrak{v}_{\mathfrak{a}}}$ be the part of Bob's output expressible in Alice's vocabulary. Alice **means** protosymbol $\alpha \in \mathfrak{s}_{\mathfrak{a}}$ by choosing u when she intends all three conditions below.

1. Bob interprets his input and $o_{\mathfrak{b}}^{\mathfrak{a}} \in O_\alpha$.
2. Bob recognises Alice's intention in the operational sense that Alice's output affects his input and hence his interpretation.
3. Bob's production of an output registered in O_α occurs because of that recognition. Had Alice not chosen u, Bob would not have produced an output whose expressible part is in O_α.

Meaningful communication therefore requires the organisms to affect one another, Bob's response to be legible in Alice's vocabulary, and their learned protosymbol systems to permit the intended response. With similar bodies, experiences, and preferences, these conditions are easier to meet.

Proposition 29 (Cooperation with advance detection)**.**
An infinitely repeated Prisoner's Dilemma has payoffs $T > R > P > S$, discount factor $\beta \in [0, 1)$ and probability $q \in [0, 1]$ that planned betrayal is inferred before the other player cooperates. Both players cooperate until betrayal or exploitative intent is detected and then defect forever, with no false detection of cooperative intent. Cooperation forever is a best response when

$$R - P \geq (1 - \beta)(1 - q)(T - P).$$

Proof. The discounted payoff from cooperation is $V_C = R/(1-\beta)$. A first planned defection yields expected payoff

$$V_D = qP + (1-q)T + \frac{\beta P}{1-\beta} = \frac{P}{1-\beta} + (1-q)(T-P).$$

Thus $V_C \geq V_D$ is the stated inequality. At a later first defection, the same payoff difference is discounted, and mutual defection is a best response after detection because $P > S$. □

Mirror Symbol Hypothesis. An organism uses its own preferences and protosymbol system as a prior for another organism's causal identity, then adjusts that prediction through interaction. Shared embodied preferences make this reuse a proposed basis for empathy and social learning [Bennett and Maruyama, 2022].

Definition 39 (A two-stage trust/commitment game)**.**
Fix a commitment cost c with $0 < c < 1$. Two agents $\mathfrak{a}$ (Alice, the actor) and $\mathfrak{b}$ (Bob, the audience) play the following extensive-form game.

1. **Stage 0 (self-binding).** $\mathfrak{a}$ chooses either *Bind* (B) or *Free* (F). Choosing B incurs an immediate cost c to $\mathfrak{a}$ and removes $\mathfrak{a}$'s ability to *exploit* in Stage 2. Choosing F incurs no cost and leaves Stage 2 unconstrained.

2. **Stage 1 (audience response).** $\mathfrak{b}$ observes $\mathfrak{a}$'s choice (B or F) and chooses either *Trust* (Y) or *Do not trust* (N). If $\mathfrak{b}$ chooses N, the game ends with payoff $(0, 0)$ after F and $(-c, 0)$ after B, because the binding cost was already paid.

3. **Stage 2 (act).** If $\mathfrak{b}$ chose Y, then if $\mathfrak{a}$ chose B in Stage 0, $\mathfrak{a}$ must play *Honour* (K). If $\mathfrak{a}$ chose F in Stage 0, $\mathfrak{a}$ chooses either K or *Exploit* (E). Payoffs before subtracting the binding cost are $u_{\mathfrak{a}}(Y, K) = 1$, $u_{\mathfrak{b}}(Y, K) = 1$, $u_{\mathfrak{a}}(Y, E) = 2$, $u_{\mathfrak{b}}(Y, E) = -1$. If $\mathfrak{a}$ chose B in Stage 0, then c is subtracted from $\mathfrak{a}$'s payoff.

Theorem 22 (Separating audience modelling from self-binding)**.**
A subgame-perfect equilibrium is a choice of strategy for each player such that, from every stage of the game onward, no player can gain by changing strategy alone. In the game of Definition 39, the following hold.

1. If the self-binding move B is not available ($\mathfrak{a}$ can only play F), then the unique subgame-perfect equilibrium outcome is N with payoff $(0, 0)$.

2. If B is available and $0 < c < 1$, then in the unique subgame-perfect equilibrium $\mathfrak{a}$ plays B, $\mathfrak{b}$ plays Y after observing B (and N after observing F), and the Stage 2 outcome is necessarily K, yielding payoff $(1 - c, 1)$.

Proof. (1) If B is unavailable, then after $\mathfrak{b}$ chooses Y the actor $\mathfrak{a}$ chooses between K and E and strictly prefers E because $2 > 1$. Anticipating this, $\mathfrak{b}$ strictly prefers N to Y because $0 > -1$. Hence the unique subgame-perfect equilibrium outcome is N.

(2) Now allow B. In the subgame following (F, Y), $\mathfrak{a}$ plays E because $2 > 1$, so $\mathfrak{b}$'s best response after observing F is N. In the subgame following $(B, \cdot)$, $\mathfrak{a}$ must play K, so $\mathfrak{b}$'s best response after observing B is Y because $1 > 0$. Anticipating these best responses, $\mathfrak{a}$ compares Stage 0 payoffs. Choosing F yields 0 (since $\mathfrak{b}$ chooses N), whereas choosing B yields $1 - c$. Because $0 < c < 1$, we have $1 - c > 0$, so $\mathfrak{a}$ strictly prefers B. Every best response in this induction is strict, so the subgame-perfect equilibrium is unique. □ □

19 Collective identity and embodied goals

The parts of a collective have $\mathfrak{v}$-tasks $\omega_1, \ldots, \omega_m$ and correct-policy sets $\Pi_{\omega_j} \subseteq L_{\mathfrak{v}}$.

Definition 40 (Restricted policies, over-constraint, and splinters)**.**
The **collective policies** are $\Pi_\omega := \bigcap_{j=1}^m \Pi_{\omega_j}$. The collective has an identity at $\mathfrak{v}$ iff $\Pi_\omega \neq \emptyset$. To model top-down boundary conditions, fix restricted policy sets $\Pi'_{\omega_j} \subseteq \Pi_{\omega_j}$ and define the restricted collective policies $\Pi'_\omega := \bigcap_{j=1}^m \Pi'_{\omega_j}$. The system is **over-constrained at $\mathfrak{v}$** if $\Pi'_\omega = \emptyset$. A **splinter** is a nonempty index set $S \subseteq \{1, \ldots, m\}$ such that $\bigcap_{j \in S} \Pi'_{\omega_j} \neq \emptyset$.

Proposition 30 (Over-constraint implies splintering)**.**
Assume at least one part still has a correct policy ($\Pi'_{\omega_j} \neq \emptyset$ for some j). A policy is feasible for part j when it lies in Π'_{ω_j}. If the system is over-constrained ($\Pi'_\omega = \emptyset$), then no policy is feasible for every part. At least one proper splinter exists, meaning a splinter that excludes at least one part. For any realised policy $\pi \in L_{\mathfrak{v}}$ with $S(\pi) := \{j : \pi \in \Pi'_{\omega_j}\} \neq \emptyset$, the set $S(\pi)$ is a proper splinter.

Proof. If some π were feasible for all parts then $\pi \in \bigcap_{j=1}^m \Pi'_{\omega_j} = \Pi'_\omega$, contradicting $\Pi'_\omega = \emptyset$. Thus no realised policy is feasible for every part. Since at least one part still has a correct policy, choose j with $\Pi'_{\omega_j} \neq \emptyset$ and choose $\pi \in \Pi'_{\omega_j}$. Then $S(\pi)$ is nonempty and $\pi \in \bigcap_{j \in S(\pi)} \Pi'_{\omega_j}$, so its restricted policy intersection is nonempty. Since no policy is feasible for every part, $S(\pi)$ is a proper subset of $\{1, \ldots, m\}$. Hence $S(\pi)$ is a proper splinter. □

Corollary 24 (Incompatible maximal splinters)**.**
Suppose the parts of a collective share no policy after restriction, so $\Pi'_\omega = \emptyset$, and every part still has at least one policy of its own, so $\Pi'_{\omega_j} \neq \emptyset$ for each j. Then every part lies in a maximal splinter that excludes at least one part of the former collective. This means there are at least two maximal splinters, and no policy is shared across the members of any two maximal splinters, so for distinct maximal splinters S and R,

$$\bigcap_{j \in S \cup R} \Pi'_{\omega_j} = \emptyset.$$

Proof. Each part has a policy of its own, so every singleton $\{j\}$ is a splinter. I begin with any singleton and add parts for as long as the members still share a policy. There are finitely many parts, so this stops, and it stops at a maximal splinter. The maximal splinter leaves someone out, since a splinter holding every part would require a policy in Π'_ω, which is empty. Were there only one maximal splinter, every part would lie in it, so it would hold every part, which has just been ruled out. Finally, if two different maximal splinters S and R shared a policy across all their members, then $S \cup R$ would be a splinter larger than both, and neither could have been maximal. □

Proposition 31 (Shared representations need every distinction)**.**
Let parts $1, \ldots, m$ of a collective receive the same inputs Z, and suppose each part's task has one correct output for each input. A single representation of those inputs serves every part, each interpreting using a decoder of its own. A part must discriminate between two inputs when they are associated with different correct outputs. If part j is correct on inputs z and z' and must discriminate between them, then the representation distinguishes z from z'. Decoders under which each and every part is correct on every input exist if and only if the representation is sufficient to distinguish between any two inputs that one or more parts must discriminate between.

Proof. Suppose the representation failed to distinguish between two inputs z and z' that part j must discriminate between. Part j's decoder has only the representation to go on, so it produces

the same output for z and z', and the resulting output can be correct for at most one of them. Part j is therefore wrong on z or on z'. If every part is correct on every input, the representation distinguishes every pair that some part must discriminate between. Conversely, suppose the representation distinguishes every pair that some part must discriminate between. Give part j the decoder that returns, for each representation, the correct output of any input having that representation. If two inputs with the same representation had different correct outputs for part j, part j would have to discriminate between them and the representation would distinguish them. So the decoder is well defined, and it is correct on every input. □

Proposition 32 (Embodied success blocks strong orthogonality)**.**
Let $\mathcal{C}$ be a nonempty set of software states, let Γ be a nonempty set of behaviours, let $f_1 \in \mathcal{C}$ be a software state, let $f_2 : \mathcal{C} \to \Gamma$ be the body or interpreter, and let $f_3 : \Gamma \to \{0, 1\}$ be the goal environment. Assume f_3 is nontrivial, so some behaviour succeeds and some behaviour fails. Then success is a function neither of f_1 alone, nor of (f_1, f_3), nor of (f_1, f_2). It is a property of the whole triple (f_1, f_2, f_3).

Proof. I choose behaviours γ_+ and γ_- with $f_3(\gamma_+) = 1$ and $f_3(\gamma_-) = 0$. There is a body f_2^+ with $f_2^+(c) = \gamma_+$ for every $c \in \mathcal{C}$, so $f_3(f_2^+(f_1)) = 1$. There is also a body f_2^- with $f_2^-(c) = \gamma_-$ for every $c \in \mathcal{C}$, so $f_3(f_2^-(f_1)) = 0$. Neither the software state f_1 nor the goal environment f_3 has changed, but success has changed because the body changed. Thus success is a function neither of f_1 alone nor of (f_1, f_3). Conversely, for any realised behaviour $\gamma = f_2(f_1)$, the goal environments under which it succeeds are those $f : \Gamma \to \{0, 1\}$ with $f(\gamma) = 1$. The admissible goals therefore depend on the realised behaviour. Goal environments with $f(\gamma) = 1$ and with $f(\gamma) = 0$ both exist, so with f_1 and f_2 fixed, success changes with the goal environment and is not a function of (f_1, f_2). Success is determined by software, body, and goal environment together. □

No selves, no alignment. The necessity claims concern conduct whose correctness depends on causal responsibility, audience-relative interpretation, or commitments maintained across change. Theorems 20, 21 and 22 establish the respective separations. The embodied orthogonality proposition addresses the strong factorisation of successful goals from bodily feasibility.

Recursive self impairment (conjecture). Self-binding in an unhealthy collective may propagate corrupted shared meanings and higher-order identities into its splinters and their parts, recursively impairing coordination.

20 Consciousness and temporal conditions

Definition 41 (levels of consciousness)**.**

1. an organism that acts but does not learn, meaning $\mathfrak{p}_\mathfrak{o}$ is fixed from birth.
2. an organism that learns, but $\mathfrak{o}^1 \notin \mathfrak{p}_\mathfrak{o}$ either because $\mathcal{C}(INT_\mathfrak{o}, OBS_\mathfrak{o}) = \emptyset$ (failing the "representation precondition") or because the organism is not incentivised to construct $\mathfrak{o}^1$ (failing the "incentive precondition").
3. reafference and phenomenal or core consciousness are achieved when $\mathfrak{o}^1 \in \mathfrak{p}_\mathfrak{o}$ is learned by an organism as a consequence of attraction to and repulsion from statements in $L_{\mathfrak{v}_\mathfrak{o}}$.
4. (a) access or self reflexive consciousness is achieved when $\mathfrak{o}^2 \in \mathfrak{p}_\mathfrak{o}$.
   (b) hard consciousness [Boltuc, 2012] is achieved when a phenomenally conscious organism learns a 2ND-order-self (an organism is consciously aware of the contents of 2ND-order-selves, which must have quality if learned through phenomenal consciousness).
5. meta self reflexive consciousness (human level hard consciousness) is achieved when $\mathfrak{o}^3 \in \mathfrak{p}_\mathfrak{o}$.

Proposition 33 (Arpeggio costs persistence and viability)**.**
Fix a finite vocabulary $\mathfrak{v}$, a viability statement μ, a policy $\pi \in L_{\mathfrak{v}}$ with $\Omega_\pi \neq \emptyset$, and a moment $\ell = \{p_1, \ldots, p_n\} \in L_{\mathfrak{v}}$ with $n \geq 2$. A **record** of a content p_j is a program $r_j \in \mathfrak{v}$ with $r_j \not\subseteq p_j$, so that r_j can be true while p_j is false. A chordal realisation of ℓ is a tick at which every content is true. When the contents become true at the same tick, I take it to add no program to π, because they are true together and nothing need be held across ticks. An arpeggiated realisation of ℓ over a fragment $\tau_{[t_0,t_1]}$ is one in which no tick of the fragment satisfies every content. The system therefore holds a record r_j of each content p_j other than the last to occur. Each record is held from the tick at which p_j occurs to the tick at which the last content occurs. Let $R = \{r_1, \ldots, r_{n-1}\}$ and let $\pi_A = \pi \cup R$ be the arpeggiating policy, read with $T(\pi_A) = \emptyset$ when $\pi_A \notin L_{\mathfrak{v}}$. Then, with R the records or any other finite subset of $\mathfrak{v}$,

1. for every timeline τ and interval $[u_1, u_2]$, $\rho_{\tau,[u_1,u_2]}(\pi_A) \leq \rho_{\tau,[u_1,u_2]}(\pi)$, with strict inequality if and only if the timeline visits a state in $T(\pi) \setminus T(\pi_A)$ during the interval;

2. $\Omega_{\pi_A} \subseteq \Omega_\pi$, so $\Omega_{\pi_A} = \emptyset$ or $F_2(\pi_A) \geq F_2(\pi)$. If some $r \in R$ is not required by viability inside π, meaning some $o \in \Omega_\pi$ has $T(o) \not\subseteq r$, then $\Omega_{\pi_A} = \emptyset$ or $F_2(\pi_A) > F_2(\pi)$;

3. the inequality of the first part is unchanged when π and R are replaced by their groundings at any lower layer of a finite chain.

Hence a system able to hold every content of ℓ at the same tick must have concurrency capacity of at least n (Proposition 35). Such a system is never worse off realising ℓ as a chord whose contents become true at the same tick, and is strictly better off whenever either inequality is strict.

Proof. The first part is Corollary 4 with $l = \pi$ and $c = R$, since $\pi_A = \pi \cup R$. For the second part, the set Ω_{π_A} is empty when $\pi_A \notin L_{\mathfrak{v}}$. Every member of this set is a statement containing π and μ, so $\Omega_{\pi_A} \subseteq \Omega_\pi$ and $F_2(\pi_A) \geq F_2(\pi)$ whenever $\Omega_{\pi_A} \neq \emptyset$. For every $r \in R$, $T(R) \subseteq r$, so the $o \in \Omega_\pi$ with $T(o) \not\subseteq r$ has $T(o) \not\subseteq T(R)$, which is to say R is not required by viability inside π, and Theorem 18 with the finite bundle $c = R$ gives $\Omega_{\pi_A} = \emptyset$ or $F_2(\pi_A) > F_2(\pi)$. The third part is Corollary 2 with $l = \pi$ and the finite bundle $c = R$. For the final claim, a chordal realisation whose contents become true at the same tick adds nothing to π, and Proposition 35 shows that a capacity below n forbids it, so a system that can hold every content at the same tick has capacity at least n and nothing forces the records. The first two parts then rank π at least as high as π_A by persistence and by F_2, and strictly higher whenever either inequality is strict. □

Definition 42 (Arpeggio and Chord)**.**
Let $\ell = \{p_1, \ldots, p_n\} \subseteq \mathfrak{v}$ be a moment with $n \geq 1$, let $\tau : I \to \Phi$ be a timeline, and let $\tau_{[t_0,t_1]}$ be a finite timeline fragment of τ. Arpeggio holds when

$$\forall i \in \{1, \ldots, n\}\ \exists u_i \in \{t_0, \ldots, t_1\} \quad \tau(u_i) \in p_i.$$

I call ℓ **co-instantiated** in $\tau_{[t_0,t_1]}$ when

$$\exists u \in \{t_0, \ldots, t_1\} \quad \tau(u) \in T(\ell).$$

Co-instantiation requires $T(\ell) \neq \emptyset$, so a co-instantiated moment is a statement, $\ell \in L_{\mathfrak{v}}$, while an Arpeggio need not be. Chord holds when the following three conditions are true:

1. The moment ℓ is co-instantiated in $\tau_{[t_0,t_1]}$.

2. The size of ℓ satisfies $|\ell| \leq c_{\text{conc}}$, where c_{conc} is the concurrency capacity.

3. The sites of ℓ interact within the time window.

Concurrency capacity and interaction within the time window are defined in Definition 43 below.

Definition 43 (Sites and interaction)**.**
Let $\ell \subseteq \mathfrak{v}$ be a **moment**, a finite bundle of programs. Sites_ℓ is the nonempty finite set of physical sites that instantiate the contents of ℓ at an arbitrarily chosen abstraction layer, the one at which the test is made. The **interaction graph** $\mathrm{Graph}_\ell = (\mathrm{Sites}_\ell, \mathrm{Edges}_\ell)$ has vertex set Sites_ℓ. An edge identifies two sites that the tested binding mechanism requires to **interact**, one sending an outward signal and the other a reply caused by its receipt. Let $\tau_{[t_0,t_1]}$ be a finite timeline fragment in the sense of Definition 2, the restriction of the timeline τ to the ticks $t_0 \le u \le t_1$. A **physical model** is a layer with the clock of Theorem 10 and a unit for that clock, in the sense of Section 27. The model assigns a duration to each span of its own clock, while an objective tick is a change, and thus has no duration. The **time window** of the fragment is the span of that layer time across $\tau_{[t_0,t_1]}$. The window's duration is θ, calculated by multiplying the number of ticks of the model's clock across the fragment by the unit. A required edge is completed within the time window when both transmissions occur inside it. The sites of ℓ **interact within the time window** when Graph_ℓ is connected and every required edge is completed within the time window.

The **concurrency capacity** c_{conc} is the largest number of contents the physical model can instantiate together at that abstraction layer.

Proposition 34 (Temporal Gap separation)**.**
For every ℓ, τ and $\tau_{[t_0,t_1]}$, co-instantiation implies Arpeggio. Some ℓ, τ and $\tau_{[t_0,t_1]}$ satisfy Arpeggio without co-instantiation, and so without Chord.

Proof. If some tick u satisfies every p_i, choose $u_i = u$ for every content, and Arpeggio follows. For the converse, Theorem 23 gives a moment that satisfies Arpeggio without co-instantiation in any fragment that visits two distinct states while a third state exists, and the fragment (a, b) with $p_1 = \{a, c\}$ and $p_2 = \{b, c\}$ is its case $n = 2$. □

Proposition 35 (Persistence and capacity)**.**
Let $\ell = \{p_1, \ldots, p_n\} \subseteq \mathfrak{v}$ be a moment, let $\tau : I \to \Phi$ be a timeline, and let $\tau_{[t_0,t_1]}$ be a finite timeline fragment of τ.

1. Suppose each p_i is true at some tick $u_i \in \{t_0, \ldots, t_1\}$ and, once true, remains true through $u^* = \max_i u_i$, the last of these ticks. Then every content is true at u^*.

2. If at each tick of the fragment at most $c < n$ contents of ℓ are true, then there is no tick of the fragment at which every content is true.

Proof. For persistent contents, let $u^* = \max_i u_i$. Every content has become true by u^*, and under persistence each one remains true through u^*, so $\tau(u^*) \in T(\ell)$. Under concurrency limit $c < n$, any tick satisfying all n contents would violate that limit, so no such tick exists. □

For a moment $g \subseteq \mathfrak{v}$ and a present statement $r \in L_{\mathfrak{v}}$, if $T(r) \subseteq T(g)$, then every content of g is presently true. If $T(r) \nsubseteq T(g)$, some state satisfies r without g. An inert record at such a state instantiates a different moment. If $T(g) = \emptyset$, no satisfiable r can entail g. Persistence of the contents themselves, as in Proposition 35, can close the temporal gap; coexistence of inert records alone does not.

Corollary 25 (Subject and moment)**.**
Fix a timeline $\tau : I \to \Phi$ and a finite timeline fragment $\tau_{[t_0,t_1]}$. Fix a finite chain as in Theorem 1. Let g_s be the grounding of a 1ST-order-self at the lowest layer of the chain and let g_c be the grounding of a moment at that layer. Then

$$T(g_s \cup g_c) = T(g_s) \cap T(g_c).$$

The joint claim that this subject currently undergoes this moment is co-instantiated in $\tau_{[t_0,t_1]}$ when

$$\exists u \in \{t_0, \ldots, t_1\} \quad \tau(u) \in T(g_s) \cap T(g_c).$$

Separate occurrence of g_s and g_c at different ticks does not imply $\exists u \in \{t_0, \ldots, t_1\} : \tau(u) \in T(g_s) \cap T(g_c)$. Let r be the grounding at that layer of an inert representation of the moment. The state at a tick $u \in \{t_0, \ldots, t_1\}$ can then instantiate the subject together with the representation, while failing to instantiate the subject together with the original moment. This happens if and only if

$$\tau(u) \in T(g_s) \cap T(r) \setminus T(g_c).$$

Replacing g_s, g_c and r by the self, the moment and the representation, or by their groundings at any layer of the chain, changes the truth value of neither condition.

Proof. By Definition 3, $T(g_s \cup g_c) = T(g_s) \cap T(g_c)$ for the finite bundles g_s and g_c, so co-instantiation means that some tick $u \in \{t_0, \ldots, t_1\}$ has $\tau(u) \in T(g_s) \cap T(g_c)$. By Proposition 34, separate occurrence is insufficient. Membership $\tau(u) \in T(g_s) \cap T(r) \setminus T(g_c)$ means subject and representation occur at u while the original moment does not. By Theorem 1, the self, the moment and the representation have the truth sets $T(g_s)$, $T(g_c)$ and $T(r)$, as do their groundings at every layer of the chain, so both conditions have the same truth value at every layer. □

Theorem 23 (Arpeggio collapse)**.**
Let a finite timeline fragment $\tau_{[t_0,t_1]}$ visit $n \geq 2$ distinct states $x_1, \ldots, x_n$, suppose there is a state x_* that it does not visit, and let the vocabulary contain the programs $p_i = \{x_i, x_*\}$. Then the moment $\ell = \{p_1, \ldots, p_n\}$ has $T(\ell) = \{x_*\}$, so $\ell \in L_{\mathfrak{v}}$, and satisfies Arpeggio in $\tau_{[t_0,t_1]}$ while co-instantiation fails. So if Arpeggio suffices, ℓ qualifies as an Arpeggiated consciousness, the contents of which are not instantiated by any one tick within the fragment.

Proof. For each visited state x_i, the program $p_i = \{x_i, x_*\}$ is in the vocabulary by hypothesis, so ℓ is a moment. At the tick at which the fragment visits x_i, program p_i is true, so every content of ℓ is true at some tick of the fragment, which is Arpeggio. Because the visited states are distinct and $n \geq 2$, no visited state lies in all programs p_i, while x_* lies in all of them, so

$$\bigcap_{i=1}^{n} p_i = \{x_*\}.$$

So $T(\ell) = \{x_*\}$ and $\ell \in L_{\mathfrak{v}}$. Since the fragment does not visit x_*, no tick of the fragment satisfies every content, so co-instantiation fails, while if Arpeggio suffices ℓ qualifies as an Arpeggiated consciousness. □

Layer A has finite vocabulary $\mathfrak{v}_A$, intervention and matched-observation sets $INT_A, OBS_A \subseteq L_{\mathfrak{v}_A}$, and $\mathcal{C}_A := \mathcal{C}(INT_A, OBS_A)$ as in Definition 30. Enlarging to $\mathfrak{v} = \mathfrak{v}_A \cup \{q_X\}$ preserves old statements and causal identities, because satisfiability and the identity conditions depend only on truth sets. Freedom is recomputed in the enlarged language, so its maximising identity may change.

Theorem 24 (Arpeggiated inclusion drift)**.**
Let A have a causal identity $c_A \in \mathcal{C}_A$, let $q_X \in \mathcal{P}$ be a program of another part X, and let $\mathfrak{v} := \mathfrak{v}_A \cup \{q_X\}$. Choose

$$c^\star \in \arg \max_{c \in \mathcal{C}(INT_A, OBS_A)} |\operatorname{Ext}(c)|,$$

with the causal identities and extensions taken in $L_{\mathfrak{v}}$. Under the representation and incentive preconditions of Definition 33, $c^\star$ is a 1ST-order-self. Let a timeline τ visit $T(int_A)$ for some

$int_A \in INT_A$ and visit q_X. Then for every $c \in \mathcal{C}(INT_A, OBS_A)$, and in particular for $c^\star$, the moment

$$\ell := c \cup \{q_X\} \subseteq \mathfrak{v}$$

satisfies Arpeggio in every finite fragment of τ containing both visits. Hence, if Arpeggio suffices, X is included in an Arpeggiated consciousness, although X need not have a 1ST-order-self and q_X need never be true at a tick at which c is true.

Proof. Since $L_{\mathfrak{v}_A} \subseteq L_{\mathfrak{v}}$ and the conditions of Definition 30 depend only on truth sets, c_A is a causal identity in $L_{\mathfrak{v}}$. The set of causal identities is therefore nonempty. It is finite because $\mathfrak{v}$ is finite, so $c^\star$ exists. By the first condition of Definition 30, $T(int_A) \subseteq T(c)$ for every $c \in \mathcal{C}(INT_A, OBS_A)$, so at the tick u_1 at which τ visits $T(int_A)$ every program of c is true, and at the tick u_2 at which τ visits q_X the program q_X is true. Any finite fragment $\tau_{[t_0,t_1]}$ with $t_0 \leq \min\{u_1, u_2\}$ and $t_1 \geq \max\{u_1, u_2\}$ contains both ticks, so every content of ℓ is true at some tick of the fragment, which is Arpeggio. If Arpeggio suffices, ℓ qualifies as an Arpeggiated consciousness. No step uses a causal identity for X, and no step uses a tick at which c and q_X are true together. □

Definition 44 (Distances and signal speed)**.**
Let ℓ be a moment with sites Sites_ℓ and interaction graph $\mathrm{Graph}_\ell = (\mathrm{Sites}_\ell, \mathrm{Edges}_\ell)$ from Definition 43. Let $d(x, y)$ be the physical distance between sites. The diameter of ℓ is

$$D(\ell) = \max_{x,y \in \mathrm{Sites}_\ell} d(x, y).$$

Let $v > 0$ be the maximum signal speed. If Graph_ℓ is connected, let $\mathrm{hops}(x, y)$ be the length in edges of a shortest path from x to y. The hop diameter is

$$\mathrm{hop}(\mathrm{Graph}_\ell) = \max_{x,y \in \mathrm{Sites}_\ell} \mathrm{hops}(x, y).$$

The length of the longest required edge is

$$L_{\max}(\mathrm{Graph}_\ell, d) := \begin{cases} 0, & \mathrm{Edges}_\ell = \emptyset, \\ \max_{\{x,y\} \in \mathrm{Edges}_\ell} d(x, y), & \mathrm{Edges}_\ell \neq \emptyset. \end{cases}$$

The architecture factor, which depends only on the shape of the interaction graph, is

$$\varepsilon(\mathrm{Graph}_\ell) := \frac{\mathrm{hop}(\mathrm{Graph}_\ell)}{2}.$$

Theorem 25 (Edge length and spacetime diameter bounds)**.**
Let ℓ be a moment over a finite timeline fragment $\tau_{[t_0,t_1]}$, and let θ be the duration of the fragment's window in the chosen physical model. Suppose the following conditions are true:

1. Every required edge in Graph_ℓ supports an outward signal and a reply caused by the signal's receipt. Both occur inside the fragment's window.
2. The physical model registers the emission and receipt of each such signal.
3. No relevant signal travels faster than v in the model's units.

Then

$$L_{\max}(\mathrm{Graph}_\ell, d) \leq v\theta/2,$$

equivalently

$$\theta \geq \frac{2L_{\max}(\mathrm{Graph}_\ell, d)}{v}.$$

If Graph_ℓ is connected, then using only the shape of the interaction graph I obtain

$$D(\ell) \leq \varepsilon(\mathrm{Graph}_\ell) v\theta.$$

Proof. The reply cannot begin before the outward signal is received. The two transit intervals are therefore sequential and their durations add. For any edge $\{a, b\} \in \mathrm{Edges}_\ell$, inside the time window a sends an outward signal and b sends a reply caused by its receipt. So there are travel times $t_{a\to b}, t_{b\to a} \geq 0$ such that

$$t_{a\to b} + t_{b\to a} \leq \theta.$$

Since no signal travels faster than v, each one-way trip takes at least $d(a, b)/v$. Hence

$$2d(a, b)/v \leq t_{a\to b} + t_{b\to a} \leq \theta,$$

so the edge length satisfies

$$d(a, b) \leq v\theta/2.$$

Thus $L_{\max}(\mathrm{Graph}_\ell, d) \leq v\theta/2$ whenever $\mathrm{Edges}_\ell \neq \emptyset$. Now suppose Graph_ℓ is connected. For any two sites $x, y \in \mathrm{Sites}_\ell$, let

$$x = x_0, x_1, \ldots, x_k = y$$

be a shortest path in Graph_ℓ, so $k = \mathrm{hops}(x, y) \leq \mathrm{hop}(\mathrm{Graph}_\ell)$. By the triangle inequality,

$$d(x, y) \leq \sum_{j=0}^{k-1} d(x_j, x_{j+1}) \leq k\, v\theta/2 \leq \mathrm{hop}(\mathrm{Graph}_\ell) v\theta/2.$$

Taking the maximum over $x, y \in \mathrm{Sites}_\ell$, I obtain

$$D(\ell) \leq \frac{\mathrm{hop}(\mathrm{Graph}_\ell)}{2} v\theta = \varepsilon(\mathrm{Graph}_\ell) v\theta.$$

□

For a complete interaction graph with at least two sites, $\mathrm{hop}(\mathrm{Graph}_\ell) = 1$ and $D(\ell) \leq v\theta/2$. For a hub graph with at least three sites, $\mathrm{hop}(\mathrm{Graph}_\ell) = 2$ and $D(\ell) \leq v\theta$. A Chordal moment's required sites must belong to one connected interaction component within the window.

Corollary 26 (Feasible window)**.**
Let ℓ be a moment over a finite timeline fragment $\tau_{[t_0,t_1]}$ in a chosen physical model, and let θ be the duration of the fragment's window. Let $\mathcal{R}$ be a set of permitted timings, each recording when every content first becomes true and how long it stays true. Suppose Chord is required for every $k \in \mathcal{R}$. Suppose $\theta_{\mathrm{pers}}(\ell)$ is a guaranteed persistence duration for every content of ℓ once it becomes true. Assume $\mathcal{R}$ includes a timing in which one content first becomes true at window start, remains true through $\theta_{\mathrm{pers}}(\ell)$, and is false afterwards, while another first becomes true at window end. Then the window must satisfy

$$\theta_{\min}(\ell) \leq \theta \leq \theta_{\mathrm{pers}}(\ell),$$

where $\theta_{\min}(\ell) = 2L_{\max}(\mathrm{Graph}_\ell, d)/v$. Conversely, suppose the time window lasts no longer than $\theta_{\mathrm{pers}}(\ell)$. This means a content that becomes true in the window will persist through to its end. Hence by the end of the window, all of the contents are true, so ℓ is co-instantiated by the final tick.

Proof. By Theorem 25, $\theta \geq 2L_{\max}(\mathrm{Graph}_\ell, d)/v$. To prove $\theta \leq \theta_{\mathrm{pers}}(\ell)$, choose a permitted history with onsets and expiry as specified. If $\theta > \theta_{\mathrm{pers}}(\ell)$, the early content in that history is false by the time the late content becomes true, while the late content is false at every earlier tick of the fragment. No tick of the fragment contains both. Chord fails in this permitted history, contradicting the requirement that Chord hold for every $k \in \mathcal{R}$. Hence $\theta \leq \theta_{\mathrm{pers}}(\ell)$. When $\theta \leq \theta_{\mathrm{pers}}(\ell)$, a content whose onset lies inside the window remains true for at least $\theta_{\mathrm{pers}}(\ell) \geq \theta$ after its onset, hence through the end of the window, so every earlier content whose onset lies inside the window remains true when the last one becomes true. Under a permitted timing in which every content first becomes true inside the window, Proposition 35 then gives co-instantiation at the tick at which the last content becomes true. □

Corollary 27 (Same program above, different control below)**.**
Let π^i be the policy at layer i and let $R \in \mathfrak{v}^{i+1}$ be a program at layer $i+1$. Suppose $T(\pi^i) \setminus R$ has at least two and finitely many states, let k be their number, and suppose $\mathfrak{v}^i$ contains the program R and the program $\Phi \setminus \{s\}$ for each state $s \in T(\pi^i) \setminus R$. For the states $s_1, \ldots, s_k$ of $T(\pi^i) \setminus R$, let $p_j = \Phi \setminus \{s_j\}$. Then the layer-$i$ statements $x = \pi^i \cup \{R\}$ and $y = \pi^i \cup \{p_1, \ldots, p_k\}$ are completions of π^i with

$$T(x) = T(y) = R,$$

such that removing any p_j from y gives a statement with truth set strictly larger than R, and for each j the state s_j satisfies every program of y except p_j. Either statement is a witness for R in the sense of Definition 6, and $x \neq y$. The two groundings therefore agree on R at layer $i+1$. At layer i, the truth of R in the grounding x depends on the one further program R alone, because $R \subseteq T(\pi^i)$. In the grounding y, the truth of R depends on π^i and each of k further programs.

Proof. Since $R \in \mathfrak{v}^{i+1}$, there is a completion o of π^i with $T(o) = R$, so R is nonempty and $R \subseteq T(\pi^i)$. Hence $T(x) = T(\pi^i) \cap R = R$. The states $s_1, \ldots, s_k$ are the states of $T(\pi^i)$ outside R, so $T(y) = T(\pi^i) \cap \bigcap_j p_j = T(\pi^i) \setminus \{s_1, \ldots, s_k\} = R$. Both x and y are finite, contain π^i and have truth set R. Since R is nonempty, both are completions of π^i with truth set R. Either therefore satisfies the witness condition of Definition 6. Removing p_j from y gives a statement with truth set $T(\pi^i) \setminus \{s_m : m \neq j\} = R \cup \{s_j\}$, which strictly contains R because $s_j \in T(\pi^i)$. The state s_j satisfies π^i and satisfies p_m for every $m \neq j$, while falsifying p_j. Since $k \geq 2$, y has more programs than x, so $x \neq y$. At layer $i+1$ both groundings give the program R with the same truth set. At layer i, R is true under the grounding x if and only if the program R is true, because $R \subseteq T(\pi^i)$ leaves π^i redundant. Under the grounding y, R is true if and only if π^i and all of $p_1, \ldots, p_k$ are true. The state s_j falsifies p_j alone. □

Proposition 36 (Occurrence does not determine order)**.**
Fix a finite vocabulary $\mathfrak{v}$ and an admissible finite fragment $h = (\phi_0, \ldots, \phi_N)$, meaning one whose consecutive states differ and which the transition constraints permit. Let r permute $\{0, \ldots, N\}$, and suppose $h' = (\phi_{r(0)}, \ldots, \phi_{r(N)})$ is also admissible. For every $p \in \mathfrak{v}$,

$$\exists j \in \{0, \ldots, N\}\ \phi_j \in p \quad \Longleftrightarrow \quad \exists j \in \{0, \ldots, N\}\ \phi_{r(j)} \in p.$$

Consequently Arpeggio has identical truth value for every finite bundle $\ell \subseteq \mathfrak{v}$ on h and h'. Co-instantiation considered alone also has identical truth value. Nevertheless, the order of events can differ. Which contents occur need not determine their order. For an actual timeline, layer clocks instead satisfy $u \leq u' \Rightarrow \kappa^\tau_\mathfrak{v}(u) \leq \kappa^\tau_\mathfrak{v}(u')$.

Proof. Since r is bijective, every visited state appears at a position before and after permutation. This proves equivalence for each p, and hence for Arpeggio, the conjunction over contents of these occurrence conditions. Applying that observation to $T(\ell)$ proves invariance of co-instantiation alone.

For event order, let $\Phi = \{a, b, c\}$, $p = \{a, c\}$, $q = \{b, c\}$ and $\mathfrak{v} = \{p, q\}$. Let both fragments (a, b) and (b, a) be permitted. Each content occurs in one fragment if and only if it occurs in the other, and $\{p, q\}$ is satisfiable at c. Yet first occurrence of p precedes first occurrence of q in only one fragment. Any function whose inputs specify only which contents occur receives identical arguments on both fragments and cannot reproduce that difference. Finally, each increment of a layer clock is zero or one, so its value cannot decrease along objective time. Admissibility of a permutation is a premise. □

Psychophysical interpretation. The Psychophysical Principle identifies represented objects and properties with causes of evaluation; evaluation precedes representation. Identifying felt quality with the constitutive evaluations, and a subject with the learned 1ST-order-self, gives the

book's interpretation of efficient biological access as implying feeling. These identifications connect the formal efficiency results to phenomenal consciousness. Conscious attention is identified with the contents of active 2ND-order-selves; 3RD-order-selves support projected social interactions and an impelling narrative whose anticipated evaluations are themselves felt. The levels of Definition 41 assume representations built from evaluations; assignments of biological taxa to those levels remain conjectural.

Human-like access. Here access requires audience-relative Gricean report. With variable admitted audience decoders and no universally successful utterance, Theorem 21 requires an audience model; its active probing construction also uses causal self-discrimination. Generic availability for reasoning or broadcast alone is not this report condition.

Multilayer functionalism. Equivalence of an experience requires preservation of its constitutive evaluations and causal relations under grounding. By Proposition 7, one finer-layer constitutive condition on which implementations differ refutes the proposed equivalence. Whole-stack equality is stronger than equality of one experience. On the identification of experience with these evaluations, a system preserving them but lacking that experience is contradictory; matching only higher-layer reports does not preserve them.

Conscious contraptions. The proposed necessary architecture is delegated, collective, self-organising and freedom-maximising, with the persistent structure of a solid brain, evaluation preceding representation, first-, second- and third-order selves satisfying the representation and incentive preconditions, and enough concurrency for the intended contents. The proposed sufficiency hypothesis for human-like consciousness adds the artificial scientist's representational, inferential, causal, communicative, planning and experimental capacities, together with mutual interaction of grounded contents in a feasible window satisfying $\theta \geq 2L_{\max}/v$. The sufficiency claim is a hypothesis; the signal-speed and capacity bounds are conditional mathematical results. Audience-relative report and a global workspace architecture do not logically entail one another. Dedicated routes supplied with an adequate audience model can support intended report without a workspace; broadcasting to a speech process lacking that model does not overcome the decoder separation in Theorem 21 [Bennett, 2026j]. **IIT: necessity and choice of units.** The sets H and P contain, respectively, hosts of experience and systems satisfying proposed physical postulates. The premises $H \subseteq P$ and that every experience has a host do not imply $P \subseteq H$: adjoining $x \in P \setminus H$ preserves those premises. An identity or sufficiency premise is additional [Bennett, 2026i]. The book's two-switch example has $A' = A$, $B' = A \operatorname{XOR} B$. Units (A, B) and $(B, A \operatorname{XOR} B)$ preserve the whole-system transition structure but specify different independent interventions. Under the 2023 IIT calculation, the pair's system integrated information is respectively 0 and 2 ibits; the 2026 deterministic-zero revision removes this deterministic example. The cited software study gives a noisy variant with distinct unit verdicts and a corresponding m-switch construction. A no-overlap rule relative to selected micro-units does not itself choose those units [Bennett, 2026i, Albantakis et al., 2023, Mayner et al., 2026, Marshall et al., 2026].

21 Vocabulary-relative computation

Proposition 37 (Every task is a satisfiability problem)**.**
Let $i, \pi \in L_{\mathfrak{v}}$. Then

$$\operatorname{Ext}(i) \cap \operatorname{Ext}(\pi) = \{o \in L_{\mathfrak{v}} : i \cup \pi \subseteq o\},$$

which is $\operatorname{Ext}(i \cup \pi)$ when $i \cup \pi \in L_{\mathfrak{v}}$ and empty otherwise. In particular, for a $\mathfrak{v}$-task α and $i \in I_\alpha$, inference under π on i selects a statement containing the input and the policy, and it is defined if and only if the input and the policy together are a statement.

Proof. By Definition 4, $\operatorname{Ext}(i) \cap \operatorname{Ext}(\pi)$ is the set of $o \in L_{\mathfrak{v}}$ with $i \subseteq o$ and $\pi \subseteq o$. If $i \cup \pi \in L_{\mathfrak{v}}$ then the set is $\operatorname{Ext}(i \cup \pi)$ by the same definition. If some o lies in the set then $T(i \cup \pi) \supseteq T(o) \neq \varnothing$,

and $i \cup \pi$ is finite, so $i \cup \pi \in L_{\mathfrak{v}}$ by Definition 3. □

Definition 45 (run on an input)**.**
Let α be a $\mathfrak{v}$-task, let τ be a timeline and let u be a tick at which an input $i \in I_\alpha$ is true, meaning $\tau(u) \in T(i)$. The **run** of α on i along τ from u is the finite timeline fragment $\tau_{[u,u+n]}$ for the least $n \geq 0$ with $u + n \in I$ such that some correct output $o \in O_\alpha$ with $i \subseteq o$ is true at $\tau(u + n)$. If no such n exists, the run is undefined. Its **run time** is $\kappa_{\mathfrak{v}}(\tau_{[u,u+n]})$.

Definition 46 (complexity classes relative to a vocabulary)**.**
Let $\mathfrak{v}$ be a finite vocabulary and let $\mathcal{T}$ be a set of timelines, the **permitted timelines**, those a physical model permits from a presented input in Section 27. A finite bundle $x \subseteq \mathfrak{v}$ is **presented** on $\tau \in \mathcal{T}$ when $\tau(0) \in T(x)$, and $|x|$, its number of programs, is its description length. A $\mathfrak{v}$-task α is **completed within** $f : \mathbb{N} \to \mathbb{N}$ under $\mathcal{T}$ when both of the following conditions hold. Every input is presented on some permitted timeline. On every permitted timeline τ on which an input i is presented, the run of α on i along τ from tick 0 is defined with run time at most $f(|i|)$. A task is in **P relative to $\mathfrak{v}$ under $\mathcal{T}$** when it is completed within some polynomial. Let $a, r \in \mathfrak{v}$ be two programs with $T(a) \cap T(r) = \varnothing$ that a permitted timeline, once in either, does not leave, meaning that $\tau(t) \in T(a)$ implies $\tau(t') \in T(a)$ for every later tick t' of $\tau \in \mathcal{T}$, and likewise for r. These are the **verdicts**. A **decision task** is a $\mathfrak{v}$-task whose inputs are divided into yes-inputs and no-inputs, with no input a completion of another. Its correct outputs are the completions of yes-inputs that contain a and the completions of no-inputs that contain r. A decision task α is in **NP relative to $\mathfrak{v}$ under $\mathcal{T}$** when there is a polynomial p satisfying both of the following requirements. For every yes-input i, there is a bundle $c \subseteq \mathfrak{v}$ with $|c| \leq p(|i|)$, called a **certificate**, such that $i \cup c$ is presented on some permitted timeline. For that certificate, the run of α on i along τ from tick 0 is defined with run time at most $p(|i|)$ on every permitted timeline τ on which $i \cup c$ is presented. For every no-input i, every bundle c with $|c| \leq p(|i|)$ and every permitted timeline τ on which $i \cup c$ is presented, $\tau(t) \notin T(i \cup \{a\})$ at every tick t with $\kappa_{\mathfrak{v}}(\tau_{[0,t]}) \leq p(|i|)$. The vocabulary and the permitted timelines may form a family indexed by a tape bound $S \in \mathbb{N}$, as the configuration vocabularies of Proposition 38 do. A task is in P or NP relative to such a family when a single polynomial bound, or a single p in the NP case, works for every S large enough for the runs in question.

Proposition 38 (P and NP are the classes of the Turing machine)**.**
Let M be an embodied S-bounded machine as in Definition 8 that halts on every input and is **complete**, meaning that the map $\phi \mapsto (q(\phi), i(\phi), T_\phi, u_\phi)$ from $\mathcal{M}_{M,S}$ to $Q \times \{1, \ldots, S\} \times \Gamma^S \times \{0,1\}^S$ is surjective, which extends the configuration completeness of Corollary 7 to the usage programs, and let $\# \in \Sigma$ be a separator symbol. The classes **relative to** M are the classes relative to $\mathfrak{v}^{\mathrm{conf}}_{M,S}$ under the legal bounded runs of M that end at a halting control state, with $Q_{q_{\mathrm{acc}}}$ and $Q_{q_{\mathrm{rej}}}$ as verdicts and S as large as the runs in question require, one polynomial serving for every such S as in Definition 46, and a task is completed within f relative to M when it is completed within f under those runs. For $L \subseteq (\Sigma \setminus \{\#\})^*$ let α_L be the decision task whose inputs are the statements

$$i_x := \{C_{j,x_j} : 1 \leq j \leq |x|\} \cup \{C_{|x|+1,\#}\},$$

that $x\#$ is written from cell 1, with i_x a yes-input when $x \in L$. Then

1. on a legal run of M that begins with $x\#$ on the tape, the run of α_L on i_x along it from tick 0 is defined if and only if the legal run reaches a halting control state with the verdict of x and with $x\#$ still in cells 1 to $|x| + 1$, and it is then the legal run itself, with run time the number of steps of M;
2. $L \in$ P if and only if α_L is in P relative to some M;
3. $L \in$ NP if and only if α_L is in NP relative to some M.

Proof. For the first item, i_x is true at tick 0. Every correct output contains $Q_{q_{\text{acc}}}$ or $Q_{q_{\text{rej}}}$, so none is true while the control state is nonhalting, and at the halting state a correct output is true if and only if i_x is true there, which is $x\#$ in cells 1 to $|x|+1$, and the control state is the verdict of x. The run is then the whole legal run, and its run time is the number of steps by Proposition 1.

For the second, suppose α_L is completed within p relative to M. The input i_x is presented on the legal run on $x\#$, which ends at a halting control state because M halts on every input. By the first item, M halts on $x\#$ within $p(|x|+1)$ steps with the verdict of x. A machine that writes # after its input and then runs M decides L in polynomial time. Conversely let M' decide L in polynomial time. Let M halt on every tape that does not begin with some $x\#$. On a tape beginning $x\#$, the machine writes a marker in cell $|x|+2$, copies x to the right of the marker and runs M' on the copy. The marker moves one cell to the right and a blank is written in the cell it leaves whenever M' needs a further cell. Thus cells 1 to $|x|+1$ are unchanged and nothing written after the separator affects the run. This costs polynomially many further steps, every i_x is presented on the run on $x\#$, and by the first item α_L is completed within a polynomial relative to M.

For the third, let $L \in \text{NP}$ with a verifier V that decides within a polynomial in $|x|$ whether a string y over $\Sigma \setminus \{\#\}$ with $|y| \le q(|x|)$ is a valid certificate for x, where $x \in L$ if and only if some certificate is valid [Sipser, 2012]. Let M halt on every tape that does not begin with some $x\#$, and, on a tape beginning $x\#$, copy x and then the symbols after the separator up to the next separator or blank, at most $q(|x|)$ of them, behind a marker as above, run V on the copy and halt with its verdict. For $x \in L$ with a valid y, the bundle $c := \{C_{|x|+1+j,y_j} : 1 \le j \le |y|\} \cup \{C_{|x|+2+|y|,\#}\}$ has at most $q(|x|)+1$ programs and is presented with i_x on the run on $x\#y\#$. On every run on which $i_x \cup c$ is presented, the symbols after the first separator begin $y\#$. Hence M accepts within polynomially many steps with $x\#$ in place. The run of α_L on i_x is defined by the first item. For $x \notin L$ no certificate is valid, so V rejects whatever M copies and M never accepts. So α_L is in NP relative to M. Conversely suppose α_L is in NP relative to M with polynomial p, and let V on x and a string y with $|y| < p(|x|+1)$ run M on $x\#y$ for $p(|x|+1)$ steps and accept if M has accepted with $x\#$ still in cells 1 to $|x|+1$. If $x \in L$, some bundle c with at most $p(|x|+1)$ programs is presented with i_x on some legal run, whose tape begins $x\#$, and on that run a correct output is true within $p(|x|+1)$ steps, so M accepts with $x\#$ in place. In those steps the head reads cells 1 to $p(|x|+1)$ at most, so with y the symbols of that tape after the separator up to cell $p(|x|+1)$ and before the first blank, the run on $x\#y$ is the same for those steps, and V accepts. If $x \notin L$ and $|y| < p(|x|+1)$, the bundle $\{C_{|x|+1+j,y_j} : 1 \le j \le |y|\} \cup \{C_{|x|+2+|y|,b}\}$ has at most $p(|x|+1)$ programs and is presented with i_x on the run on $x\#y$. That run ends at a halting control state because M halts on every input. On that run, the no-input condition implies that no state within $p(|x|+1)$ steps has both $x\#$ in place and the accepting control state, so V rejects. So V is a polynomial-time verifier for L. □

Remark 3 (Finite-state machines as tasks)**.**
A deterministic finite-state machine (FSM) is a tuple

$$f = \langle Q, \Sigma, \delta, q_0, A \rangle$$

with finite state set Q, finite input alphabet Σ, transition function $\delta : Q \times \Sigma \to Q$, start state $q_0 \in Q$, and accepting set $A \subseteq Q$.

I can encode f as a finite $\mathfrak{v}_f$-task in Stack Theory by choosing the (finite) environment

$$\Phi_f := Q \times \Sigma \times Q,$$

whose microstates are triples (q, σ, q') interpreted as "current state q, read symbol σ, next state q'."

Define the following programs (subsets of Φ_f):

$$\mathrm{Curr}_q := \{(q,\sigma,q') : \sigma \in \Sigma,\ q' \in Q\}, \qquad \mathrm{Sym}_\sigma := \{(q,\sigma,q') : q \in Q,\ q' \in Q\},$$

$$\mathrm{Next}_{q'} := \{(q,\sigma,q') : q \in Q,\ \sigma \in \Sigma\}, \qquad \mathrm{Step}_f := \{(q,\sigma,q') : q' = \delta(q,\sigma)\}.$$

Let $\mathrm{Acc}_f := \{(q,\sigma,q') : q' \in A\}$ and $\mathrm{Rej}_f := \Phi_f \setminus \mathrm{Acc}_f$.

Let $\mathfrak{v}_f$ be the finite vocabulary consisting of all programs Curr_q ($q \in Q$), Sym_σ ($\sigma \in \Sigma$), $\mathrm{Next}_{q'}$ ($q' \in Q$), and the three global programs $\mathrm{Step}_f, \mathrm{Acc}_f, \mathrm{Rej}_f$.

For each pair $(q,\sigma) \in Q \times \Sigma$ define an input statement

$$i_{q,\sigma} := \{\mathrm{Curr}_q, \mathrm{Sym}_\sigma, \mathrm{Step}_f\}$$

and an output statement

$$o_{q,\sigma} := i_{q,\sigma} \cup \{\mathrm{Next}_{\delta(q,\sigma)}\} \cup \begin{cases} \{\mathrm{Acc}_f\} & \text{if } \delta(q,\sigma) \in A, \\ \{\mathrm{Rej}_f\} & \text{otherwise.} \end{cases}$$

Then $\rho_f := \langle I_f, O_f \rangle$ with $I_f := \{i_{q,\sigma} : (q,\sigma) \in Q \times \Sigma\}$ and $O_f := \{o_{q,\sigma} : (q,\sigma) \in Q \times \Sigma\}$ is a $\mathfrak{v}_f$-task: each input $i_{q,\sigma}$ has a completion in the outputs, and each output refines its corresponding input.

The task determines the transition relation encoded by Step_f and the accepting set encoded by Acc_f. The initial state q_0 must be specified separately.

Definition 47 (oracle program)**.**
Let $p \subseteq \Phi$ be a program, and let M have three further control states $q_?$, q_{yes} and q_{no}, with δ defined on $(Q \setminus \{q_{\mathrm{acc}}, q_{\mathrm{rej}}, q_?\}) \times \Gamma$. A pair $\phi, \psi \in \mathcal{M}_{M,S}$ with $q(\phi) = q_?$ is a **legal query step on** p when $i(\psi) = i(\phi)$, $T_\psi = T_\phi$, $u_\psi = u_\phi$, and $q(\psi) = q_{\mathrm{yes}}$ if $\phi \in p$ while $q(\psi) = q_{\mathrm{no}}$ otherwise. For a set of strings $A \subseteq \Gamma^*$ and $\phi \in \mathcal{M}_{M,S}$, the **query string** of ϕ is the tape contents $T_\phi(1)T_\phi(2)\cdots$ up to the first blank, and the **oracle program** of A is

$$O_A := \{\phi \in \mathcal{M}_{M,S} : \text{the query string of } \phi \text{ lies in } A\}.$$

A **legal run of** M **with oracle** A is a finite timeline fragment in $\mathcal{M}_{M,S}$ that begins as a legal bounded run does and ends at a halting control state, and whose consecutive pairs are legal bounded machine steps or legal query steps on O_A. The **oracle vocabulary** is $\mathfrak{v}^{\mathrm{conf}}_{M,S} \cup \{O_A\}$. The usual convention gives the oracle a separate query tape. The two conventions simulate each other with polynomial overhead, so the classes below are the same under either.

Lemma 17 (oracle steps are layer time)**.**
Let $r = (\phi_0, \ldots, \phi_n)$ be a finite timeline fragment in $\mathcal{M}_{M,S}$ whose consecutive pairs are each a legal bounded machine step or a legal query step on some program, and let $\mathfrak{v} \supseteq \mathfrak{v}^{\mathrm{conf}}_{M,S}$ be a finite vocabulary. Then

$$\kappa_{\mathfrak{v}}(r) = n.$$

If r is a legal run of M with oracle A, the sequence $(q(\phi_t), i(\phi_t), T_{\phi_t})_{t \le n}$ is a computation of the oracle Turing machine M^A whose query string is the tape up to the first blank. Conversely, suppose the implementation is complete as in Proposition 38. Then every computation of M^A whose head stays within cells 1 to S is realised by such a run.

Proof. At an ordinary step the head moves, so $H_{i(\phi_{t-1})} \neq H_{i(\phi_t)}$ and the encoding changes, as in Proposition 1. At a query step the control program changes from $Q_{q_?}$ to $Q_{q_{\mathrm{yes}}}$ or $Q_{q_{\mathrm{no}}}$, so the encoding changes. Each of the n consecutive pairs is therefore a layer tick of $\mathfrak{v}^{\mathrm{conf}}_{M,S}$, so $\kappa_{\mathfrak{v}^{\mathrm{conf}}_{M,S}}(r) = n$, and $\kappa_{\mathfrak{v}}(r) = n$ since $\kappa_{\mathfrak{v}^{\mathrm{conf}}_{M,S}}(r) \le \kappa_{\mathfrak{v}}(r)$ by Proposition 3 and $\kappa_{\mathfrak{v}}(r) \le n$. On a legal run with oracle A, ordinary steps follow δ by Definition 8, and a query step moves to q_{yes}

when the query string lies in A and to q_{no} otherwise, which is the transition of M^A at $q_?$, so the projected sequence is a computation of M^A. Conversely, under completeness each configuration of such a computation, with the usage programs produced by the computation's history, is attained by some state of $\mathcal{M}_{M,S}$. Consecutive states chosen this way satisfy the conditions of a legal bounded machine step or a legal query step, so the computation is realised by a legal run. □

Proposition 39 (Satisfiability on the tape)**.**
Let M be an embodied S-bounded machine as in Definition 8, complete as in Proposition 38, with a separator $\# \in \Sigma$. For $L \subseteq (\Sigma \setminus \{\#\})^*$ let α_L be the decision task of Proposition 38 and let O_L be the oracle program of the set of strings beginning $x\#$ with $x \in L$. Let each formula F in conjunctive normal form be encoded as a string $\langle F \rangle \in (\Sigma \setminus \{\#\})^m$ by a standard encoding. This means that distinct formulas have distinct strings, that encoding and decoding are computable in polynomial time, and that the length of $\langle F \rangle$ is polynomial in the size of F. The task α_{SAT} and oracle O_{SAT} are α_L and O_L for the language of encodings of satisfiable formulas. The inputs of α_{SAT} include the statements

$$i_F := \{C_{j,\langle F \rangle_j} : 1 \leq j \leq m\} \cup \{C_{m+1,\#}\},$$

that $\langle F \rangle\#$ is written from cell 1, and the task's correct outputs are the statements $o \supseteq i_F$ with $Q_{q_{\text{acc}}} \in o$ when F is satisfiable and $Q_{q_{\text{rej}}} \in o$ when it is not. Then

1. for every L, every input of α_L has a correct output, and when α_L has a yes-input and a no-input, no statement of any vocabulary containing $\mathfrak{v}^{\text{conf}}_{M,S}$ is a correct policy for it;
2. α_{SAT} is in NP relative to some M, and it is in P relative to some M if and only if satisfiability lies in P;
3. for every L, O_L is true at a state if and only if the input written there has a correct output of α_L containing $Q_{q_{\text{acc}}}$, and α_L is completed within the constant 3 relative to a machine with a query step on O_L, under its legal runs with that oracle.

Proof. By completeness each of i_x, $i_x \cup \{Q_{q_{\text{acc}}}\}$ and $i_x \cup \{Q_{q_{\text{rej}}}\}$ is a statement, so every input of α_L has a correct output. Suppose $\pi \in L_{\mathfrak{v}}$ for a vocabulary $\mathfrak{v} \supseteq \mathfrak{v}^{\text{conf}}_{M,S}$ satisfies $\text{Ext}(I_{\alpha_L}) \cap \text{Ext}(\pi) = O_{\alpha_L}$. Then every correct output contains π. Let i_x be a yes-input and $i_{x'}$ a no-input. The correct outputs $i_x \cup \{Q_{q_{\text{acc}}}\}$ and $i_{x'} \cup \{Q_{q_{\text{rej}}}\}$ both contain π, so $\pi \subseteq i_x \cap i_{x'}$. The statement $i_{x'} \cup \{Q_{q_{\text{acc}}}\}$ contains $i_{x'}$ and π, so it lies in $\text{Ext}(I_{\alpha_L}) \cap \text{Ext}(\pi) = O_{\alpha_L}$. But the only input it contains is $i_{x'}$, since $i_{x''} \subseteq i_{x'} \cup \{Q_{q_{\text{acc}}}\}$ puts the separator of $i_{x''}$ in the cell of the separator of $i_{x'}$ and then $x'' = x'$, and $x' \notin L$, so it is not a correct output. This contradiction proves the first item.

The second item is Proposition 38 for the language of encodings of satisfiable formulas, which lies in NP, since a machine given an assignment after the separator checks each clause against it in polynomially many steps [Sipser, 2012].

For the third, the query string of ϕ begins $x\#$ if and only if i_x is true at ϕ, since $x\#$ has no blank, and only one x has i_x true at ϕ, since the separator marks the end of x. So O_L is true at ϕ if and only if the string written there lies in L, which is when $i_x \cup \{Q_{q_{\text{acc}}}\}$ is a correct output. Let $\delta(q_0, a) = (q_?, a, R)$, $\delta(q_{\text{yes}}, a) = (q_{\text{acc}}, a, R)$ and $\delta(q_{\text{no}}, a) = (q_{\text{rej}}, a, R)$ for every a. On every tape beginning $x\#$ the run is an ordinary step to $q_?$, a query step to q_{yes} or q_{no} according to O_L, and an ordinary step to q_{acc} or q_{rej}, the verdict of x. It leaves cells 1 to $|x| + 1$ unchanged, so the run of α_L on i_x is defined, and each step changes the control program or moves the head, so its run time is 3. □

Proposition 40 (A vocabulary in which P ≠ NP)**.**
There is a relation $R \subseteq \bigcup_n \{0,1\}^n \times \{0,1\}^n$ such that $L_R := \{x : (x, y) \in R \text{ for some } y \text{ with } |y| = |x|\}$ lies in NP^R and not in P^R. Relative to the vocabulary $\mathfrak{v}^{\text{conf}}_{M,S} \cup \{O_R\}$, with O_R the oracle

program of the strings $x\#y$ with $(x, y) \in R$ and the tape bound large enough for the runs in question, P $\neq$ NP [Baker et al., 1975].

Proof. $L_R \in \mathrm{NP}^R$ for every R, since a certificate y is checked by one query. Let $(M_i, p_i)_{i\geq 1}$ enumerate every pair of an oracle machine and a polynomial, so that every polynomial-time oracle machine appears with a clock that bounds its running time. I build R in stages, and at each stage the membership of finitely many strings is fixed. At stage i I choose n_i larger than the length of every string fixed so far and with $2^{n_i} > p_i(n_i)$, and run M_i on 0^{n_i} for $p_i(n_i)$ steps, answering each query by the membership fixed so far and by no for every string not yet fixed. Fewer than 2^{n_i} of the queries have the form $0^{n_i}\#y$, so some $y^\star$ of length n_i is never queried. If the run rejects or fails to halt in time, I put $(0^{n_i}, y^\star)$ into R, and if it accepts I put nothing. I then fix the membership of every string queried and of every string of the form $0^{n_i}\#y$ as it now stands. Later stages add only pairs whose strings are longer than every string fixed before, so they change nothing fixed, and the run of M_i^R on 0^{n_i} within $p_i(n_i)$ steps is the run simulated. If some polynomial-time oracle machine decided L_R, it would be M_i with running time bounded by p_i for some i, so it would halt on 0^{n_i} within $p_i(n_i)$ steps, rejecting when $0^{n_i} \in L_R$ or accepting when $0^{n_i} \notin L_R$. So $L_R \notin \mathrm{P}^R$. By Lemma 17 the step counts are layer time in the oracle vocabulary, and for a complete machine with a separator that halts on every input, Proposition 38 relative to M^R puts the decision task α_{L_R} in NP and not in P relative to that vocabulary, which is the last claim. □

In the full vocabulary $\mathcal{P}$ with oracle querying allowed, O_L is available for every language L. Proposition 39 then gives a constant-time decision for each L, so P = NP = PSPACE is the class of all languages in this oracle model. This collapse and the oracle separation concern different permitted vocabularies. They do not settle P versus NP for the configuration vocabulary. Adding a PSPACE-complete oracle yields $\mathrm{P}^A = \mathrm{NP}^A$; therefore properties shared before and after that addition cannot by themselves prove separation. Finiteness, vocabulary-relative description length and layer time are insufficient. The conclusion is compatible with an extended Church–Turing simulation thesis [Baker et al., 1975, Bernstein and Vazirani, 1997].

22 Assembly observables

Definition 48 (Assembly observables)**.**
Fix finite basic object types and permitted binary joins. A type produced by a join is added to the available types and may be reused in later joins. An object type's *assembly space* is the collection of recursive assembly pathways that produce it. For observed object type i, its *assembly index* a_i is the number of joins in a shortest pathway from the basic types. Its *copy number* n_i is the number of observed copies. If an ensemble contains N observed types and N_{T} objects in total, its ensemble assembly is

$$A = \sum_{i=1}^{N} \mathrm{e}^{a_i} \left(\frac{n_i - 1}{N_{\mathrm{T}}} \right),$$

where e is Euler's number [Sharma et al., 2023].

Proposition 41 (Layer-relative assembly index)**.**
One physical bearer can have different assembly indices at two layers that have different distinctions and are related by a join-preserving map, meaning a map sending each join of two parts to a join of their images. Its ensemble assembly also differs at every copy number $n \geq 2$.

Proof. Let the fine layer consist of nonempty ordered strings of length at most four over the basic block types `a` and `b`. Let concatenation be the permitted join, and represent the bearer by

`aabb`.

The pathway

$$\mathtt{a}+\mathtt{a}=\mathtt{aa}, \qquad \mathtt{b}+\mathtt{b}=\mathtt{bb}, \qquad \mathtt{aa}+\mathtt{bb}=\mathtt{aabb}$$

uses three joins. Two joins cannot suffice. After one join, only one composite type of length two has been constructed. Reaching length four on the second join would require joining that type to itself, producing a string with two identical halves. The two halves of `aabb` are `aa` and `bb`. Its fine-layer assembly index is therefore three.

Let the coarse layer replace both fine block types `a` and `b` with one basic block type `x`, so block order survives while block type is lost. Map both `a` and `b` to `x` and extend the map through concatenation. The same bearer is then represented by `xxxx`. The pathway

$$\mathtt{x}+\mathtt{x}=\mathtt{xx}, \qquad \mathtt{xx}+\mathtt{xx}=\mathtt{xxxx}$$

uses two joins, while one join from basic blocks cannot produce length four. Its coarse-layer assembly index is two. For any $n \geq 2$ observed copies of the bearer, fine and coarse ensemble assemblies are

$$\mathrm{e}^3\frac{n-1}{n} \qquad \text{and} \qquad \mathrm{e}^2\frac{n-1}{n}.$$

□

A permitted production pair is an unordered pair of distinct active occurrences whose types may be joined; a join consumes them and creates one active occurrence. Equiprobable production assigns equal conditional probability to every permitted pair. Preferential production assigns different positive probabilities to two pairs at a state reached with positive probability. A stopping rule defines the endpoint ensemble. Endpoint identification requires different process parameters to induce different endpoint distributions.

Theorem 26 (Endpoint non-identification)**.**
For every pair of integers $a, n \geq 2$, there is one finite assembly space on which every choice of production probabilities produces

$$N=1, \qquad a_1=a, \qquad n_1=N_{\mathrm{T}}=n, \qquad A=\mathrm{e}^a\frac{n-1}{n}$$

with probability one. The inventory is n noninteracting compartments that each begin with 2^a monomer occurrences, and the stopping rule is that production continues while a permitted pair exists. Hence no property of the production probabilities is identifiable from the endpoint over any class of processes on this space. For each finite preference ratio $\nu > 0$[4], joining is equiprobable when $\nu = 1$ and preferential when $\nu \neq 1$. Production may also be restricted so that every chosen pair contains the newest aggregate. In particular neither ν nor whether $\nu \neq 1$ is identifiable from the endpoint.

Proof. I use unary chain types with lengths $1, \ldots, 2^a$. Length one is basic. Joining lengths r and s produces length $r + s$ whenever $r + s \leq 2^a$. Repeated doubling constructs length 2^a in a joins. After j type-level joins, no available type can have length greater than 2^j. The chain of length 2^a therefore has assembly index a.

A compartment begins with 2^a active monomer occurrences. Each production join consumes two active occurrences and inserts their joined chain. Total length remains 2^a, while each join reduces the number of active occurrences by one. Every active pair is permitted because its two positive lengths sum to at most the conserved total. Every history therefore ends after $2^a - 1$ joins with one chain of length 2^a.

[4]The first pair is chosen uniformly. After each join, each pair containing the newest aggregate, meaning the chain that join formed, has weight ν, while each other pair has weight one. The next pair is chosen with probability proportional to its weight.

The first pair is chosen uniformly. After each join, every pair containing the newest aggregate is given weight ν and every other pair weight one, and the weights are normalised. At $\nu = 1$ all pairs are equiprobable. After the first join there are $2^a - 1 \geq 3$ active occurrences. There is therefore both a pair containing the newest aggregate and a pair not containing it. For every finite $\nu > 0$ with $\nu \neq 1$, these pairs have different positive conditional probabilities while every history still has positive probability. Production may instead be restricted to pairs containing that aggregate. By conservation, every choice of production probabilities has the same terminal chain. Running n noninteracting compartments and observing their terminal chains, I obtain $a_1 = a$, $n_1 = N_{\mathrm{T}} = n$, and $A = \mathrm{e}^a(n-1)/n$ with probability one. □

In the equiprobable construction with $n = 2$, $A = \mathrm{e}^a/2 \to \infty$ as $a \to \infty$. Thus no fixed threshold on endpoint A alone excludes equiprobable production over this class. A specified transition model, resource bound or temporal observation can restrict the class [Bennett, 2026k].

23 Macro descriptions and scale

A macro map is a map $q : \Phi \to Z$, with fibres $q^{-1}(\{z\})$. For a finitely supported probability mass function P, write $H(P) := -\sum_\phi P(\phi) \log_2 P(\phi)$, with $0 \log_2 0 := 0$.

Lemma 18 (Truth sets are unions of macro fibres)**.**
Fix an environment Φ and a finite vocabulary $\mathfrak{v} \subseteq 2^\Phi$. For any statement $l \in L_\mathfrak{v}$ and any $\phi_1, \phi_2 \in \Phi$, if $\mathrm{Enc}_\mathfrak{v}(\phi_1) = \mathrm{Enc}_\mathfrak{v}(\phi_2)$ then $\phi_1 \in T(l) \iff \phi_2 \in T(l)$.

Proof. By Definition 7, $\phi \in T(l) \iff l \subseteq \mathrm{Enc}_\mathfrak{v}(\phi)$. If $\mathrm{Enc}_\mathfrak{v}(\phi_1) = \mathrm{Enc}_\mathfrak{v}(\phi_2)$, then $l \subseteq \mathrm{Enc}_\mathfrak{v}(\phi_1) \iff l \subseteq \mathrm{Enc}_\mathfrak{v}(\phi_2)$. □

Proposition 42 (Truth sets are pullbacks of encoding images)**.**
For every statement $l \in L_\mathfrak{v}$, $T(l) = \mathrm{Enc}_\mathfrak{v}^{-1}(\mathcal{M}(l))$, where $\mathcal{M}(l) = \{\mathrm{Enc}_\mathfrak{v}(\phi) \mid \phi \in T(l)\}$. If $\Phi_\star \subseteq \Phi$ is finite, write $\mathcal{M}_\star(l) := \{\mathrm{Enc}_\mathfrak{v}(\phi) \mid \phi \in T(l) \cap \Phi_\star\}$ and $S_{\mathrm{state}}(l) := \log_2 |T(l) \cap \Phi_\star|$. Then

$$|T(l) \cap \Phi_\star| = \sum_{m \in \mathcal{M}_\star(l)} |\mathrm{Enc}_\mathfrak{v}^{-1}(\{m\}) \cap \Phi_\star|.$$

If all fibres have the same size C, then $S_{\mathrm{state}}(l) = \log_2 |\mathcal{M}_\star(l)| + \log_2 C$.

Proof. Fix l. For any ϕ, $\phi \in \mathrm{Enc}_\mathfrak{v}^{-1}(\mathcal{M}(l)) \iff \mathrm{Enc}_\mathfrak{v}(\phi) \in \mathcal{M}(l) \iff \exists\, \psi \in T(l) : \mathrm{Enc}_\mathfrak{v}(\psi) = \mathrm{Enc}_\mathfrak{v}(\phi)$. By Lemma 18, this implies $\phi \in T(l)$. Conversely, $\phi \in T(l)$ implies $\mathrm{Enc}_\mathfrak{v}(\phi) \in \mathcal{M}(l)$. This proves $T(l) = \mathrm{Enc}_\mathfrak{v}^{-1}(\mathcal{M}(l))$.

For the counting identity, the fibres $\mathrm{Enc}_\mathfrak{v}^{-1}(\{m\})$ are disjoint, so $|T(l) \cap \Phi_\star| = \sum_{m \in \mathcal{M}_\star(l)} |\mathrm{Enc}_\mathfrak{v}^{-1}(\{m\}) \cap \Phi_\star|$. If all fibre sizes equal C, then $|T(l) \cap \Phi_\star| = C \cdot |\mathcal{M}_\star(l)|$. Taking $\log_2$ gives the entropy identity. □

Proposition 43 (Entropy chain rule for a macro map)**.**
Let P be a probability mass function on Φ with finite support $\Phi_\star$. Let $q : \Phi \to Z$ be surjective and define

$$P_Z(z) := \sum_{\phi : q(\phi) = z} P(\phi).$$

For each z with $P_Z(z) > 0$, define the conditional distribution on the fibre by

$$P(\phi \mid z) := \frac{P(\phi)}{P_Z(z)}.$$

Then

$$H(P) = H(P_Z) + \sum_{z \,:\, P_Z(z) > 0} P_Z(z)\, H(P(\cdot \mid z)).$$

If for every z with $P_Z(z) > 0$ the conditional is uniform on $q^{-1}(\{z\})$ and the positive-mass fibres have sizes $C_z := |q^{-1}(\{z\})|$, then

$$H(P) = H(P_Z) + \sum_{z\,:\,P_Z(z)>0} P_Z(z) \log_2 C_z.$$

In particular, if all positive-mass fibres have the same size C, then $H(P) = H(P_Z) + \log_2 C$.

Proof. For every ϕ with $P(\phi) > 0$, the macro state $z = q(\phi)$ satisfies $P_Z(z) > 0$, so

$$\log P(\phi) = \log P_Z(q(\phi)) + \log P(\phi \mid q(\phi)).$$

Multiply by $-P(\phi)$ and sum over $\Phi_\star$. Grouping terms by $z = q(\phi)$ gives

$$H(P) = H(P_Z) + \sum_{z\,:\,P_Z(z)>0} P_Z(z)\, H(P(\cdot \mid z)).$$

If the conditional distribution on each fibre of positive mass is uniform on $q^{-1}(\{z\})$, then $H(P(\cdot \mid z)) = \log_2 C_z$ for every occupied fibre. Substituting yields the displayed weighted formula. If all occupied fibres have the same size C, the weighted average reduces to $\log_2 C$. □

Lemma 19 (Relabel-invariant vocabularies cannot separate relabelings)**.**
Let G be a finite group acting on Φ. Say $\mathfrak{v}$ is **relabel-invariant** if for every $p \in \mathfrak{v}$ and every $g \in G$, $g \cdot p := \{g(\phi) : \phi \in p\} \in \mathfrak{v}$. Then for every $g \in G$ and every $\phi \in \Phi$, $\mathrm{Enc}_{\mathfrak{v}}(g(\phi)) = g \cdot \mathrm{Enc}_{\mathfrak{v}}(\phi) := \{g \cdot p : p \in \mathrm{Enc}_{\mathfrak{v}}(\phi)\}$.

If additionally $g \cdot p = p$ for all $p \in \mathfrak{v}$ and all $g \in G$ (the vocabulary is literally invariant, not just covariant), then $\mathrm{Enc}_{\mathfrak{v}}(g(\phi)) = \mathrm{Enc}_{\mathfrak{v}}(\phi)$ for all g, ϕ. In this case, the entire orbit $G \cdot \phi$ maps to the same encoding. The encoding fibre has size at least $|G \cdot \phi|$.

Proof. $p \in \mathrm{Enc}_{\mathfrak{v}}(\phi) \iff \phi \in p \iff g(\phi) \in g \cdot p$. So $p \in \mathrm{Enc}_{\mathfrak{v}}(\phi) \iff g \cdot p \in \mathrm{Enc}_{\mathfrak{v}}(g(\phi))$. If $g \cdot p = p$ for all p, this gives $\mathrm{Enc}_{\mathfrak{v}}(g(\phi)) = \mathrm{Enc}_{\mathfrak{v}}(\phi)$. □

Corollary 28 (The $N!$ term in the Gibbs paradox)**.**
Let Φ consist of N labelled particles in k boxes, with $k \geq N$, and let the symmetric group S_N act by permuting particle labels. If the vocabulary is literally invariant under S_N (it encodes only occupation numbers, not which particle is where), then for any configuration with all particles in distinct boxes, the encoding fibre has size $N!$. If moreover the microstate distribution is uniform on that orbit, then the within-fibre entropy contribution in Proposition 43 is $\log_2(N!)$. The Gibbs correction is the subtraction of this relabelling entropy.

Proof. If all N particles are in distinct boxes, every permutation of labels gives a different microstate but the same occupation-number encoding. The orbit under S_N has size $N!$ because the stabiliser is trivial. By Lemma 19, all $N!$ microstates in that orbit share the same encoding, so the fibre size is $N!$. If P is uniform on the orbit, then the positive-mass conditional on that fibre is uniform on $N!$ points. Proposition 43 therefore gives a within-fibre contribution of $\log_2(N!)$. □

Definition 49 (Covering and box dimension)**.**
Let $A \subseteq \mathbb{R}^d$ be bounded. For $\varepsilon > 0$, the **covering number** $N(A, \varepsilon)$ is the minimum number of closed ε-balls needed to cover A. The **(upper) box-counting dimension** is

$$\overline{\dim}_B(A) = \limsup_{\varepsilon \to 0} \frac{\log N(A, \varepsilon)}{-\log \varepsilon}.$$

Proposition 44 (Scale as a macro map)**.**
Fix $\varepsilon > 0$ and let $\mathcal{G}_\varepsilon$ be the standard grid of half-open cubes of side length ε in $\mathbb{R}^d$. Let $\mathcal{G}_\varepsilon(A)$ be the finite set of grid cells that intersect A, and define

$$q_\varepsilon : A \to \mathcal{G}_\varepsilon(A)$$

by sending each point to the unique grid cell that contains it. This is a finite macro map on A. Two points that land in the same cell are indistinguishable at resolution ε. Write

$$N_\varepsilon(A) := |\mathcal{G}_\varepsilon(A)|$$

for the number of occupied cells. Then $\log_2 N_\varepsilon(A)$ is the scale entropy at resolution ε. Furthermore,

$$\overline{\dim}_B(A) = \limsup_{\varepsilon \to 0} \frac{\log N_\varepsilon(A)}{-\log \varepsilon}.$$

Proof. Because A is bounded, only finitely many ε-grid cells meet A, so $\mathcal{G}_\varepsilon(A)$ is finite. The map q_ε sends each point of A to its cell, so it is a macro map on the bounded support A. Any two points in the same cell are at Euclidean distance at most $\sqrt{d}\,\varepsilon$, so those points are indistinguishable at that resolution. The number of occupied cells is exactly $N_\varepsilon(A)$, and the corresponding scale entropy is $\log_2 N_\varepsilon(A)$. The grid-cell count and the ball-cover count differ by at most dimension-dependent constants, so they yield the same upper box dimension. □

Proposition 45 (Scale entropy diverges on bounded infinite sets)**.**
If $A \subseteq \mathbb{R}^d$ is bounded and infinite, then

$$N_\varepsilon(A) \to \infty \qquad \text{and hence} \qquad \log_2 N_\varepsilon(A) \to \infty$$

as $\varepsilon \to 0$. In particular, if $\overline{\dim}_B(A) > 0$, then the same conclusion holds.

Proof. Suppose not. Then there exists $M \in \mathbb{N}$ and a sequence $\varepsilon_n \to 0$ such that

$$N_{\varepsilon_n}(A) \le M \qquad \text{for all } n.$$

For each n, choose at most M occupied ε_n-grid cells that cover A, and choose one point

$$x_{n,1}, \dots, x_{n,M}$$

from the closure of each chosen cell, repeating points if fewer than M cells are needed. Because A is bounded, after passing to a subsequence I may assume that for each $j = 1, \dots, M$,

$$x_{n,j} \to x_j \qquad \text{as } n \to \infty$$

for some points $x_1, \dots, x_M \in \mathbb{R}^d$.

Now fix any $a \in A$. For each n, the point a lies in one of the chosen cells, so

$$\|a - x_{n,j_n}\| \le \sqrt{d}\,\varepsilon_n$$

for some index $j_n \in \{1, \dots, M\}$. By passing to a further subsequence I may assume $j_n = j$ is constant. Then

$$\|a - x_j\| \le \|a - x_{n,j}\| + \|x_{n,j} - x_j\| \to 0,$$

so $a = x_j$. Since $a \in A$ was arbitrary, I get

$$A \subseteq \{x_1, \dots, x_M\},$$

which contradicts that A is infinite.

Therefore $N_\varepsilon(A) \to \infty$, and the logarithmic conclusion follows immediately. Positive upper box dimension implies that A is infinite, so the final claim is a corollary. □

24 Feature vocabularies

Definition 50 (Region-class vocabulary)**.**
Fix a finite set of regions $\mathcal{R} = \{r_1, \dots, r_R\}$ and a class set $C = \{0, \dots, K-1\}$. Define the environment

$$\Phi_{\mathcal{R}} := C^{\mathcal{R}},$$

the set of all total assignments of classes to regions. For each region $r \in \mathcal{R}$ and class $c \in C$, define the program

$$p_{r,c} := \{g \in \Phi_{\mathcal{R}} : g(r) = c\}.$$

The resulting vocabulary is

$$\mathfrak{v}_{\text{reg}} := \{p_{r,c} : r \in \mathcal{R},\ c \in C\}.$$

Proposition 46 (Structure of the region-class language)**.**
$L_{\mathfrak{v}_{\text{reg}}}$ is exactly the set of partial functions from $\mathcal{R}$ to C, encoded as consistent subsets of $\mathfrak{v}_{\text{reg}}$. Consequently

$$|L_{\mathfrak{v}_{\text{reg}}}| = (K+1)^R,$$

because each region can be assigned one of K classes or left unspecified.

Proof. A subset $l \subseteq \mathfrak{v}_{\text{reg}}$ is consistent iff no region is assigned two different classes. That is exactly the condition for l to encode a partial function from $\mathcal{R}$ to C. Each of the R regions has $K+1$ independent options. It can be sent to one of the K classes or left unspecified. □

Definition 51 (Training-label policy and free regions)**.**
Fix training data with inputs $x_1, \dots, x_n$. Let $\mathcal{R}_{\text{train}} \subseteq \mathcal{R}$ be the set of regions containing at least one training point. For each $r \in \mathcal{R}_{\text{train}}$, let y_r be a majority training label in that region, breaking ties arbitrarily. Define the **training-label policy**

$$\pi_{\text{train}} := \{p_{r,y_r} : r \in \mathcal{R}_{\text{train}}\}.$$

Let

$$\mathcal{R}_{\text{free}} := \mathcal{R} \setminus \mathcal{R}_{\text{train}}, \qquad k_{\text{free}} := |\mathcal{R}_{\text{free}}|.$$

Theorem 27 (Weakness of the training-label policy)**.**
Under $\mathfrak{v}_{\text{reg}}$,

$$\text{ff}(\pi_{\text{train}}) = |\operatorname{Ext}(\pi_{\text{train}})| = (K+1)^{k_{\text{free}}}.$$

In the usual ten-class case, this is $11^{k_{\text{free}}}$.

Proof. The extension $\operatorname{Ext}(\pi_{\text{train}})$ consists of all partial classifications that agree with π_{train} on the training regions. Every region in $\mathcal{R}_{\text{train}}$ is already fixed. Every region in $\mathcal{R}_{\text{free}}$ still has $K+1$ independent choices. It can be assigned one of the K classes or left unspecified. Multiplying those choices gives $(K+1)^{k_{\text{free}}}$. □

Remark 4 (Approximation and shared-weight caveat)**.**
This vocabulary is intentionally crude. Inside a fixed ReLU region, the final affine layer can still classify different inputs differently. So the one-class-per-region encoding is exact only when each relevant region contains at most one input. It also ignores the shared-weight constraint. A single final linear layer must realise all assignments at once. The next vocabulary repairs that.

Definition 52 (Feature-classifier vocabulary)**.**
Fix a finite evaluation set $S = \{x_1, \dots, x_m\}$, a feature map $\varphi : S \to \mathbb{R}^d$, and a class set $C = \{0, \dots, K-1\}$. Let the environment be the parameter space

$$\Phi_{\varphi} := \mathbb{R}^{Kd+K},$$

whose elements are the class weight vectors and biases (W, b). For each $x \in S$ and class $c \in C$, define

$$p_{x,c} := \{(W, b) \in \Phi_\varphi : W_c \cdot \varphi(x) + b_c > W_{c'} \cdot \varphi(x) + b_{c'} \text{ for all } c' \neq c\}.$$

The resulting vocabulary is

$$\mathfrak{v}_\varphi := \{p_{x,c} : x \in S,\ c \in C\}.$$

Proposition 47 (Consistency is linear feasibility)**.**
A subset $l \subseteq \mathfrak{v}_\varphi$ lies in $L_{\mathfrak{v}_\varphi}$ iff the strict linear system

$$W_{c_i} \cdot \varphi(x_i) + b_{c_i} > W_c \cdot \varphi(x_i) + b_c$$

for every program $p_{x_i,c_i} \in l$ and every $c \neq c_i$ is feasible. Equivalently, $l \in L_{\mathfrak{v}_\varphi}$ iff for some $\varepsilon > 0$ the LP with constraints

$$(W_{c_i} - W_c) \cdot \varphi(x_i) + (b_{c_i} - b_c) \geq \varepsilon$$

for every such pair is feasible.

Proof. By Definition 52, $\bigcap_{p \in l} p$ is the set of all (W, b) satisfying the displayed strict inequalities. If l assigns two classes to the same input, the system contains a strict inequality and its reverse, so it is infeasible. Otherwise $l \in L_{\mathfrak{v}_\varphi}$ iff the strict system has a solution.

For the LP reformulation, suppose $(W^\star, b^\star)$ satisfies all strict inequalities. There are only finitely many of them, so the minimum positive margin

$$\varepsilon^\star := \min_{i,\ c \neq c_i} \left[(W^\star_{c_i} - W^\star_c) \cdot \varphi(x_i) + (b^\star_{c_i} - b^\star_c)\right]$$

is strictly positive. Then $(W^\star, b^\star)$ is feasible for every $0 < \varepsilon \leq \varepsilon^\star$. Conversely, any LP solution with $\varepsilon > 0$ satisfies the original strict system. □

Definition 53 (Training policy and pointwise pair proxy)**.**
Given training data $(x_1, y_1), \ldots, (x_n, y_n)$ with each $x_i \in S$, define

$$\pi := \{p_{x_i,y_i} : i = 1, \ldots, n\}.$$

Assume $\pi \in L_{\mathfrak{v}_\varphi}$. Let

$$S_{\text{train}} := \{x_1, \ldots, x_n\}, \qquad S_{\text{unseen}} := S \setminus S_{\text{train}}.$$

For each $x_j \in S_{\text{unseen}}$, define the **pointwise extension count**

$$\text{ext}(x_j) := |\{c \in C : \pi \cup \{p_{x_j,c}\} \in L_{\mathfrak{v}_\varphi}\}|.$$

Define the **feature pair proxy**

$$\text{PP}(\varphi) := \sum_{x_j \in S_{\text{unseen}}} \text{ext}(x_j).$$

Theorem 28 (Feature-classifier reduction and affine invariance)**.**
Fix φ, S, and a training policy $\pi \in L_{\mathfrak{v}_\varphi}$.

1. For each unseen $x_j \in S_{\text{unseen}}$, $\text{ext}(x_j)$ equals the number of classes $c \in C$ for which $p_{x_j,c}$ appears in some completion $l \in \text{Ext}(\pi)$.
2. $\text{PP}(\varphi)$ is computable by solving $K \cdot |S_{\text{unseen}}|$ LP feasibility problems.
3. If $A \in \mathbb{R}^{d \times d}$ is invertible, $a \in \mathbb{R}^d$, and $\varphi'(x) = A\varphi(x) + a$, then $\text{PP}(\varphi') = \text{PP}(\varphi)$.
4. Every $l \in L_{\mathfrak{v}_\varphi}$ is realised by a single shared parameter vector (W, b).

Proof. For Part (1), suppose first that $\pi \cup \{p_{x_j,c}\} \in L_{\mathfrak{v}_\varphi}$. Then this set is itself a completion of π, so $p_{x_j,c}$ appears in some $l \in \mathrm{Ext}(\pi)$. Conversely, if $p_{x_j,c} \in l$ for some $l \in \mathrm{Ext}(\pi)$, then $\pi \cup \{p_{x_j,c}\} \subseteq l$. Consistency is inherited by subsets because intersecting fewer programs cannot make a nonempty intersection empty. So $\pi \cup \{p_{x_j,c}\} \in L_{\mathfrak{v}_\varphi}$.

Part (2) is immediate from Proposition 47. For each unseen input there are K candidate classes, and each candidate class requires one LP feasibility check.

For Part (3), let $\varphi'(x) = A\varphi(x) + a$ with A invertible. Define

$$\Psi : \Phi_\varphi \to \Phi_{\varphi'}, \qquad \Psi(W, b) := (WA^{-1},\, b - WA^{-1}a).$$

Its inverse is

$$\Psi^{-1}(W', b') = (W'A,\, b' + W'a),$$

so Ψ is a bijection. If I write

$$\Psi(W, b) =: (W', b'),$$

then for any $x \in S$ and class $c \in C$,

$$W'_c \cdot \varphi'(x) + b'_c = (WA^{-1})_c \cdot (A\varphi(x) + a) + \big(b - WA^{-1}a\big)_c = W_c \cdot \varphi(x) + b_c.$$

Hence

$$(W, b) \in p_{x,c} \iff \Psi(W, b) \in p'_{x,c}.$$

So Ψ sends each program in $\mathfrak{v}_\varphi$ bijectively to the corresponding program in $\mathfrak{v}_{\varphi'}$ and preserves consistency. Therefore every pointwise extension count is unchanged, and so is PP.

Part (4) is built into the definition of the language. If $l \in L_{\mathfrak{v}_\varphi}$, then by definition

$$\bigcap_{p \in l} p \neq \emptyset.$$

Any point in that nonempty intersection is a single shared parameter vector (W, b) realising every program in l at once. □

Corollary 29 (Architecture-independence on a fixed sample)**.**
Let φ_1 and φ_2 be two frozen feature maps, possibly from different architectures, that agree on every input in S. Then $\mathfrak{v}_{\varphi_1} = \mathfrak{v}_{\varphi_2}$ and $\mathrm{PP}(\varphi_1) = \mathrm{PP}(\varphi_2)$.

Proof. If $\varphi_1(x) = \varphi_2(x)$ for all $x \in S$, then the defining inequalities for every program $p_{x,c}$ are identical under the two feature maps. So the vocabularies, languages, extensions, and proxies coincide. □

For a frozen feature map φ_f in Definition 52, let X_{child}, X_{anchor} and X_{probe} be disjoint. With observed child labels y_{child} and the trained network's anchor predictions $\hat{y}_f$, put

$$\pi_f = \{p^{(f)}_{x,y_{\mathrm{child}}(x)} : x \in X_{\mathrm{child}}\} \cup \{p^{(f)}_{a,\hat{y}_f(a)} : a \in X_{\mathrm{anchor}}\}.$$

For an indexed list $H = (s_1, \ldots, s_m)$ of joint probe demands, with $s_j^{(f)}$ its translation to the feature vocabulary, the score is

$$J_H(f) = \frac{1}{m} \sum_{j=1}^{m} \mathbf{1}\{\pi_f \cup s_j^{(f)} \in L_{\mathfrak{v}_{\varphi_f}}\}.$$

Each check uses one shared hypothetical affine head for all child, anchor and requested probe labels. Different checks may use different heads. With unrestricted heads and finitely many strict inequalities, positive rescaling makes every positive score gap at least one. This rescaling is unavailable when output constraints bound score changes. J_H is a pass fraction under the specified demand list, not the count of the full extension.

Proposition 48 (finite joint completion score)**.**
Under a uniform draw from the m indexed demands, $J_H(f)$ is the probability that the drawn joint demand is feasible with π_f. Every union $\pi_f \cup s_j^{(f)}$ that is feasible is a completion of π_f.

Proof. Quantity $J_H(f) = m^{-1} \sum_{j=1}^{m} \mathbf{1}\{\pi_f \cup s_j^{(f)} \in L_{\mathfrak{v}_{\varphi_f}}\}$ is the empirical probability of success under a uniform draw from the demand list H. Feasibility means $\pi_f \subseteq \pi_f \cup s_j^{(f)}$ and the union is a statement in $L_{\mathfrak{v}_{\varphi_f}}$, so the union lies in $\mathrm{Ext}(\pi_f)$. □

Proposition 49 (common-task compatibility)**.**
Let $\pi_1, \ldots, \pi_M \in L_{\mathfrak{v}}$. If a $\mathfrak{v}$-task α has $O_\alpha \neq \emptyset$ and every $\pi_i \in \Pi_\alpha$, then $\bigcup_i \pi_i \in \Pi_\alpha$. Conversely, if $\bigcup_i \pi_i \in L_{\mathfrak{v}}$, then there exists a $\mathfrak{v}$-task with nonempty correct outputs for which every π_i is correct.

Proof. For necessity choose $y \in O_\alpha$. By correctness, $y \in \mathrm{Ext}(\pi_i)$ for every i, so $\bigcup_i \pi_i \subseteq y$ and the union is satisfiable. Since $\mathrm{Ext}(\bigcup_i \pi_i) = \bigcap_i \mathrm{Ext}(\pi_i)$, I get $\mathrm{Ext}(I_\alpha) \cap \mathrm{Ext}(\bigcup_i \pi_i) = \bigcap_i (\mathrm{Ext}(I_\alpha) \cap \mathrm{Ext}(\pi_i)) = O_\alpha$, so $\bigcup_i \pi_i \in \Pi_\alpha$. For sufficiency put $\bar{\pi} := \bigcup_i \pi_i$ and define

$$I_\alpha := \{\bar{\pi}\}, \qquad O_\alpha := \mathrm{Ext}(\bar{\pi}).$$

Since $\pi_i \subseteq \bar{\pi}$, one has $\mathrm{Ext}(\bar{\pi}) \subseteq \mathrm{Ext}(\pi_i)$ and therefore

$$\mathrm{Ext}(I_\alpha) \cap \mathrm{Ext}(\pi_i) = \mathrm{Ext}(\bar{\pi}) = O_\alpha.$$

□

In the common external language of total labellings $g : X \to \mathcal{C}$, child and anchor policies are jointly satisfiable iff their labels agree wherever inputs coincide; with a shared child, this requires agreement at every anchor. Agreement establishes compatibility with some common task, not correctness for a specified child task or equality of the disembodied tasks in different feature vocabularies. Archived aggregate results do not certify anchor agreement. For $\varphi'_f = A\varphi_f + a$ with A invertible, the head bijection

$$W' = WA^{-1}, \qquad b' = b - WA^{-1}a$$

preserves all class scores, so J_H and every feasibility check with constraints on class scores are unchanged when reference inputs and sampling rules remain fixed. Numerical tolerances can still affect borderline cases.

25 Transformer vocabularies

Definition 54 (Transformer backbone, units, and selectors)**.**
Fix a transformer backbone $\mathcal{T}_\theta$. For an input sequence $x = (x_1, \ldots, x_{|x|})$, let

$$h_t^\ell(x) \in \mathbb{R}^{d_\ell}$$

denote the contextual hidden state at layer ℓ and token position t.

An **evaluation unit** u is one of the following.

1. a sequence unit $u = x$
2. a token unit $u = (x, t)$
3. a prefix unit $u = x_{<t}$ for restricted next-token prediction

A **readout selector** σ maps each unit u to a contextual feature vector

$$\varphi_{\theta,\sigma}(u) \in \mathbb{R}^d.$$

When θ and σ are fixed I write $\varphi(u)$.

Let $\mathcal{U}$ be a finite set of evaluation units. I assume the support is task-homogeneous, meaning that all units in $\mathcal{U}$ belong to one task family and that label meanings are consistent across units. For each $u \in \mathcal{U}$, let C_u be a finite admissible label set. I use

$$Y := \bigcup_{u \in \mathcal{U}} C_u$$

for the finite label universe induced by the support.

Remark 5 (Default selectors)**.**
The default selector is task-dependent.

1. For sequence classification, use a pooled final-layer vector or a designated classification token.
2. For token classification, use the final-layer hidden state at the labelled token position.
3. For restricted next-token prediction, use the final hidden state at the last prefix position in a causal model.

The theory does not depend on which selector is chosen. It depends only on the resulting feature map φ on the finite support.

Remark 6 (Why the next-token case is restricted)**.**
For sequence and token classification, the label set is naturally finite. For next-token prediction, the full tokenizer vocabulary is large and makes the first operationalisation unnecessarily expensive. I therefore begin with a finite candidate set C_u for each prefix unit u. The cleanest choices are the gold token plus the top $k-1$ distractors under the frozen model, or an existing multiple-choice answer set. This keeps the embodied language finite while preserving the local commitment structure that matters for weakness.

Definition 55 (Transformer feature-classifier vocabulary)**.**
Fix a finite support $\mathcal{U}$, a feature map $\varphi : \mathcal{U} \to \mathbb{R}^d$, and the induced finite label universe Y. Let the environment be the parameter space for a shared affine readout

$$\Phi_{\varphi,Y} := \mathbb{R}^{|Y|d+|Y|},$$

whose elements are class weight vectors and biases (W, b).

For each unit $u \in \mathcal{U}$ and label $c \in C_u$, define the program

$$p_{u,c} := \{(W,b) \in \Phi_{\varphi,Y} : W_c \cdot \varphi(u) + b_c > W_{c'} \cdot \varphi(u) + b_{c'} \text{ for all } c' \in C_u \setminus \{c\}\}.$$

The resulting vocabulary is

$$\mathfrak{v}_{\varphi,Y} := \{p_{u,c} : u \in \mathcal{U},\ c \in C_u\}.$$

Proposition 50 (Consistency is LP feasibility)**.**
A subset $l \subseteq \mathfrak{v}_{\varphi,Y}$ lies in $L_{\mathfrak{v}_{\varphi,Y}}$ if and only if the strict linear system with inequalities

$$W_{c_i} \cdot \varphi(u_i) + b_{c_i} > W_c \cdot \varphi(u_i) + b_c$$

for every program $p_{u_i,c_i} \in l$ and every $c \in C_{u_i} \setminus \{c_i\}$ is feasible. Equivalently, $l \in L_{\mathfrak{v}_{\varphi,Y}}$ if and only if for some $\varepsilon > 0$ the LP with constraints

$$(W_{c_i} - W_c) \cdot \varphi(u_i) + (b_{c_i} - b_c) \geq \varepsilon$$

for every such pair is feasible.

Proof. By Definition 55, $\bigcap_{p\in l} p$ is the set of all (W, b) satisfying the displayed strict inequalities. So $l \in L_{\mathfrak{v}_{\varphi,Y}}$ if and only if that strict system is feasible. The standard positive-margin reformulation is equivalent because the system is finite. □

In this unconstrained affine-head construction, the finite strict system is also equivalent to the system with $\varepsilon = 1$: a strictly feasible head with minimum gap $\varepsilon_0 > 0$ rescales to $(W/\varepsilon_0, b/\varepsilon_0)$.

Corollary 30 (Architecture-independence on finite support)**.**
Let φ_1 and φ_2 be two transformer feature maps on the same finite support $\mathcal{U}$. Assume they are paired with the same admissible label family $\{C_u\}_{u\in\mathcal{U}}$ and hence the same label universe $Y = \bigcup_{u\in\mathcal{U}} C_u$. If $\varphi_1(u) = \varphi_2(u)$ for every $u \in \mathcal{U}$, then

$$\mathfrak{v}_{\varphi_1,Y} = \mathfrak{v}_{\varphi_2,Y}.$$

Hence they induce the same satisfiable language and the same correct-policy sets for any task posed on that support. In particular, for any fixed choice of training units, probe units, and training labels on that support, they induce the same operational proxies defined below.

Proof. For each unit u and admissible label $c \in C_u$, the program $p_{u,c}$ is defined by inequalities against labels in the same set $C_u \setminus \{c\}$. So $p_{u,c}$ depends only on the realised feature value $\varphi(u)$ and the admissible label family $\{C_u\}$ on the chosen support. If those coincide for φ_1 and φ_2, then every program $p_{u,c}$ coincides, and therefore the two vocabularies are equal. □

Definition 56 (Transformer support concept)**.**
Fix a $\mathfrak{v}_{\varphi,Y}$-task α in the existing Stack Theory sense. Two correct policies $\pi_1, \pi_2 \in \Pi_\alpha$ are **concept-equivalent** when

$$\mathrm{Ext}(\pi_1) = \mathrm{Ext}(\pi_2).$$

A **transformer support concept** is an equivalence class

$$[\pi]_{\mathrm{Ext}} := \{\pi' \in \Pi_\alpha : \mathrm{Ext}(\pi') = \mathrm{Ext}(\pi)\}$$

of correct policies under extension equality. This relation is distinct from equal freedom in Definition 4. The support concept's weakness is

$$\mathrm{ff}([\pi]_{\mathrm{Ext}}) := |\,\mathrm{Ext}(\pi)|,$$

which is well-defined because all representatives have the same extension.

Theorem 29 (Exact transformer concept optimality)**.**
Fix a finite support $\mathcal{U}$, a feature map φ, and its vocabulary $\mathfrak{v}_{\varphi,Y}$. Let α be a task in this vocabulary with $\Pi_\alpha \neq \emptyset$, and let P_α be a distribution of future demands on

$$U := L_{\mathfrak{v}_{\varphi,Y}} \setminus \mathrm{Ext}(I_\alpha).$$

For $\pi \in \Pi_\alpha$, $B_\pi := \mathrm{Ext}(\pi) \cap U$ and $G_{P_\alpha}(\pi) = P_\alpha(S \subseteq B_\pi)$ as in Definition 17. If P_α is exchangeable, every correct policy of maximal freedom maximises G_{P_α}. If $P_\alpha(|S| = 1) > 0$, every policy maximising G_{P_α} has maximal freedom. When $I_\alpha \neq \emptyset$, policy R_α of Lemma 7 maximises G_{P_α} under every distribution of demands.

Proof. The vocabulary is finite by Definition 55. Every correct policy satisfies $\mathrm{Ext}(\pi) = O_\alpha \dot\cup B_\pi$, so freedom and buffer size induce the same ordering. Theorem 2 proves the exchangeable result and its strict case. Corollary 6 proves the final claim. □

Corollary 31 (Concept-level formulation)**.**
Under the exchangeable-demand assumptions of Theorem 29, every support concept $[\pi]_{\mathrm{Ext}}$ with maximal freedom is generalisation-optimal. If $P_\alpha(|S| = 1) > 0$, optimality holds if and only if its freedom is maximal.

Proof. All representatives share an extension, so they share a buffer, freedom and generalisation probability. Theorem 29 therefore applies to support concepts. □

Corollary 32 (Restricted next-token LLM corollary)**.**
Take the evaluation units to be prefixes $u = x_{<t}$, and let each C_u be a finite candidate next-token set. Under the assumptions of Theorem 29, any language model that selects a $<_{\mathrm{ff}}$-maximal correct policy in $\mathfrak{v}_{\varphi,Y}$ learns an optimally generalisable next-token concept on that finite support.

Proof. This is Theorem 29 specialised to prefix units. □

Remark 7 (What remains open)**.**
Full freedom requires joint feasibility across completions. The proxies below evaluate one or two new assignments at a time. Joint-completion measurements and current experiments appear in Sections 24 and 28.3.

Definition 57 (Training policy and pointwise extension counts)**.**
Fix labelled observed units

$$\mathcal{U}_{\mathrm{train}} \subseteq \mathcal{U}, \qquad y(u) \in C_u \text{ for each } u \in \mathcal{U}_{\mathrm{train}}.$$

Define the **training policy**

$$\pi_{\mathrm{train}} := \{p_{u,y(u)} : u \in \mathcal{U}_{\mathrm{train}}\}.$$

Assume $\pi_{\mathrm{train}} \in L_{\mathfrak{v}_{\varphi,Y}}$. Let

$$\mathcal{U}_{\mathrm{probe}} := \mathcal{U} \setminus \mathcal{U}_{\mathrm{train}}.$$

For each probe unit $u \in \mathcal{U}_{\mathrm{probe}}$, define the **pointwise extension count**

$$\mathrm{ext}(u) := \left|\{c \in C_u : \pi_{\mathrm{train}} \cup \{p_{u,c}\} \in L_{\mathfrak{v}_{\varphi,Y}}\}\right|.$$

Define the **transformer pair proxy**

$$\mathrm{PP}(\varphi) := \sum_{u \in \mathcal{U}_{\mathrm{probe}}} \mathrm{ext}(u).$$

Define the **normalised transformer pair proxy**

$$\widehat{\mathrm{PP}}(\varphi) := \frac{1}{|\mathcal{U}_{\mathrm{probe}}|} \sum_{u \in \mathcal{U}_{\mathrm{probe}}} \frac{\mathrm{ext}(u)}{|C_u|}.$$

Proposition 51 (Exact single-point reduction)**.**
Fix φ, $\mathcal{U}$, Y, and a training policy $\pi_{\mathrm{train}} \in L_{\mathfrak{v}_{\varphi,Y}}$.

1. For each probe unit $u \in \mathcal{U}_{\mathrm{probe}}$, $\mathrm{ext}(u)$ equals the number of labels $c \in C_u$ for which $p_{u,c}$ appears in some completion $l \in \mathrm{Ext}(\pi_{\mathrm{train}})$.
2. $\mathrm{PP}(\varphi)$ is computable by solving $\sum_{u \in \mathcal{U}_{\mathrm{probe}}} |C_u|$ LP feasibility problems.
3. Every $l \in L_{\mathfrak{v}_{\varphi,Y}}$ is realised by a single shared parameter vector (W, b).

Proof. Part (1) is immediate from the definition of extension and the fact that consistency is inherited by subsets. Part (2) follows from Proposition 50. Part (3) is just the meaning of $L_{\mathfrak{v}_{\varphi,Y}}$ in this vocabulary. □

Definition 58 (Two-point proxy)**.**
Fix a finite set of unordered probe pairs

$$\mathcal{P}_2 \subseteq \{\{u, v\} : u, v \in \mathcal{U}_{\text{probe}},\ u \neq v\}.$$

For each pair $\{u, v\} \in \mathcal{P}_2$, define

$$\text{ext}^{(2)}(u, v) := \big|\{(c, d) \in C_u \times C_v : \pi_{\text{train}} \cup \{p_{u,c}, p_{v,d}\} \in L_{\mathfrak{v}_{\varphi,Y}}\}\big|.$$

The **sampled two-point proxy** is

$$\text{PP}^{(2)}(\varphi) := \frac{1}{|\mathcal{P}_2|} \sum_{\{u,v\} \in \mathcal{P}_2} \frac{\text{ext}^{(2)}(u, v)}{|C_u||C_v|}.$$

Proposition 52 (Affine invariance)**.**
Let $A \in \mathbb{R}^{d \times d}$ be invertible, let $a \in \mathbb{R}^d$, and define $\varphi'(u) := A\varphi(u) + a$ for every $u \in \mathcal{U}$. Then

$$\text{PP}(\varphi') = \text{PP}(\varphi) \qquad \text{and} \qquad \text{PP}^{(2)}(\varphi') = \text{PP}^{(2)}(\varphi).$$

Proof. Define

$$\Psi(W, b) := (WA^{-1},\ b - WA^{-1}a).$$

Its inverse is $\Psi^{-1}(W', b') = (W'A,\ b' + W'a)$, so it is a bijection between readout parameter spaces. For every unit u and every class c,

$$(WA^{-1})_c \cdot (A\varphi(u) + a) + \big(b - WA^{-1}a\big)_c = W_c \cdot \varphi(u) + b_c.$$

So a class assignment is feasible for φ if and only if the corresponding class assignment is feasible for φ'. The feasible one-point and two-point assignment sets therefore agree exactly, which proves the two proxy equalities. □

Remark 8 (Relation to exact weakness)**.**
The proxies above are not the full extension cardinality $|\text{Ext}(\pi_{\text{train}})|$. They are tractable summaries based on counts of feasible assignments at one or two probe units at a time. That is deliberate. The feature-classifier vocabulary gives exact single-point extension counts, but computing the full joint extension remains open Bennett [2026c]. The transformer case should therefore be framed as a staged operationalisation, not as a completed measurement theorem.

26 Embodied reinforcement learning

Definition 59 (EGRL)**.**
Fix a vocabulary $\mathfrak{v}$, a reward predicate $r : L_{\mathfrak{v}} \to \{1, 0\}$, and a finite horizon $len \in \mathbb{N}$ with $len > 0$.

The reward predicate induces a $\mathfrak{v}$-task $\mu = \langle I_\mu, O_\mu \rangle \in \Gamma_{\mathfrak{v}}$ by

$$O_\mu := \{\, l \in L_{\mathfrak{v}} \mid r(l) = 1 \,\}, \qquad I_\mu := \{\, i \in L_{\mathfrak{v}} \mid \exists o \in O_\mu\ (i \subseteq o) \,\}.$$

Assume $\Pi_\mu \neq \emptyset$.

I model interaction up to time t using a history task $\mathfrak{h}_{<t} \in \Gamma_{\mathfrak{v}}$. Intuitively, $I_{\mathfrak{h}_{<t}}$ are the inputs encountered so far and $O_{\mathfrak{h}_{<t}}$ are the outputs produced so far.

Given $\mathfrak{h}_{<t}$, I split it into a positive and negative subhistory by applying r to past outputs. Define

$$O_{\mathfrak{h}^1_{<t}} := \{\, o \in O_{\mathfrak{h}_{<t}} \mid r(o) = 1 \,\}, \qquad O_{\mathfrak{h}^0_{<t}} := O_{\mathfrak{h}_{<t}} \setminus O_{\mathfrak{h}^1_{<t}}.$$

For $b \in \{0, 1\}$, let

$$I_{\mathfrak{h}^b_{<t}} := \{\, i \in I_{\mathfrak{h}_{<t}} \mid \exists o \in O_{\mathfrak{h}^b_{<t}} (i \subseteq o) \,\}, \qquad \mathfrak{h}^b_{<t} := \langle I_{\mathfrak{h}^b_{<t}}, O_{\mathfrak{h}^b_{<t}} \rangle.$$

A **pair proxy** $<^{\mathfrak{pp}}$ is a binary relation on the set of admissible policy pairs

$$\mathfrak{pp} \;:=\; \{(m,n) \in \Pi_{\mathfrak{h}^1_{<t}} \times \Pi_{\mathfrak{h}^0_{<t}} \mid \mathrm{Ext}(m) \cap \mathrm{Ext}(n) = \emptyset\}.$$

Define the unseen region of the current history by

$$U_{\mathfrak{h}_{<t}} \;:=\; L_{\mathfrak{v}} \setminus \mathrm{Ext}(I_{\mathfrak{h}_{<t}}).$$

For such pairs, **learning** means choosing $(\pi, n) \in \mathfrak{pp}$ that is maximal under $<^{\mathfrak{pp}}$. Given a current input $i_t \in I_\mu$, **inference** means choosing an output $o_t \in \mathrm{Ext}(i_t) \cap \mathrm{Ext}(\pi)$.

Given $(\pi, n), (j, k) \in \mathfrak{pp}$, the **freedom pair proxy** is

$$(j,k) <^{\mathfrak{pp}}_{\mathrm{ff}} (\pi,n) \quad \text{iff} \quad |(\mathrm{Ext}(j) \cup \mathrm{Ext}(k)) \cap U_{\mathfrak{h}_{<t}}| < |(\mathrm{Ext}(\pi) \cup \mathrm{Ext}(n)) \cap U_{\mathfrak{h}_{<t}}|\,.$$

Proposition 53 (Weakness pair proxy optimality)**.**
Let $\mathfrak{h}_{<t}$ be a history-task and let $\mathfrak{h}^1_{<t}$ and $\mathfrak{h}^0_{<t}$ denote the induced positive and negative history-tasks (as in Definition 59). Consider the admissible set of policy pairs

$$\mathfrak{pp} \;:=\; \{(\pi,n) \in \Pi_{\mathfrak{h}^1_{<t}} \times \Pi_{\mathfrak{h}^0_{<t}} \mid \mathrm{Ext}(\pi) \cap \mathrm{Ext}(n) = \varnothing\}.$$

Define the *effective completion set* of a pair as $E_{\pi,n} := \mathrm{Ext}(\pi) \cup \mathrm{Ext}(n)$, and write

$$U_{\mathfrak{h}_{<t}} \;:=\; L_{\mathfrak{v}} \setminus \mathrm{Ext}(I_{\mathfrak{h}_{<t}})$$

for the unseen region of the current history. Under the same maximally uninformative extension model as in Proposition 8, apply the model to the effective completion set $E_{\pi,n}$. Then the probability that a future demand set lies in $E_{\pi,n} \cap U_{\mathfrak{h}_{<t}}$ is maximised by choosing $(\pi, n) \in \mathfrak{pp}$ with maximal $|E_{\pi,n} \cap U_{\mathfrak{h}_{<t}}|$. Equivalently, the freedom pair proxy is optimal under that model. If, in addition, $(E_{\pi,n} \cap \mathrm{Ext}(I_{\mathfrak{h}_{<t}}))$ is the same set for every admissible pair, then maximising $|E_{\pi,n} \cap U_{\mathfrak{h}_{<t}}|$ is equivalent to maximising $|E_{\pi,n}|$.

Proof. The proof is the same counting argument as in Proposition 8, with $\mathrm{Ext}(\pi)$ replaced by the effective completion set $E_{\pi,n} := \mathrm{Ext}(\pi) \cup \mathrm{Ext}(n)$.

Under the maximally uninformative model, the parent selects an additional set $S \subseteq U_{\mathfrak{h}_{<t}}$ uniformly at random. An admissible pair (π, n) is consistent with that parent extension precisely when all selected unseen outputs lie inside the effective completion set, i.e., when $S \subseteq E_{\pi,n} \cap U_{\mathfrak{h}_{<t}}$. Therefore,

$$\mathbb{P}\big((\pi,n) \text{ generalises} \mid \mathfrak{h}_{<t}\big) \;=\; \frac{2^{|E_{\pi,n} \cap U_{\mathfrak{h}_{<t}}|}}{2^{|U_{\mathfrak{h}_{<t}}|}},$$

which is strictly increasing in $|E_{\pi,n} \cap U_{\mathfrak{h}_{<t}}|$. That proves optimality of the freedom pair proxy.

For the final equivalence, suppose the seen component $E_{\pi,n} \cap \mathrm{Ext}(I_{\mathfrak{h}_{<t}})$ is the same set for every admissible pair. Then

$$|E_{\pi,n}| = |E_{\pi,n} \cap \mathrm{Ext}(I_{\mathfrak{h}_{<t}})| + |E_{\pi,n} \cap U_{\mathfrak{h}_{<t}}|$$

differs across admissible pairs only through the unseen term. So maximising $|E_{\pi,n} \cap U_{\mathfrak{h}_{<t}}|$ is equivalent to maximising $|E_{\pi,n}|$. □

The algorithms below assume nonempty sets wherever a choice is taken. A failed search returns $\emptyset$ for unconstrained exploration.

Algorithm 1 EGRL system (schematic)

```
If S is a set, then S.Choose() returns an element of S.
If S is a set, then S.Add(x) := S ∪ {x}.
function EGRL_Loop(μ, <^pp, r, len)
    t ← 0
    r_{h<t} ← 0
    h_{<t} ← ⟨∅, ∅⟩
    i_t ← I_μ.Choose()
    while t < len do
        π ← learning(h_{<t}, <^pp, i_t, r)          ▷ see learning methods
        o_t ← inference(π, i_t)
        r_{h<t} ← r_{h<t} + r(o_t)                  ▷ feedback
        I_{h<t} ← I_{h<t}.Add(i_t)
        O_{h<t} ← O_{h<t}.Add(o_t)
        t ← t + 1
        i_t ← I_μ.Choose()
    end while
    return r_{h<t}
end function

function Inference(π, i_t)
    E ← Ext(π) ∩ Ext(i_t)
    return E.Choose()
end function
```

27 Physical quantities

A chosen physical model restricts $\mathcal{C}_{\tau,t}$ and assigns measured durations to its ordered changes. Durations need not be constant. The control volume $\mathcal{R}$ is bounded, with clock readings $a < b$. Over $[a, b]$, with energy in joules and $\Delta Y := Y(b) - Y(a)$,

$$\Delta E_{\mathcal{R}} = Q_{\text{in}} - Q_{\text{out}} + W_{\text{in}} - W_{\text{out}} + E_{\text{mat,in}} - E_{\text{mat,out}}, \qquad E_{\text{draw}} = Q_{\text{in}} + W_{\text{in}} + E_{\text{mat,in}}.$$

Here stored energy comprises internal, kinetic and potential energy, excluding rest energy; W is non-flow work, and matter-stream energy includes enthalpy (and thus flow work), kinetic and potential energy. Average power is energy transferred divided by $b - a$. For each conserved constituent,

$$\Delta m_i = m_{i,\text{in}} - m_{i,\text{out}}.$$

Species transformed by reactions require the corresponding stoichiometric balances. The reference environment has fixed temperature $\Theta_0 > 0$, pressure and composition. With exergy Ex (maximum work excluding ambient displacement), entropy S and generated entropy $S_{\text{gen}} \geq 0$,

$$\Delta \text{Ex}_{\mathcal{R}} = \text{Ex}_{\text{in}} - \text{Ex}_{\text{out}} - \Theta_0 S_{\text{gen}}, \qquad \Delta S_{\mathcal{R}} = S_{\text{in}} - S_{\text{out}} + S_{\text{gen}}.$$

Exergy is in joules and entropy in joules per kelvin. Gross input E_{draw} and irreversible exergy loss $\Theta_0 S_{\text{gen}}$ are different quantities. Admissible continuations must satisfy these balances and the stipulated power, storage, heat and material bounds. For a physical observable $Y(c; J_t)$, its expectation conditional on persistence is

$$\mathbb{E}_{c \sim P_{\mathcal{C}}}[Y(c; J_t) \mid \rho_{c,J_t}(\pi) = |J_t|],$$

provided the conditioning event has positive probability. For equiprobable microstates under a fixed coarse-graining, $S_{\text{phys}} = k_B \ln N_{\text{micro}}$; generally $S_{\text{phys}} = -k_B \sum_i p_i \ln p_i$. The identity

Algorithm 2 LEARNING METHODS

For a $\mathfrak{v}$-task α, $\alpha.expand() := \{\omega \in \Gamma_{\mathfrak{v}} : \omega \sqsubset \alpha, \neg\exists\zeta \in \Gamma_{\mathfrak{v}}(\omega \sqsubset \zeta \sqsubset \alpha)\}$ gets the set of all α's immediate children (i.e. children which are only one generation away from α, meaning I exclude grandchildren etc)

$\mathfrak{q}$ is a prioritised queue. If (n, obj) is in $\mathfrak{q}$, then n is inverse priority (smaller n is higher priority). $\mathfrak{q}.put(n, obj)$ places (n, obj) in the queue, and $\mathfrak{q}.get()$ returns the highest priority item. $\mathfrak{q}.notempty()$ returns true iff $\mathfrak{q}$ is not empty.

If $\mathcal{V}$ is a set then $\mathcal{V}.add(\mathfrak{c}) := \mathcal{V} \cup \{\mathfrak{c}\}$.

function CLASSIFICATION($\mathfrak{h}_{<t}$, r) ▷ used by LEARNING()

  $O_{\mathfrak{h}^1_{<t}} \leftarrow \{o \in O_{\mathfrak{h}_{<t}} : r(o) = 1\}$ ▷ relatively "attractive" memories

  $O_{\mathfrak{h}^0_{<t}} \leftarrow O_{\mathfrak{h}_{<t}} - O_{\mathfrak{h}^1_{<t}}$

  $I_{\mathfrak{h}^1_{<t}} \leftarrow \{i \in I_{\mathfrak{h}_{<t}} : \exists o \in O_{\mathfrak{h}^1_{<t}}(i \subseteq o)\}$

  $I_{\mathfrak{h}^0_{<t}} \leftarrow \{i \in I_{\mathfrak{h}_{<t}} : \exists o \in O_{\mathfrak{h}^0_{<t}}(i \subseteq o)\}$

  $\mathfrak{h}^0_{<t} \leftarrow \langle I_{\mathfrak{h}^0_{<t}}, O_{\mathfrak{h}^0_{<t}}\rangle$

  $\mathfrak{h}^1_{<t} \leftarrow \langle I_{\mathfrak{h}^1_{<t}}, O_{\mathfrak{h}^1_{<t}}\rangle$

  return $\mathfrak{h}^1_{<t}, \mathfrak{h}^0_{<t}$

end function

Example 1 : A simple example showing how learning works without noise.

function LEARNING($\mathfrak{h}_{<t}$, $<^{\mathfrak{pp}}$, i_t, r)

  $\mathfrak{h}^1_{<t}, \mathfrak{h}^0_{<t}$ = CLASSIFICATION($\mathfrak{h}_{<t}, r$)

  $N_{\mathfrak{h}^1_{<t}} \leftarrow \{j \in \Pi_{\mathfrak{h}^1_{<t}} : \mathrm{Ext}(j) \cap \mathrm{Ext}(i_t) \neq \emptyset\}$ ▷ discard policies that don't apply

  $\Pi_{\mathfrak{h}^{(1,0)}_{<t}} \leftarrow \{(j,k) \in N_{\mathfrak{h}^1_{<t}} \times \Pi_{\mathfrak{h}^0_{<t}} : \mathrm{Ext}(j) \cap \mathrm{Ext}(k) = \emptyset\}$ ▷ remove ambiguity

  return π s.t. $\exists(\pi, n) \in \Pi_{\mathfrak{h}^{(1,0)}_{<t}}$ that maximises $<^{\mathfrak{pp}}$

end function

Algorithm 3 LEARNING METHODS CONTINUED...

Example 2 : Learning is possible even with noise (the only drawback being that it is harder to understand) by selectively "forgetting" outputs when there is no valid policy. Given only noise, there would be a policy for each output (rote learning means $|\,\mathrm{Ext}(\pi)\,| \to 1$*).*

function LEARNING($\mathfrak{h}_{<t}$, $<^{\mathfrak{pp}}$, i_t, r)
  $\mathfrak{q} \leftarrow \mathfrak{q}.put(0, \mathfrak{h}_{<t})$
  $\mathcal{V} \leftarrow \{\mathfrak{h}_{<t}\}$ ▷ set of visited nodes
  while $\mathfrak{q}.notempty()$ **do**
    $n, \mathfrak{h} \leftarrow \mathfrak{q}.get()$
    $\mathfrak{h}^1_{<t}, \mathfrak{h}^0_{<t} =$ CLASSIFICATION$(\mathfrak{h}, r)$
    $N_{\mathfrak{h}^1_{<t}} \leftarrow \{j \in \Pi_{\mathfrak{h}^1_{<t}} : \mathrm{Ext}(j) \cap \mathrm{Ext}(i_t) \neq \emptyset\}$
    $\Pi_{\mathfrak{h}^{(1,0)}_{<t}} \leftarrow \{(j,k) \in N_{\mathfrak{h}^1_{<t}} \times \Pi_{\mathfrak{h}^0_{<t}} : \mathrm{Ext}(j) \cap \mathrm{Ext}(k) = \emptyset\}$
    if $\Pi_{\mathfrak{h}^{(1,0)}_{<t}} \neq \emptyset$ **then**
      return π s.t. $\exists(\pi, n) \in \Pi_{\mathfrak{h}^{(1,0)}_{<t}}$ that maximises $<^{\mathfrak{pp}}$
    end if
    for all $\mathfrak{c} \in \mathfrak{h}.expand()$ **do**
      if $\mathfrak{c} \notin \mathcal{V}$ **then**
        $\mathfrak{q}.put(n+1, \mathfrak{c})$
        $\mathcal{V} \leftarrow \mathcal{V}.add(\mathfrak{c})$
      end if
    end for
  end while
  return $\emptyset$ ▷ try uninformed exploration when there is no valid policy
end function

$S_{\text{phys}} = k_B \ln |\operatorname{Ext}(\pi)|$ requires the physical model to identify completions with those equiprobable microstates. Compatibility $S \subseteq B_\pi$ imposes no additional energy expenditure by itself; action, maintenance and repair costs belong to the physical model.

28 Experimental results

Table 4: Additional reported finite-world and controlled-policy results. These are empirical illustrations, not further theorem hypotheses.

Study	Reported outcome and scope
Controlled classifiers	Ten benchmark families, 48 constructed child-correct policies each: Spearman correlations 0.541–0.883, pooled within-benchmark $\rho = 0.725$; freest selection exceeds the 48-policy mean in every family. Responses beyond the child were deliberately constructed [Bennett, 2026f].
Prediction with memory	Freedom-guided selection reduces normalised regret by 58% for delayed cues and 69% for longer delays relative to selection by trained-task regret [Bennett, 2026b].
Causal selves	Finite-world setting: freedom regret 0, simplicity 13.042 bits. Under Beta$(0.7, 2)$: 0.225 versus 7.274 bits, with optimal choices in 92.3% versus 69.7% of cases [Bennett, 2026a].
Audience decoders	With two probes allowed, stationary-decoder success is 0.185 without probing, 0.926 with probing and 0.961 for an oracle. With episode-varying decoders: 0.282, 0.928 and 0.959 respectively [Bennett, 2026a].
Trust and commitment	At horizon one, silence/nonbinding trust 0 versus binding 0.442; binding reaches 0.950 at horizon five. Informative promises give 0.633 at horizon one and binding 0.750 at horizon two [Bennett, 2026a].
Alignment ablations	In the matched constructions, removing the relevant self caps success at 1/2; the intact construction attains 1 [Bennett, 2026a,g].
Collective restriction	Three-part synthetic collectives: cancer rates 94.0% under maximal restriction, 80.8% under random selection and 59.9% under freest selection. Surveillance plus termination reduces persistent cancer from 79.1% to 7.4% [Bennett, 2026h].
Collective lineages	Twelve generations, controls succeeding with probability 0.8: unseen survival rises from 0.3% to 95.4%, with 2.19 inherited changes on average. Survival requires every part viable and a common policy; transient cancer still occurs in almost every controlled lineage [Bennett, 2026h].
Finite novelty	Freedom selection gains 1.674 bits in mean base-two log survival over random viable choice under sampled novelty demands.

28.1 Neural freedom

The frozen-feature cohorts use 100 trained networks per dataset, 250 labelled child examples, 32 anchors and 128 probes. The common demand bank has 128 independent paths, each comprising 16 distinct probe inputs with uniformly sampled hypothetical labels; prefixes of sizes 2, 4, 8 and 16 give 512 indexed demands. Prefixes within a path are dependent; paths are the independent units. Settings were fixed without calibration against accuracy. Unresolved feasibility checks give lower and upper pass fractions; these coincide in the reported cohorts. Table 6 reports the joint score, controls and survival certificates [Bennett, 2026c]. Certificates concern expected path score under this design, not test accuracy. The data-scale experiment varies batch size and learning rate together; the large-batch accuracy advantage peaks at 2,000 examples and shrinks thereafter, while its Hessian trace is lower at every scale. Function-preserving rescaling between adjacent ReLU layers changes raw Hessian trace by up to 99×, showing that raw trace is not determined by the input–output function. This does not refute curvature measures adjusted for

recoding. On Fashion-MNIST, paired intervals resolve a stronger accuracy correlation for joint completion score than for raw trace and final-layer relative flatness; no other paired difference is resolved. Random-label networks have higher mean joint score than natural-label MNIST networks without a resolved accuracy association, so these scores provide no universal cross-task quality scale [Bennett, 2026c]. For the reported survival certificates, $n = 128$ independent paths each score the mean feasibility of four nested prefixes. A uniform prior over 100 frozen networks is chosen independently of path sampling; selection of one network gives a point posterior with $\mathrm{KL} = \log 100$. Theorem 8 with $\delta = 0.025$ per cohort gives the two expected path-score bounds in Table 6, jointly at confidence at least 0.95.

Table 5: The large-batch test-accuracy advantage on MNIST, in percentage points, at the reported training-set sizes.

Training examples	Large-batch advantage
2,000	1.6
24,000	0.08
60,000	0.02

Table 6: Joint completion score in matched 100-network cohorts. Correlation ρ is with test accuracy, with its 95% interval and Holm-corrected p-value q. Reliability is split-half Spearman–Brown reliability. Margin ρ is spectral-product margin's correlation with test accuracy. Mean score is the mean joint completion score. Bound is the PAC-Bayes lower bound on expected path score, and the two bounds hold jointly at confidence at least 0.95.

Cohort	ρ	95% interval	q	Reliability	Margin ρ	Mean score	Bound
MNIST	0.290	$[0.094, 0.468]$	0.004	0.940	0.504	0.287	0.241
Fashion-MNIST	0.468	$[0.301, 0.607]$	4.00×10^{-5}	0.956	0.546		0.269
Random labels	0.058		0.567			0.420	

28.2 Freedom-guided XM

For finite nonempty contexts C and finite output set Y, let Φ_{gen} contain all permission profiles $F : C \to 2^Y \setminus \{\emptyset\}$. The exclusion vocabulary and its policy are

$$e_{c,y} := \{G \in \Phi_{\mathrm{gen}} : y \notin G(c)\}, \qquad \mathfrak{v}_{\mathrm{perm}} := \{e_{c,y} : c \in C, y \in Y\}, \qquad \pi_F := \{e_{c,y} : y \notin F(c)\}.$$

Writing $a_c = |F(c)|$, each completion retains a nonempty subset of $F(c)$ independently at each context, whence

$$\mathrm{ff}(\pi_F) = \prod_{c \in C} (2^{a_c} - 1).$$

The measured log freedom uses estimated output support; K instead counts sampled training candidates [Bennett, 2026e]. XM control-efficiency results are from Gladstone et al. [2026]. The freedom-guided study [Bennett, 2026e] uses 144 jobs in Experiment 1 (six K values, two target functions, twelve repetitions); best-of-eight hit rate peaks at $K = 8$ despite increasing measured freedom. Experiment 2 uses thirty worlds, 48 candidates per world and a separately calibrated selector with child-fit threshold 0.970. Its primary comparison permits different K; the fixed-setting comparison is post hoc. Experiment 3 pairs standard and freedom-guided ImageNet training at $K = 25$, matched architecture, data order, seed, optimiser, sampler and 200,000 updates; evaluation uses 50,000 images. Selection-rule calibration precedes the paired final runs. Tables 8–10 report the outcomes and calibration.

Table 7: Efficiency gains reported for XM on control tasks.

Quantity	Reported gain
Data efficiency	6.2×
FLOP efficiency	4.1×
Parameter efficiency	47%
Inference steps	up to 256× fewer

Table 8: Mean log freedom of the induced permission profile in XM Experiment 1, under a context-dependent target function, at the reported values of K.

K	Mean log freedom
1	approximately 0
8	5.03
32	8.21

Table 9: Outcomes of XM Experiments 2 and 3. The baseline is child-validation selection in Experiment 2 and standard XM in Experiment 3. Image metrics are from the final evaluation at classifier-free guidance 1.5, meaning sampling with a strengthened class condition.

Experiment	Quantity	Baseline	Freedom
2	Balanced-parent hit rate	0.3169	0.3892
2	K chosen, and in how many of the 30 worlds	8, in all 30	16, in 29
2	Worlds won by freedom		29 of 30
2	Paired sign-flip p-value		1.86×10^{-9}
2	Post hoc worlds with positive gain		30 of 30
2	Post hoc mean gain		0.00646
3	FID	46.9803	43.7753
3	Inception Score	18.17	19.02
3	Recall	0.214	0.239
3	Precision	0.586	0.562
3	Mean eligible-set size		3.76
3	Added mean reconstruction loss		0.00559

Table 10: Calibration stages of the ImageNet candidate-choice rule in XM Experiment 3.

Stage	Formulas	Updates	Seed
1	12	5,000	33
2	4	10,000	44

28.3 Learned-vocabulary selection

The learned-vocabulary study runs twelve trials per benchmark, with 48 initially trained networks and up to twenty selection generations. Admission requires perfect training classification with positive margin; four parents each produce twelve offspring inheriting specifications, not weights. Calibration uses four-fold splits of the trial training samples, then fixes one rule per benchmark; it adds no separate labels but reuses labels across final trials [Bennett, 2026f, 2025c]. Measurement freezes features and jointly constrains one replacement head. Scores are centred and divided by a common standard deviation. Training, label-anchor and requested probe classes satisfy $z_y - z_k \geq 10^{-5}$. On reference inputs,

$$\left|(z_{\hat{y}_f} - z_k) - (z_{f,\hat{y}_f} - z_{f,k})\right| \leq 0.35.$$

Score anchors instead satisfy $|(z_{\hat{y}_f} - z_k) - g_0| \leq 10^{-6} + 0.025|g_0|$, where g_0 is the trained gap; near zero, this may permit a reversal. There is no separate weight bound. For $N = 256$ checks in one setting, with verified passes n_+, failures n_- and unresolved outcomes $n_?$, use

$$q = \frac{n_+}{n_+ + n_-}, \qquad q_{\text{low}} = \frac{n_+}{N}, \qquad q_{\text{high}} = \frac{n_+ + n_?}{N},$$

when $n_+ + n_- > 0$. These are unresolved-outcome bounds, not sampling-confidence bounds. The calibrated selection rule is

$$V = q + c \min\{q, 1-q\} \tanh\left(\sum_j a_j \frac{e_j - m_j}{\sigma_j}\right),$$

with fixed calibration centres m_j, scales $\sigma_j > 0$, weights a_j and cap c. Tables 18 and 19 give the retained rules. The marginal-likelihood comparator uses the fitted last-layer Laplace score [Immer et al., 2021]

$$\mathbf{A} = \mathbf{X}^\top \mathbf{X}/n, \quad \mathbf{B} = \left[\operatorname{diag}(\textstyle\sum_i \mathbf{P}_{i,:}) - \mathbf{P}^\top \mathbf{P}\right]/n, \quad \mathbf{H} = n(\mathbf{B} \otimes \mathbf{A}),$$

$$E(\xi) = -\ell(\theta) - \tfrac{\xi}{2}\|\theta\|^2 + \tfrac{r}{2}\log\xi - \tfrac{1}{2}\sum_{j=1}^{r} \log(h_j + \xi),$$

maximised over 49 logarithmically spaced $\xi \in [10^{-6}, 10^{6}]$. Here $\mathbf{X}$ includes the bias coordinate, $\mathbf{P}$ contains training softmax probabilities, $r = K(d+1)$, h_j are eigenvalues of $\mathbf{H}$, and ℓ is training negative log likelihood. Parameters are not refitted for each precision. Evidence is invariant if the prior transforms with coordinates; general affine feature recoding takes this isotropic prior outside its chosen family. GdScore uses unlabelled official test inputs, pseudo-labels $\mathbf{Y}$ and feature matrix $\mathbf{X}$ without bias [Xie et al., 2025]:

$$\mathbf{G} = (\mathbf{P} - \mathbf{Y})^\top \mathbf{X}/m, \qquad \|\mathbf{G}\|_{0.3} = \Big(\sum_{j,k} |\mathbf{G}_{jk}|^{0.3}\Big)^{1/0.3}.$$

Smaller batch-averaged values are preferred. For $c > 0$, $(\mathbf{X}, W_f) \mapsto (c\mathbf{X}, W_f/c)$ preserves scores and pseudo-labels but multiplies this gradient score by c, whereas the freedom checks remain unchanged. These are descriptive paired results from an under-review study. Shared samples, calibration and reused test sets do not supply independent trajectories. Some arms differ in hardware or library versions; matching seeds and specifications does not imply identical initial networks. Generation-matched comparisons use each competitor's stopping generation. Mean final accuracy exceeds both comparators on all five benchmarks, while higher measured freedom need not coincide with higher accuracy at every generation; a once-selected model using 512 additional validation labels wins on three benchmarks. Applying freest-correct optimality across networks additionally requires compatible commitments and the task construction.

Table 11: Official test accuracy in percent, as means over twelve matched trials per benchmark, with sample standard deviations in parentheses. Random is mean accuracy of all 48 networks in freedom's generation zero. Other columns give accuracy of each criterion's final pick.

Benchmark	Random	Freedom	Marginal likelihood	GdScore
KMNIST	44.28 (2.49)	47.02 (3.60)	46.86 (3.20)	32.24 (2.95)
Fashion-MNIST binary	78.23 (1.64)	78.80 (1.57)	78.13 (2.37)	78.14 (1.91)
Rotten Tomatoes	51.85 (0.50)	55.99 (1.54)	53.82 (2.05)	55.46 (1.88)
AG News binary	50.24 (0.19)	59.84 (2.98)	50.00 (0.01)	58.23 (3.34)
20NG binary	60.65 (1.07)	67.22 (2.39)	66.56 (3.83)	66.72 (2.85)

Table 12: Initial-population mean accuracy in percent, per arm, on the two benchmarks whose arms differ.

Benchmark	Freedom arm	Competitor arm
KMNIST	44.2798	44.2107
20NG binary	60.6526	60.6629

28.4 Persistence and repair

Each update applies background damage, attempts a task with further possible damage, adds resource for a completed task, and restores disrupted programs at one resource unit each while resource remains (this "resource" distinction is for illustration. All resources amount to freedom of function, and all resource gathering or learning is the sacrifice of present freedom of function for future freedom of function). Viability is checked after repair.

In the twelve-program simulation, a demand requires three uniformly chosen programs, giving success probability $\binom{n}{3}/\binom{12}{3}$ when n are intact. Under independent damage across programs, steps and cycles, independent of demand choice,

$$d = 1 - (1 - 0.02)(1 - 0.05) = 0.069, \qquad \mathbb{E}[C_n] = 12d = 0.828,$$
$$\gamma \geq 1.6(1 - d)^3 - 12d \approx 0.463 > 0.$$

The bound counts success only when all requested programs survive both damage steps. Repair requirement $a = 12$ restores every program. From an intact body and reserve 2, seven damage-free cycles reach

$$A_7 = 2 + 7(1.6) = 13.2 = a + 1.2, \qquad p_{\text{start}} = (1 - d)^{84} > 0.$$

Proposition 23 and Corollary 19 then give positive probability of perpetual operation and the stated asymptotic reserve growth on that event. This calculation is for a fixed body; open-ended generation needs the additional hypotheses of Theorem 17.

28.5 Binary arithmetic

For each finite arithmetic task, four bits encode an input and four an addition or multiplication output. A child has six, ten or fourteen sampled correct outputs. A^* searches the same child for a freest correct and a minimum-description-length correct policy; success means reconstructing the full sixteen-input parent's output set. There are 75–256 trials per setting. The reported freedom-to-description-length generalisation-rate ratios range from 1.10 to 5.00 [Bennett, 2023a, 2025a]. For each trial and each selector $c \in \{c_{\text{ff}}, c_{mdl}\}$, define

$$O_{\text{recon}}(c) := \text{Ext}(c) \cap \text{Ext}(I_n).$$

Table 13: Freedom minus each competitor, in mean percentage points over twelve paired trials, with wins, ties and losses at final picks. First pick is selected from generation zero, before offspring. Matched is measured at each competitor's stopping generation.

Benchmark	Competitor	First pick	Final	Matched	Wins, ties, losses
KMNIST	Marginal likelihood	+1.23	+0.16	+0.16	5, 0, 7
KMNIST	GdScore	+9.60	+14.78	+14.78	12, 0, 0
Fashion-MNIST binary	Marginal likelihood	+0.80	+0.67	+0.67	7, 0, 5
Fashion-MNIST binary	GdScore	+1.05	+0.65	+0.65	8, 1, 3
Rotten Tomatoes	Marginal likelihood	+2.46	+2.17	+2.17	11, 0, 1
Rotten Tomatoes	GdScore	+0.77	+0.53	+0.53	9, 0, 3
AG News binary	Marginal likelihood	+6.33	+9.84	+9.84	12, 0, 0
AG News binary	GdScore	+3.33	+1.61	+1.61	7, 0, 5
20NG binary	Marginal likelihood	+0.45	+0.66	+0.41	7, 0, 5
20NG binary	GdScore	+0.86	+0.51	+0.48	8, 0, 4

Table 14: Freedom-selected trajectories as means over twelve trials. V is freedom's selection score. Subscripts give selection generations. Text scores are passing fractions, and image scores are passing fractions raised or lowered according to other passing fractions. Accuracy changes are in percentage points. Validation is the official test accuracy of the network selected once from freedom's generation zero using 512 separate labelled examples.

Benchmark	V_1	V_{20}	Accuracy_1	Accuracy_{20}	Change	Validation
KMNIST	0.8883	0.9000	45.56	47.02	+1.47	48.69
Fashion-MNIST binary	0.7837	0.8417	78.90	78.80	−0.10	79.25
Rotten Tomatoes	0.6386	0.6979	56.32	55.99	−0.34	55.42
AG News binary	0.3138	0.4836	56.33	59.84	+3.51	56.13
20NG binary	0.8446	0.8634	66.13	67.22	+1.10	67.77

Table 15: Training and test inputs of each benchmark. The Rotten Tomatoes training split is the official one after deduplication.

Benchmark	Training inputs	Test inputs
KMNIST	60,000	10,000
Fashion-MNIST binary	12,000	2,000
Rotten Tomatoes	deduplicated	1,066
AG News binary	60,000	3,800
20NG binary	4,586	3,032

Table 16: Data and measurement sizes per trial. Pool inputs are unlabelled. Reference inputs number feature dimension plus one.

Benchmark	Classes	Labels	Pool	Features	Anchor numbers	Joint demand sizes
KMNIST	10	40	160	32	5, 9, 17	1, 2, 3
Fashion-MNIST binary	2	32	128	32	5, 9, 17	1, 2, 3
Rotten Tomatoes	2	128	512	128	17, 33, 65	3, 5, 8
AG News binary	2	32	128	64	9, 17, 33	2, 3, 5
20NG binary	2	256	1,024	128	17, 33, 65	3, 5, 8

Table 17: Calibration ranking score. A rule's score sums each term multiplied by its weight.

Term	Weight
Mean accuracy, on the fold not used in training, of a rule's first-ranked network	1
Mean accuracy of its four selected parents on that fold	0.25
Tenth percentile of parent accuracy across splits	0.10
Best-network inclusion rate	0.05
Selection stability	0.05
Rate at which all four parents reach primary score one	−0.15
Rate at which all their tie-breaking measurements also reach maximum	−0.35
Rate at which rankings contain ties within selected parents or across a selection cutoff	−0.10

Table 18: Fixed primary settings. Both image benchmarks raise or lower the passing fraction, with cap c.

Benchmark	Anchor kind	Anchors	Joint demand size	Cap c
KMNIST	Label	9	3	0.75
Fashion-MNIST binary	Label	9	2	0.5
Rotten Tomatoes	Label	65	8	0
AG News binary	Score	9	5	0
20NG binary	Label	33	5	0

Table 19: Fixed adjustment parameters, rounded. KMNIST's adjustment uses a label-anchor setting with nine anchors and joint demand size one (`label_r9o1`), and an aggregate of label-anchor settings at nine anchors (`label_r9`). Binary Fashion-MNIST's adjustment uses an aggregate lower bound from unresolved checks (`label_bound`) and an aggregate of label-anchor settings at joint demand size three (`label_o3`).

Benchmark	Measurement	a_j	m_j	σ_j
KMNIST	`label_r9o1`	0.61598866	0.89843750	0.02316562
KMNIST	`label_r9`	−0.78775502	0.82421875	0.02371077
Fashion-MNIST binary	`label_bound`	0.73652309	0.47020449	0.12973043
Fashion-MNIST binary	`label_o3`	−0.67641240	0.47716503	0.13272961

Table 20: Training-specification grids. Image settings apply to both image benchmarks. RT denotes Rotten Tomatoes. Epoch and attempt limits apply to ordinary training.

Setting	Images	RT and binary 20NG	Binary AG News
Width	192, 256, 384, 512	256, 384, 576	192, 256, 288
Learning rate	0.0003, 0.001, 0.002	0.001, 0.002, 0.003	0.0005, 0.001, 0.0015
Weight decay	0, 10^{-5}, 10^{-4}, 10^{-3}	0, 10^{-5}, 10^{-4}	0, 10^{-5}, 10^{-4}
Batch size	32, 64, 128	16, 32, 64	32, 64, 128
Dropout	0, 0.05, 0.10	0	0
Epoch limit	280	320	240
Attempt limit	8	10	10

Table 21: Stopping generations of the two competitors on binary 20NG, by trial, under patience three.

Trial	0	1	2	3	4	5	6	7	8	9	10	11
Marginal likelihood	20	14	12	8	11	16	14	6	16	16	12	15
GdScore	14	7	10	20	10	7	12	8	8	11	7	5

Table 22: Settings of the loop simulation.

Setting	Value
Programs	12
Programs a demand needs intact	3
Disruption chance per intact program per tick, background	0.02
Disruption chance per intact program per task attempt	0.05
Capacity spent per program repaired	1
Intact programs required for viability	6
Initial capacity	2
Gain per completed task, deficit	0.5
Gain per completed task, break-even	0.9
Gain per completed task, surplus	1.6
Timelines per setting	300
Horizon in ticks	2,000
Burst tick, young	5
Burst tick, mature	300
Ticks observed after a burst	200

Table 23: Outcome of the loop simulation at each rate of renewal.

Regime	Gain per task	Fraction viable at 2,000 ticks	Mean stored capacity at the end
Deficit	0.5	0.00	0.3
Break-even	0.9	0.17	13.3
Surplus	1.6	0.99	1,347

Table 24: Chance of surviving the 200 ticks after one burst, among timelines viable when the burst occurs. At break-even, 83 timelines were viable at the mature burst tick.

Programs disrupted	Break-even, young	Break-even, mature	Surplus, young	Surplus, mature
6	0.02	0.70	0.61	1.00
9	0.00	0.69	0.22	1.00

Table 25: Binary addition.

	c_{ff}				c_{mdl}			
$\lvert O_k \rvert$	Rate	±95%	AvgExt	StdErr	Rate	±95%	AvgExt	StdErr
6	.11	.039	.75	.008	.10	.037	.48	.012
10	.27	.064	.91	.006	.13	.048	.69	.009
14	.68	.106	.98	.005	.24	.097	.91	.006

Table 26: Binary multiplication.

	c_{ff}				c_{mdl}			
$\lvert O_k \rvert$	Rate	±95%	AvgExt	StdErr	Rate	±95%	AvgExt	StdErr
6	.05	.026	.74	.009	.01	.011	.58	.011
10	.16	.045	.86	.006	.08	.034	.78	.008
14	.46	.061	.96	.003	.21	.050	.93	.003

The table's Rate is the fraction of trials with $O_{\text{recon}}(c) = O_n$; its ±95% column gives the Wald confidence-interval half-width. AvgExt is the trial mean of $|O_{\text{recon}}(c) \cap O_n|/|O_n|$, and StdErr is the standard error of that mean.

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